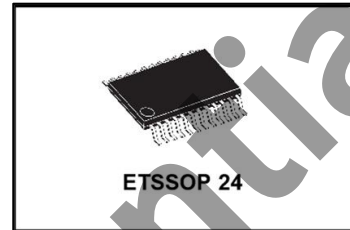




Features

- AEC-Q100 qualified for automotive applications (ongoing)
- 6, 8, 10 and 12 half-bridge outputs
- 4.5V to 32V operating voltage
- 35V dc, 40V@400ms absolute maximum voltage
- 1A RMS current for each output
- 6A maximum current for paralleled outputs
- Low-power sleep mode (1.5μA)
- Supports 3.3V and 5V logic inputs
- SPI for configuration and diagnostics
- 5MHz, 16Bit SPI communication
- Daisy chain functionality
- PWM generators programmable over SPI
- Individual half-bridge PWM operation
- Configurable for high-side, low-side, and H-bridge load driving
- Supports 8Bit duty cycle resolution
- Integrated protection features with per channel detailed diagnostics over SPI
- nFAULT pin output
- VM undervoltage lockout (UVLO)
- VM overvoltage protection (OVP)
- Logic supply undervoltage (VDDUV)

- Overcurrent protection (OCP)
- Enhanced open load detection (OLD)
- Thermal Warning and Shutdown
- Digital Thermal



Description

The HE99MD90xx is a pin-to-pin compatible family of integrated multi-channel half-bridge drivers with 6 to 12 half-bridges. The device features low on-state resistance ($R_{DS(ON)}$) for improved thermal performance during high-current operation. The device can drive brushed-DC (BDC) motors or stepper motors in independent, sequential, or parallel mode. The half-bridges are fully controllable to achieve a forward, reverse, coasting and braking operation of motor.

The device features standard 16Bit, 5MHz Serial Peripheral Interface (SPI) with daisy chain capability for complete configuration and detailed diagnostics. 4 programmable PWM generators are integrated to allow for current limiting during motor operation or LED dimming control.

The device includes numerous protection and diagnostic features including an nFAULT pin to alert the system when a fault occurs. The device features a low-current open load detection (OLD) mode to detect open-load conditions when the nominal load current is small. Digital Thermal are provided for a selective shutdown of a faulty channel. All the outputs are fully protected from short-circuit, undervoltage, and over-temperature conditions.

Part number	Package	Marking	Packing
HE99MD9012xxxx	ETSSOP24	HE99MD9012	2000 / Tape & Reel
HE99MD9010xxxx		HE99MD9010	
HE99MD9008Nxxxx		HE99MD9008N	
HE99MD9006xxxx		HE99MD9006	

* The full part number will be confirmed after the official version of the product

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

Device Comparison Table

DEVICE	UNMBER OF HALF-BRIDGES	UNMBER OF PWM GENERATORS	OPEN-LOAD DETECTION SCHEMES	LINK TO REGISTER MAP
HE99MD9012	12	4	Active OLD, Low-Current Active OLD, Negative-Current Active OLD	Table 3-15
HE99MD9010	10			Table 3-16
HE99MD9008N	8			Table 3-17
HE99MD9006	6			Table 3-18

KVC Confidential

Contents

1	Block diagram and pin description	9
1.1	Block diagram.....	9
1.2	Pin description.....	10
2	Electrical specifications	18
2.1	Absolute maximum ratings.....	18
2.2	ESD Ratings.....	18
2.3	Recommended Operating Conditions.....	18
2.4	Thermal Information	19
2.5	Electrical Characteristics.....	20
2.6	Timing Requirements	24
2.7	Typical Characteristics	25
3	Detailed Description.....	29
3.1	Overview.....	29
3.2	Feature Description	30
3.3	Device Functional Modes.....	58
3.4	Programming.....	60
3.5	Register Map	65
4	Typical Application	109
4.1	Primary Application	109
4.2	Alternative Application.....	111
5	Package and information.....	112
5.1	ETSSOP mechanical data	112
6	Recommended pad size	113
7	Customer Service.....	114
8	Revision history.....	115

List of tables

Table 1-1 Pin Functions - HE99MD9012	10
Table 1-2 Pin Functions - HE99MD9010	12
Table 1-3 Pin Functions - HE99MD9008N	14
Table 1-4 Pin Functions - HE99MD9006	16
Table 2-1 Absolute maximum ratings	18
Table 2-2 ESD Ratings	18
Table 2-3 Recommended operating conditions	18
Table 2-4 Thermal information	19
Table 3-5 Electrical characteristics	20
Table 3-6 Timing requirements	24
Table 3-1 HE99MD90xx Driver external components	30
Table 3-2 Motor Operation in Continuous Mode (Motor Connected between HB1 and HB2)	30
Table 4-3 PWM Mapping of HE99MD90xx	33
Table 4-4 PWM Frequency	34
Table 4-5 Motor Operation in Parallel Mode (Continuous Operation) (with Motor Connected between OUT1/2/3 and OUT4/5/6)	35
Table 4-6 Overcurrent Protection	50
Table 4-7 Summary of OLD features	51
Table 4-8 OLD Configuration	53
Table 4-9 Functional Modes	59
Table 4-10 Fault Action and Response	59
Table 4-11 SDI Input Data Word Format	61
Table 4-12 SDO Output Data Word Format	61
Table 4-13 Summary of HE99MD90xx Device Family	65
Table 4-14 Control Registers Access Type Codes	65
Table 3-15 HE99MD9012 Register Map	66
Table 3-16 HE99MD9010 Register Map	68
Table 3-17 HE99MD9008N Register Map	70
Table 3-18 HE99MD9006 Register Map	72
Table 4-19 Status Registers Summary Table	74
Table 4-20 IC Status Register	74
Table 4-21 IC Status Register Field Descriptions	74
Table 4-22 Overcurrent Protection (OCP) Status 1 Register	75
Table 4-23 Overcurrent Protection (OCP) Status 1 Register Field Descriptions	75
Table 4-24 Overcurrent Protection (OCP) Status 2 Register	76
Table 4-25 Overcurrent Protection (OCP) Status 2 Register Field Descriptions	76
Table 4-26 Overcurrent Protection (OCP) Status 3 Register	77
Table 4-27 Overcurrent Protection (OCP) Status 3 Register Field Descriptions	77
Table 4-28 Open-Load Detect (OLD) Status 1 Register	78
Table 4-29 Open-Load Detect (OLD) Status 1 Register Field Descriptions	78

Table 4-30 Open-Load Detect (OLD) Status 2 Register.....	79
Table 4-31 Open-Load Detect (OLD) Status 2 Register Field Descriptions.....	79
Table 4-32 Open-Load Detect (OLD) Status 3 Register.....	80
Table 4-33 Open-Load Detect (OLD) Status 3 Register Field Descriptions.....	80
Table 4-34 Additional Configuration (ADD_STAT_1) Register.....	81
Table 4-35 Additional Configuration (ADD_STAT_1) Register Field Descriptions.....	81
Table 4-36 Digital Thermal Status (TC_STAT_1) Register.....	81
Table 4-37 Digital Thermal Status (TC_STAT_1) Register Field Descriptions.....	81
Table 4-38 Control Registers Summary Table.....	82
Table 4-39 Configuration Register.....	84
Table 4-40 Configuration Register Field Descriptions.....	84
Table 4-41 Operation Control 1 Register.....	85
Table 4-42 Operation Control 1 Register Field Descriptions.....	85
Table 4-43 Operation Control 2 Register.....	86
Table 4-44 Operation Control 2 Register Field Descriptions.....	86
Table 4-45 Operation Control 3 Register.....	87
Table 4-46 Operation Control 3 Register Field Descriptions.....	87
Table 4-47 PWM Control 1 Register.....	88
Table 4-48 PWM Control 1 Register Field Descriptions.....	88
Table 4-49 PWM Control 2 Register.....	89
Table 4-50 PWM Control 2 Register Field Descriptions.....	89
Table 4-51 Free-Wheeling Control 1 Register.....	90
Table 4-52 Free-Wheeling Control 1 Register Field Descriptions.....	90
Table 4-53 Free-Wheeling Control 2 Register.....	91
Table 4-54 Free-Wheeling Control 2 Register Field Descriptions.....	91
Table 4-55 PWM Map Control 1 Register.....	92
Table 4-56 PWM Map Control 1 Register Field Descriptions.....	92
Table 4-57 PWM Map Control 2 Register.....	93
Table 4-58 PWM Map Control 2 Register Field Descriptions.....	93
Table 4-59 PWM Map Control 3 Register.....	94
Table 4-60 PWM Map Control 3 Register Field Descriptions.....	94
Table 4-61 PWM Frequency Control Register.....	95
Table 4-62 PWM Frequency Control Register Field Descriptions.....	95
Table 4-63 PWM Duty Control 1 Register.....	96
Table 4-64 PWM Duty Control 1 Register Field Descriptions.....	96
Table 4-65 PWM Duty Control 2 Register.....	96
Table 4-66 PWM Duty Control 2 Register Field Descriptions.....	96
Table 4-67 PWM Duty Control 3 Register.....	97
Table 4-68 PWM Duty Control 3 Register Field Descriptions.....	97
Table 4-69 PWM Duty Control 4 Register.....	97
Table 4-70 PWM Duty Control 4 Register Field Descriptions.....	97
Table 4-71 Slew Rate Control 1 Register.....	98

Table 4-72 Slew Rate Control 1 Register Field Descriptions	98
Table 4-73 Slew Rate Control 2 Register	99
Table 4-74 Slew Rate Control 2 Register Field Descriptions	99
Table 4-75 Open-Load Detect (OLD) Control (OLD_CTRL_1) Register	100
Table 4-76 Open-Load Detect (OLD) Control (OLD_CTRL_1) Register Field Descriptions	100
Table 4-77 Open-Load Detect (OLD) Control (OLD_CTRL_2) Register	101
Table 4-78 Open-Load Detect (OLD) Control (OLD_CTRL_2) Register Field Descriptions	101
Table 4-79 Open-Load Detect (OLD) Control (OLD_CTRL_3) Register	102
Table 4-80 Open-Load Detect (OLD) Control (OLD_CTRL_3) Register Field Descriptions	102
Table 4-81 Open-Load Detect (OLD) Control (OLD_CTRL_4) Register	103
Table 4-82 Open-Load Detect (OLD) Control (OLD_CTRL_4) Register Field Descriptions	103
Table 4-83 Additional Configuration (ADD_CONFIG_CTRL) Register	104
Table 4-84 Additional Configuration (ADD_CONFIG_CTRL) Register Field Descriptions	104
Table 4-85 Additional Control External (ADD_CTRL_EX) Register	104
Table 4-86 Additional Control External (ADD_CTRL_EX) Register Field Descriptions	104
Table 4-87 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register	106
Table 4-88 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register Field Descriptions	106
Table 4-89 Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register	107
Table 4-90 Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register Field Descriptions	107
Table 4-91 Additional Open-Load Detect Control (ADD_OLD_CTRL_3) Register	108
Table 4-92 Additional Open-Load Detect Control (ADD_OLD_CTRL_3) Register Field Descriptions	108
Table 5-1 Design Parameters	110

List of figures

Figure 1-1 Block diagram	9
Figure 1-2 HE99MD9012 Thermal Pad Top View	10
Figure 1-3 HE99MD9010 Thermal Pad Top View	12
Figure 1-4 HE99MD9008N Thermal Pad Top View	14
Figure 1-5 HE99MD9006 Thermal Pad Top View	16
Figure 3-1 SPI Timing	24
Figure 3-2 VM Sleep Mode Current (I_{VMQ}) vs Supply Voltage (V_{VM})	25
Figure 3-3 VM Sleep Mode Current (I_{VMQ}) vs Ambient Temperature (T_A)	25
Figure 3-4 VDD Sleep Mode Current (I_{VDDQ}) vs Supply Voltage (V_{VDD})	25
Figure 3-5 VDD Sleep Mode Current (I_{VDDQ}) vs Ambient Temperature (T_A)	25
Figure 3-6 VM Operating Mode Current (I_{VM}) vs Supply Voltage (V_{VM})	25
Figure 3-7 VM Operating Mode Current (I_{VM}) vs Ambient Temperature (T_A)	25
Figure 3-8 VDD Operating Mode Current (I_{VDD}) vs Supply Voltage (V_{VDD})	26
Figure 3-9 VDD Operating Mode Current (I_{VDD}) vs Ambient Temperature (T_A)	26
Figure 3-10 High Side On-State Resistance ($R_{DS(ON)}$) vs Supply Voltage (V_{VM})	26
Figure 3-11 High Side On-State Resistance ($R_{DS(ON)}$) vs Ambient Temperature (T_A)	26
Figure 3-12 Low Side On-State Resistance ($R_{DS(ON)}$) vs Supply Voltage (V_{VM})	26
Figure 3-13 Low Side On-State Resistance ($R_{DS(ON)}$) vs Ambient Temperature (T_A)	26
Figure 3-14 High-Side Open-Load Detection Current (I_{OLD}) vs Supply Voltage (V_{VM})	27
Figure 3-15 High-Side Open-load Detection Current (I_{OLD}) vs Ambient Temperature (T_A)	27
Figure 3-16 Low-Side Open-Load Detection Current (I_{OLD}) vs Supply Voltage (V_{VM})	27
Figure 3-17 Low-Side Open-load Detection Current (I_{OLD}) vs Ambient Temperature (T_A)	27
Figure 3-18 Low open-load detection current (I_{OLD_LOW}) vs supply voltage (V_{VM})	27
Figure 3-19 Low open-load detection current (I_{OLD_LOW}) vs ambient temperature (T_A)	27
Figure 3-20 Open-load pull-down current (I_{OL_VM}) vs supply voltage (V_{VM})	28
Figure 3-21 Open-load pull-up current (I_{OL_GND}) vs supply voltage (V_{VM})	28
Figure 4-1 Continuous Mode (Forward Direction)	31
Figure 4-2 Continuous Mode (Reverse Direction)	31
Figure 4-3 Continuous Mode (Coast - From Forward Direction)	31
Figure 4-4 Continuous Mode (Coast - From Reverse Direction)	31
Figure 4-5 Continuous Mode (Brake - Low-Side)	32
Figure 4-6 Continuous Mode (Brake - High-Side)	32
Figure 4-7 PWM Mode (Synchronous Rectification=OFF)	33
Figure 4-8 PWM Mode (Synchronous Rectification=ON)	33
Figure 4-9 Parallel Mode (Forward Direction)	36
Figure 4-10 Parallel Mode (Reverse Direction)	36
Figure 4-11 Parallel Mode (Coast from Forward Direction)	37
Figure 4-12 Parallel Mode (Coast from Reverse Direction)	37
Figure 4-13 Parallel Mode (Brake - Low-Side)	38
Figure 4-14 Parallel Mode (Brake - High-Side)	38

Figure 4-15 Parallel Mode (PWM with Synchronous Rectification = OFF)	40
Figure 4-16 Parallel Mode (PWM with Synchronous Rectification = ON)	40
Figure 4-17 Slew Rate Circuit Implementation	43
Figure 4-18 Slew Rate Timings	43
Figure 4-19 Cross Conduction Protection	44
Figure 4-20 Dead Time	44
Figure 4-21 Propagation Delay	45
Figure 4-22 Logic Level Input Pin Structure (nSLEEP, SCLK and SDI).....	46
Figure 4-23 Logic Level Input Pin Structure (nSCS).....	46
Figure 4-24 Open Drain Output Pin Structure (nFAULT)	47
Figure 4-25 Push Pull Output Pin (SDO) Structure	47
Figure 4-26 VM UVLO Operation.....	48
Figure 4-27 Over Voltage Protection	49
Figure 4-28 VDD UVLO Operation.....	49
Figure 4-29 Over Current Protection.....	51
Figure 4-30 Active open-load detection	52
Figure 4-31 Negative Current Flow in High-Side FET of OUT2 Channel.....	54
Figure 4-32 Negative Current Flow in OUT2 by Body Diode of High-Side FET.....	54
Figure 4-33 Waveforms Showing False OLD With Negative-Current OLD Disabled.....	55
Figure 4-34 Waveforms Showing Operation with Negative-Current OLD Enabled.....	55
Figure 4-35 Low-Current OLD.....	56
Figure 4-36 Flowchart for Enabling Low-Current OLD (Continuous Mode)	57
Figure 4-37 SPI Data Frame	60
Figure 4-38 SPI Operation Without Daisy Chain	62
Figure 4-39 SPI Operation With Daisy Chain	62
Figure 4-40 Daisy Chain SPI Operation.....	63
Figure 4-41 Header Bits	64
Figure 4-42 Daisy Chain Read Registers	64
Figure 4-43 SPI Slave Timing Diagram	65
Figure 5-1 Primary Application Schematic (Automotive HVAC Application)	109
Figure 5-2 Alternative Application Schematic (Automotive Side-Mirror Application).....	111
Figure 6-1 Mechanical data.....	112
Figure 7-1 Recommended pad size	113

1 Block diagram and pin description

1.1 Block diagram

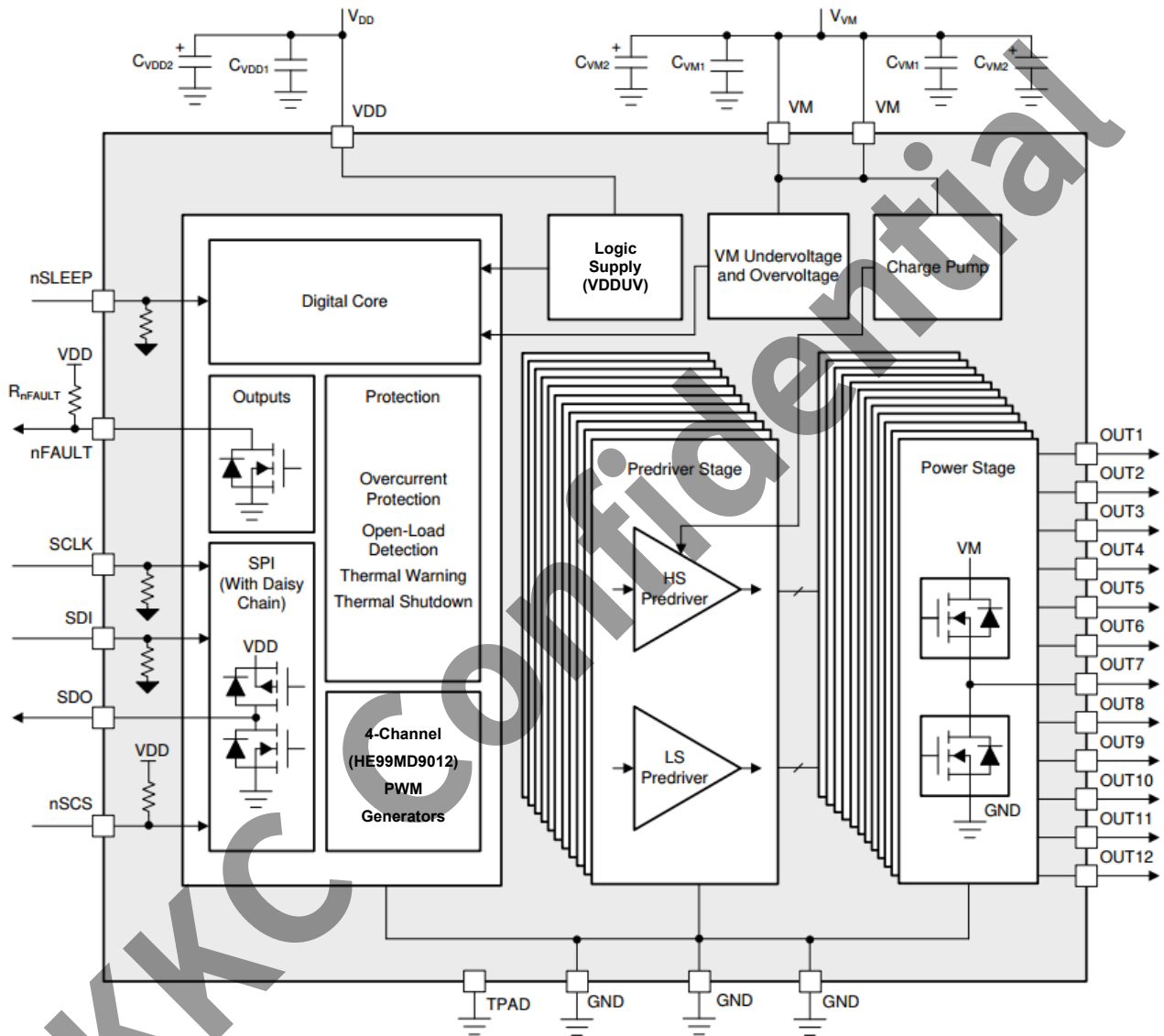


Figure 1-1 Block diagram

1.2 Pin description

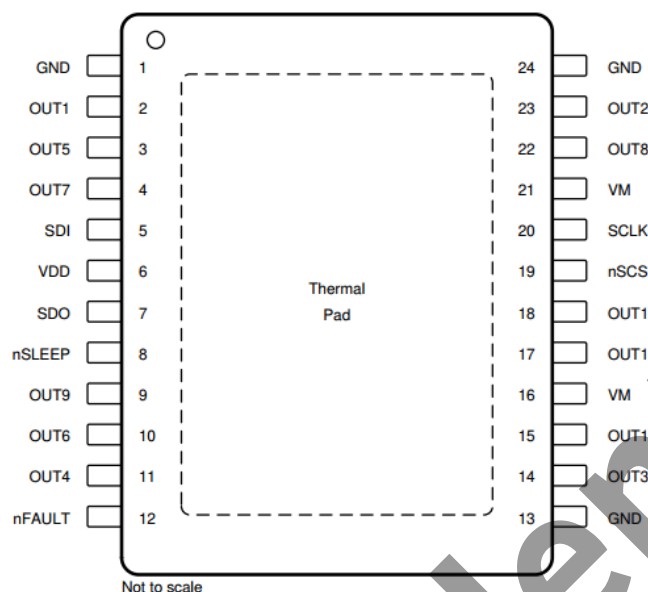


Figure 1-2 HE99MD9012 Thermal Pad Top View

Table 1-1 Pin Functions - HE99MD9012

PIN NAME	No.	TYPE	DESCRIPTION
GND	13	PWR	Device power ground. Connect the GND pin to the system ground.
GND	24	PWR	Device power ground. Connect the GND pin to the system ground.
GND	1	PWR	Device power ground. Connect the GND pin to the system ground.
nFAULT	12	OD	Fault indicator output. This pin is pulled logic low during a fault condition and requires an external pull-up resistor.
nSCS	19	I	Serial chip select. A logic low enables serial interface communication. Internal pull-up.
nSLEEP	8	I	Drivers enable pin. The device goes to low-power sleep mode when this pin is logic low. Internal pull-down.
OUT1	2	O	Half-bridge 1 output
OUT2	23	O	Half-bridge 2 output
OUT3	14	O	Half-bridge 3 output
OUT4	11	O	Half-bridge 4 output
OUT5	3	O	Half-bridge 5 output
OUT6	10	O	Half-bridge 6 output
OUT7	4	O	Half-bridge 7 output
OUT8	22	O	Half-bridge 8 output
OUT9	9	O	Half-bridge 9 output
OUT10	15	O	Half-bridge 10 output
OUT11	17	O	Half-bridge 11 output
OUT12	18	O	Half-bridge 12 output
SCLK	20	I	Serial clock input. Serial data is shifted out and captured on the corresponding rising and falling edge on this pin. Internal pull-down.
SDI	5	I	Serial data input. Data is captured on the falling edge of the SCLK pin. Internal pull-

PIN NAME	No.	TYPE	DESCRIPTION
			down.
SDO	7	O	Serial data output. Data is shifted out on the rising edge of the SCLK pin.
VDD	6	PWR	Logic power supply input. Connect a X5R or X7R, 0.1μF, VDD-rated ceramic capacitor and greater than or equal to 1μF bulk capacitance between the VDD and GND pins.
VM	16	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1μF, VM-rated ceramic capacitor and greater than or equal to 10μF bulk capacitance between the VM and GND pins.
VM	21	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1μF, VM-rated ceramic capacitor and greater than or equal to 10μF bulk capacitance between the VM and GND pins.

KKC Confidential

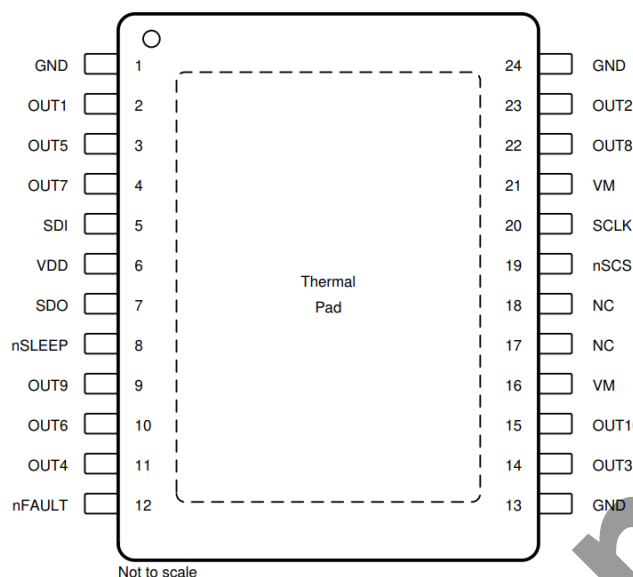


Figure 1-3 HE99MD9010 Thermal Pad Top View

Table 1-2 Pin Functions - HE99MD9010

PIN NAME	No.	TYPE	DESCRIPTION
GND	13	PWR	Device power ground. Connect the GND pin to the system ground.
GND	24	PWR	Device power ground. Connect the GND pin to the system ground.
GND	1	PWR	Device power ground. Connect the GND pin to the system ground.
NC	17	-	Not connected
NC	18	-	Not connected
nFAULT	12	OD	Fault indicator output. This pin is pulled logic low during a fault condition and requires an external pull-up resistor.
nSCS	19	I	Serial chip select. A logic low enables serial interface communication. Internal pull-up.
nSLEEP	8	I	Drivers enable pin. The device goes to low-power sleep mode when this pin is logic low. Internal pull-down.
OUT1	2	O	Half-bridge 1 output
OUT2	23	O	Half-bridge 2 output
OUT3	14	O	Half-bridge 3 output
OUT4	11	O	Half-bridge 4 output
OUT5	3	O	Half-bridge 5 output
OUT6	10	O	Half-bridge 6 output
OUT7	4	O	Half-bridge 7 output
OUT8	22	O	Half-bridge 8 output
OUT9	9	O	Half-bridge 9 output
OUT10	15	O	Half-bridge 10 output
SCLK	20	I	Serial clock input. Serial data is shifted out and captured on the corresponding rising and falling edge on this pin. Internal pull-down.
SDI	5	I	Serial data input. Data is captured on the falling edge of the SCLK pin. Internal pull-down.
SDO	7	O	Serial data output. Data is shifted out on the rising edge of the SCLK pin.

PIN NAME	No.	TYPE	DESCRIPTION
VDD	6	PWR	Logic power supply input. Connect a X5R or X7R, 0.1μF, VDD-rated ceramic capacitor and greater than or equal to 1μF bulk capacitance between the VDD and GND pins.
VM	16	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1μF, VM-rated ceramic capacitor and greater than or equal to 10μF bulk capacitance between the VM and GND pins.
VM	21	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1μF, VM-rated ceramic capacitor and greater than or equal to 10μF bulk capacitance between the VM and GND pins.

KKC Confidential

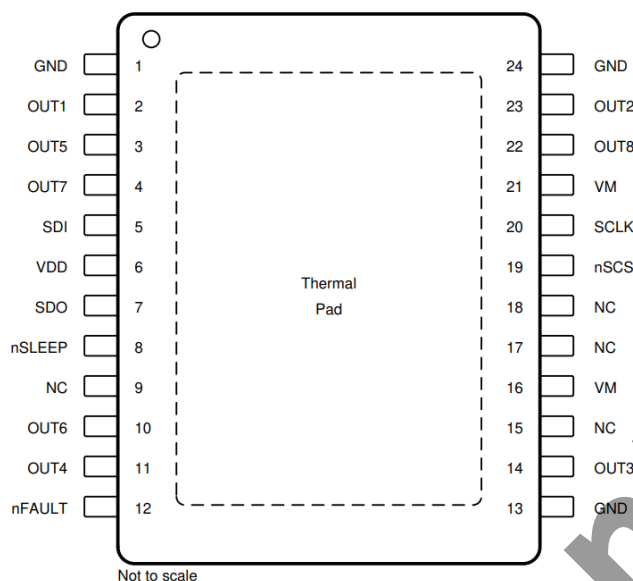


Figure 1-4 HE99MD9008N Thermal Pad Top View

Table 1-3 Pin Functions - HE99MD9008N

PIN NAME	No.	TYPE	DESCRIPTION
GND	13	PWR	Device power ground. Connect the GND pin to the system ground.
GND	24	PWR	Device power ground. Connect the GND pin to the system ground.
GND	1	PWR	Device power ground. Connect the GND pin to the system ground.
NC	9	-	Not connected
NC	15	-	Not connected
NC	17	-	Not connected
NC	18	-	Not connected
nFAULT	12	OD	Fault indicator output. This pin is pulled logic low during a fault condition and requires an external pull-up resistor.
nSCS	19	I	Serial chip select. A logic low enables serial interface communication. Internal pull-up.
nSLEEP	8	I	Drivers enable pin. The device goes to low-power sleep mode when this pin is logic low. Internal pull-down.
OUT1	2	O	Half-bridge 1 output
OUT2	23	O	Half-bridge 2 output
OUT3	14	O	Half-bridge 3 output
OUT4	11	O	Half-bridge 4 output
OUT5	3	O	Half-bridge 5 output
OUT6	10	O	Half-bridge 6 output
OUT7	4	O	Half-bridge 7 output
OUT8	22	O	Half-bridge 8 output
SCLK	20	I	Serial clock input. Serial data is shifted out and captured on the corresponding rising and falling edge on this pin. Internal pull-down.
SDI	5	I	Serial data input. Data is captured on the falling edge of the SCLK pin. Internal pull-down.

PIN NAME	No.	TYPE	DESCRIPTION
SDO	7	O	Serial data output. Data is shifted out on the rising edge of the SCLK pin.
VDD	6	PWR	Logic power supply input. Connect a X5R or X7R, 0.1µF, VDD-rated ceramic capacitor and greater than or equal to 1µF bulk capacitance between the VDD and GND pins.
VM	16	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1µF, VM-rated ceramic capacitor and greater than or equal to 10µF bulk capacitance between the VM and GND pins.
VM	21	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1µF, VM-rated ceramic capacitor and greater than or equal to 10µF bulk capacitance between the VM and GND pins.

KKC Confidential

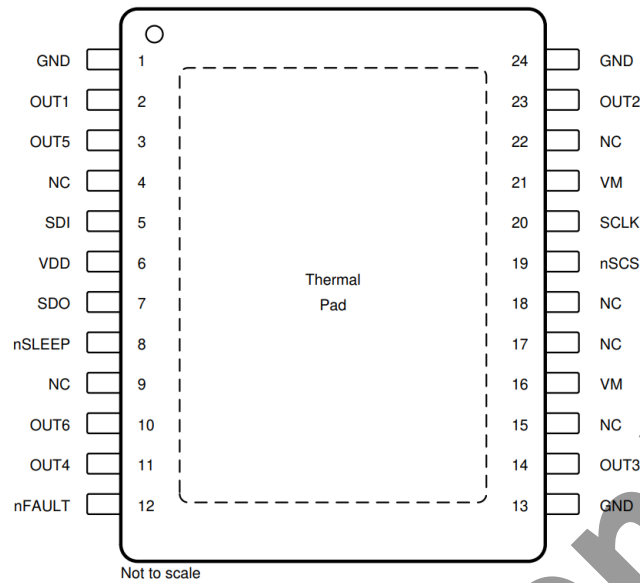


Figure 1-5 HE99MD9006 Thermal Pad Top View

Table 1-4 Pin Functions - HE99MD9006

PIN NAME	No.	TYPE	DESCRIPTION
GND	13	PWR	Device power ground. Connect the GND pin to the system ground.
GND	24	PWR	Device power ground. Connect the GND pin to the system ground.
GND	1	PWR	Device power ground. Connect the GND pin to the system ground.
NC	4	-	Not connected
NC	9	-	Not connected
NC	15	-	Not connected
NC	17	-	Not connected
NC	18	-	Not connected
NC	22	-	Not connected
nFAULT	12	OD	Fault indicator output. This pin is pulled logic low during a fault condition and requires an external pull-up resistor.
nSCS	19	I	Serial chip select. A logic low enables serial interface communication. Internal pull-up.
nSLEEP	8	I	Drivers enable pin. The device goes to low-power sleep mode when this pin is logic low. Internal pull-down.
OUT1	2	O	Half-bridge 1 output
OUT2	23	O	Half-bridge 2 output
OUT3	14	O	Half-bridge 3 output
OUT4	11	O	Half-bridge 4 output
OUT5	3	O	Half-bridge 5 output
OUT6	10	O	Half-bridge 6 output
SCLK	20	I	Serial clock input. Serial data is shifted out and captured on the corresponding rising and falling edge on this pin. Internal pull-down.
SDI	5	I	Serial data input. Data is captured on the falling edge of the SCLK pin. Internal pull-down.
SDO	7	O	Serial data output. Data is shifted out on the rising edge of the SCLK pin.

PIN NAME	No.	TYPE	DESCRIPTION
VDD	6	PWR	Logic power supply input. Connect a X5R or X7R, 0.1 μ F, VDD-rated ceramic capacitor and greater than or equal to 1 μ F bulk capacitance between the VDD and GND pins.
VM	16	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1 μ F, VM-rated ceramic capacitor and greater than or equal to 10 μ F bulk capacitance between the VM and GND pins.
VM	21	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1 μ F, VM-rated ceramic capacitor and greater than or equal to 10 μ F bulk capacitance between the VM and GND pins.

KKC Confidential

2 Electrical specifications

2.1 Absolute maximum ratings

Table 2-1 Absolute maximum ratings

Parameter	MIN	MAX	UNIT	Number
Power supply pin voltage (VM)	-0.3	40 ⁽¹⁾	V	P_2.1.1
Logic supply pin voltage (VDD)	-0.3	5.75	V	P_2.1.2
Output pin voltage (OUTx)	-0.7	VM + 0.7	V	P_2.1.3
Logic pin input voltage (nSCS, nSLEEP, SCLK, SDI)	-0.3	VDD + 0.3	V	P_2.1.4
Logic pin output voltage (nFAULT, SDO)	-0.3	VDD + 0.3	V	P_2.1.5
Continuous supply current (VM pins combined)	0	6	A	P_2.1.6
Peak output current drive (OUTx)	Internally Limited	Internally Limited	A	P_2.1.7
Continuous sink current (GND pins combined)	0	6	A	P_2.1.8
Junction temperature, T _J	-40	175	°C	P_2.1.9
Storage temperature, T _{stg}	-65	150	°C	P_2.1.10

(1) 40V for t_s400ms during load dump transient.

Note: All maximum ratings are absolute ratings. Leaving the limitation of any of these values may cause irreversible damage of the integrated circuit!

2.2 ESD Ratings

Table 2-2 ESD Ratings

Symbol	Parameter	Conditions	PIN	VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human body model (HBM), per AEC Q100-002 ⁽¹⁾	OUTx and VM pins	±4000	V
			Other pins	±2000	
		Charged device model (CDM), per AEC Q100-011	Corner pins (1, 12, 13, and 24)	±750	
			Other pins	±500	

(1) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001 specification.

2.3 Recommended Operating Conditions

over operating ambient temperature range (unless otherwise noted)

Table 2-3 Recommended operating conditions

Symbol	Parameter	MIN	NOM	MAX	UNIT	Number
V _{VM}	Power supply voltage (VM)	4.5		32	V	P_2.3.1
V _{VDD}	Logic supply voltage (VDD)	3		5.5	V	P_2.3.2
V _{IN}	Logic input voltage (nSCS, nSLEEP, SCLK, SDI)	0		5.5	V	P_2.3.3
V _{OD}	Open drain pullup voltage (nFAULT)	0		5.5	V	P_2.3.4
I _{OD}	Open drain output current (nFAULT)	0		5	mA	P_2.3.5
V _{OP}	Push-pull pullup voltage (SDO)	0		5.5	V	P_2.3.6
I _{OP}	Push-pull output current (SDO)	0		5	mA	P_2.3.7
T _A	Operating ambient temperature	-40		125	°C	P_2.3.8
T _J	Operating junction temperature	-40		175	°C	P_2.3.9

Note: All parameters are guaranteed in the junction temperature range -40 to 150°C (unless otherwise specified); the device is still operative and functional at higher temperatures (up to 175°C). Note: Parameters limits at junction temperatures higher than 150°C may change with respect to what is specified as per the standard temperature range. Note: Device functionality at high

junction temperature is guaranteed by characterization.

2.4 Thermal Information

Table 2-4 Thermal information

Symbol	Parameter	(ETSSOP)	UNIT
R _{θJA}	Junction-to-ambient thermal resistance	30.2	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	23.7	°C/W
R _{θJB}	Junction-to-board thermal resistance	10.1	°C/W

KKC Confidential

2.5 Electrical Characteristics

At $T_J = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, $V_{VM} = 4.5$ to 32V (Main Supply), $V_{VDD} = 3$ to 5.5V (Logic Supply) (unless otherwise noted). Typical limits apply for $T_A = 25^{\circ}\text{C}$, $V_{VM} = 13.5\text{V}$, $V_{VDD} = 3.3\text{V}$

Table 2-5 Electrical characteristics

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	Number
POWER SUPPLIES (VDD, VM)							
I_{VMQ}	VM sleep mode current	$V_{VM} = 13.5\text{V}$, $n\text{SLEEP} = 0_b$, $T_A = 25^{\circ}\text{C}$		0.35	1	μA	P_2.5.1
		$V_{VM} = 13.5\text{V}$, $n\text{SLEEP} = 0_b$, $T_A = 125^{\circ}\text{C}$			1	μA	P_2.5.2
I_{VDDQ}	VDD sleep mode current	$V_{VM} = 13.5\text{V}$, $V_{VDD} = 3.3\text{V}$, $n\text{SLEEP} = 0_b$, $T_A = 25^{\circ}\text{C}$		0.01	0.3	μA	P_2.5.3
		$V_{VM} = 13.5\text{V}$, $V_{VDD} = 3.3\text{V}$, $n\text{SLEEP} = 0_b$, $T_A = 125^{\circ}\text{C}$			2	μA	P_2.5.4
I_{VM}	VM operating mode current	$V_{VM} = 13.5\text{V}$, $n\text{SLEEP} = 1_b$, All High-Side FETs = 'ON', $T_A = 25^{\circ}\text{C}$		5	10	mA	P_2.5.5
		$V_{VM} = 13.5\text{V}$, $n\text{SLEEP} = 1_b$, All High-Side FETs = 'ON', $T_A = 125^{\circ}\text{C}$			10	mA	P_2.5.6
I_{VDD}	VDD operating mode current	$V_{VM} = 13.5\text{V}$, $V_{VDD} = 3.3\text{V}$, $n\text{SLEEP} = 1_b$, All High-Side FETs = 'ON', SPI = 'ON' (5 MHz), $T_A = 25^{\circ}\text{C}$			1	mA	P_2.5.7
		$V_{VM} = 13.5\text{V}$, $V_{VDD} = 3.3\text{V}$, $n\text{SLEEP} = 1_b$, All High-Side FETs = 'ON', SPI = 'ON' (5 MHz), $T_A = 125^{\circ}\text{C}$			1	mA	P_2.5.8
t_{WAKE}	Wake-up time	$n\text{SLEEP}$ high to SPI ready			200	μs	P_2.5.9
t_{SLEEP}	Turnoff time	$n\text{SLEEP}$ low to device sleep			50	μs	P_2.5.10
LOGIC-LEVEL INPUTS (nSLEEP, SCLK, SDI)							
V_{IL}	Input logic low voltage		0		$0.3 \cdot V_{DD}$	V	P_2.5.11
V_{IH}	Input logic high voltage		$0.7 \cdot V_{DD}$		VDD	V	P_2.5.12
V_{HYS}	Input logic hysteresis		0.2			V	P_2.5.13
I_{IL}	Input logic low current	$V_{IN} = 0\text{V}$	-1		1	μA	P_2.5.14
I_{IH}	Input logic high current	$V_{IN} = V_{VDD}$		34	75	μA	P_2.5.15
C_{ID}	Input capacitance				15	pF	P_2.5.16
LOGIC-LEVEL INPUTS (nSCS)							
V_{IL}	Input logic low voltage		0		$0.3 \cdot V_{DD}$	V	P_2.5.17
V_{IH}	Input logic high voltage		$0.7 \cdot V_{DD}$		VDD	V	P_2.5.18
V_{HYS}	Input logic hysteresis		0.2			V	P_2.5.19
I_{IL}	Input logic low current	$V_{IN} = 0\text{V}$		34	75	μA	P_2.5.20
I_{IH}	Input logic high current	$V_{IN} = V_{VDD}$	-1		1	μA	P_2.5.21
C_{ID}	Input				15	pF	P_2.5.22

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	Number
	capacitance						
OPEN-DRAIN OUTPUTS (nFAULT)							
V _{OL}	Output logic low voltage	I _{OD} = 5 mA	0		0.4	V	P_2.5.23
I _{OH}	Output logic high current	V _{OD} = 5 V	-1		1	μA	P_2.5.24
C _{OD}	Output capacitance				15	pF	P_2.5.25
PUSH-PULL OUTPUTS (SDO)							
V _{OL}	Output logic low voltage	I _{OP} = 5 mA	0		0.4	V	P_2.5.26
V _{OH}	Output logic high voltage	I _{OP} = 5 mA	VDD-0.6		VDD	V	P_2.5.27
I _{OL}	Output logic low current	V _{OP} = 0 V	-1		1	μA	P_2.5.28
I _{OH}	Output logic high current	V _{OP} = V _{VDD}	-1		1	μA	P_2.5.29
C _{OD}	Output capacitance				30	pF	P_2.5.30
DRIVER OUTPUTS (OUTx)							
R _{DS(ON)}	High-side MOSFET on resistance	V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 25°C		0.75	1.1	Ω	P_2.5.31
		V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 125°C			1.5	Ω	P_2.5.32
	Low-side MOSFET on resistance	V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 25°C		0.75	1.1	Ω	P_2.5.33
		V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 125°C			1.5	Ω	P_2.5.34
SR	Output rise and fall time (high-side and low-side)	V _{VM} = 13.5 V, 10-90%, R _{LOAD} = 27 Ω, HBx_SR = 0 _b		0.6		V/μs	P_2.5.35
		V _{VM} = 13.5 V, 10-90%, R _{LOAD} = 27 Ω, HBx_SR = 1 _b		2.5		V/μs	P_2.5.36
t _{DEAD}	Output dead time (high to low / low to high)	V _{VM} = 13.5 V, HBx_SR = 0 _b , HS/LS driver OFF to LS/HS driver ON	8	20	32	μs	P_2.5.37
		V _{VM} = 13.5 V, HBx_SR = 1 _b , HS/LS driver OFF to LS/HS driver ON	2	5	15	μs	P_2.5.38
t _{PD}	Propagation delay (high-side / low-side ON/OFF)	High-side ON or low-side ON command (SPI last transition) to OUTx transition from Hi-Z state, SR = 0 _b	5	12	25	μs	P_2.5.39
		High-side ON or low-side ON command (SPI last transition) to OUTx transition from Hi-Z state, SR = 1 _b	3	5	10	μs	P_2.5.40
I _{LEAK}	Leakage current low-side	V _{OUTx} = 13.5 V, nSLEEP = 1 _b , SR = 0 _b		6	10	μA	P_2.5.41
		V _{OUTx} = 13.5 V, nSLEEP = 1 _b , SR = 1 _b		6	10	μA	P_2.5.42
		V _{OUTx} = 13.5 V, nSLEEP = 0 _b		4	10	μA	P_2.5.43
	Leakage current high-side	V _{OUTx} = 0 V, nSLEEP = 1 _b			2	μA	P_2.5.44
		V _{OUTx} = 0 V, nSLEEP = 0 _b			2	μA	P_2.5.45
PWM MODE							
f _{PWM}	PWM switching frequency	PWM_CHx_FREQ = 00 _b	56	80	104	Hz	P_2.5.46
		PWM_CHx_FREQ = 01 _b	70	100	130	Hz	P_2.5.47

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	Number
		PWM_CHx_FREQ = 10 _b	140	200	260	Hz	P_2.5.48
		PWM_CHx_FREQ = 11 _b	1400	2000	2600	Hz	P_2.5.49
PROTECTION CIRCUITS							
V _{UVLO}	Supply undervoltage lockout (UVLO)	Supply rising	4		4.5	V	P_2.5.50
		Supply falling	3.8		4.3	V	P_2.5.51
V _{UVLO_HYS}	Supply undervoltage lockout hysteresis	Rising to falling threshold		200		mV	P_2.5.52
t _{UVLO}	Supply undervoltage deglitch time			10		μs	P_2.5.53
V _{OVP}	Supply overvoltage protection (OVP)	Supply rising, EXT_OVP = 0 _b	21		25	V	P_2.5.54
		Supply falling, EXT_OVP = 0 _b	20		24	V	P_2.5.55
		Supply rising, EXT_OVP = 1 _b	32.7		35	V	P_2.5.56
		Supply falling, EXT_OVP = 1 _b	32		34.3	V	P_2.5.57
V _{OVP_HYS}	Supply overvoltage protection hysteresis	Rising to falling threshold, EXT_OVP = 0 _b		1		V	P_2.5.58
		Rising to falling threshold, EXT_OVP = 1 _b		0.7		V	P_2.5.59
t _{OVP}	Supply overvoltage deglitch time			10		μs	P_2.5.60
V _{VDDUV}	Digital output supply undervoltage	Supply rising	2.45		3	V	P_2.5.61
		Supply falling	2.4		2.95	V	P_2.5.62
V _{VDDUV_HYS}	Digital output supply undervoltage hysteresis	Rising to falling threshold		75		mV	P_2.5.63
I _{OC}	Overcurrent protection trip point ^{(1) (2)}		1.3	1.8	2.3	A	P_2.5.64
t _{OC}	Overcurrent protection deglitch time	OCP_DEG = 000 _b	6	10	14	μs	P_2.5.65
		OCP_DEG = 001 _b	2.6	5	7.9	μs	P_2.5.66
		OCP_DEG = 010 _b	0.4	2.5	5.9	μs	P_2.5.67
		OCP_DEG = 011 _b	0.1	1	3.4	μs	P_2.5.68
		OCP_DEG = 100 _b	38.2	60	76.6	μs	P_2.5.69
		OCP_DEG = 101 _b	23.1	40	51.9	μs	P_2.5.70
		OCP_DEG = 110 _b	18.5	30	41.5	μs	P_2.5.71
		OCP_DEG = 111 _b	8.4	20	31.6	μs	P_2.5.72
I _{OLD}	Open load detection current	Current flowing from VM to OUTx (High-Side = ON) or OUTx to GND (Low-Side = ON)	2	9	18	mA	P_2.5.73
I _{OLD_NEG}	Negative open load detection current	Current flowing from OUTx to VM (High-Side = ON) or GND to OUTx (Low-Side = ON)	2	15	30	mA	P_2.5.74
I _{OLD_LOW}	Open load detection current in low current OLD mode	Current flowing from VM to OUTx (High-Side = ON) or OUTx to GND (Low-Side = ON)	0.2	0.8	2	mA	P_2.5.75

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	Number
tOLD	Open load deglitch time	Active OLD (Continuous Mode)	2	3	4	ms	P_2.5.76
tOLD	Open load deglitch time	Active OLD (PWM Mode)	150	200	300	μs	P_2.5.77
T _{OTW}	Thermal warning temperature	Die temperature (T _J)	120	140	170	°C	P_2.5.78
T _{OTW_HYS}	Thermal warning hysteresis	Die temperature (T _J)		20		°C	P_2.5.79
T _{OTSD}	Thermal shutdown temperature	Die temperature (T _J)	150	175	200	°C	P_2.5.80
T _{OTSD_HYS}	Thermal shutdown hysteresis	Die temperature (T _J)		20		°C	P_2.5.81

- (1) For $20V < V_{VM} < 32V$, the OCP deglitch time must be limited to $10\mu s$ (Default Deglitch Value, OCP_DEG = 000_b).
- (2) For $V_{VM} > 28V$, the OCP deglitch time must be limited to $1\mu s$ (Lowest Deglitch Value, OCP_DEG = 011_b).

2.6 Timing Requirements

Table 2-6 Timing requirements

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT	Number
SPI (nSCS, SCLK, SDI, SDO)						
t_{READY}	SPI ready after after enable	VM > V _{UVLO} , nSLEEP = 3.3 V		1	ms	P_2.6.1
t_{CLK}	SCLK minimum period	200			ns	P_2.6.2
t_{CLKH}	SCLK minimum high time	100			ns	P_2.6.3
t_{CLKL}	SCLK minimum low time	100			ns	P_2.6.4
$t_{\text{SU_SDI}}$	SDI input data setup time	40			ns	P_2.6.5
$t_{\text{HD_SDI}}$	SDI input data hold time	60			ns	P_2.6.6
$t_{\text{DLY_SDO}}$	SDO output data delay time	SCLK high to SDO valid		60	ns	P_2.6.7
$t_{\text{SU_nSCS}}$	nSCS input setup time	100			ns	P_2.6.8
$t_{\text{HD_nSCS}}$	nSCS input hold time	100			ns	P_2.6.9
$t_{\text{HI_nSCS}}$	nSCS minimum high time before active low	600			ns	P_2.6.10
$t_{\text{DIS_nSCS}}$	nSCS disable delay time	nSCS high to SDO high impedance		30	ns	P_2.6.11
$t_{\text{SC_SPI}}$	Successive SPI write gaps		2.5		μs	P_2.6.12

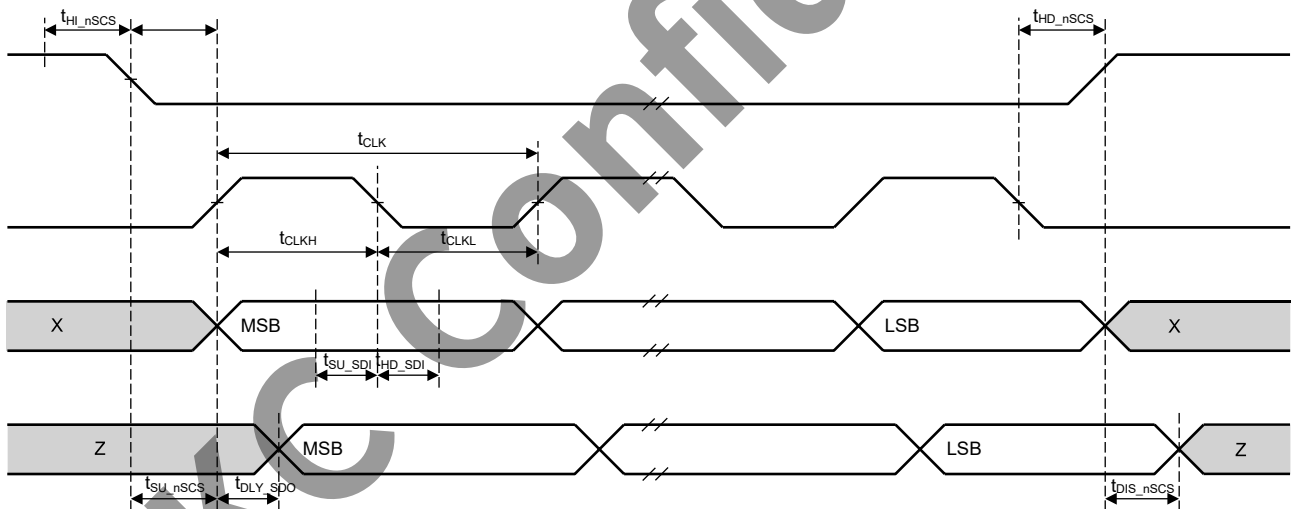


Figure 2-1 SPI Timing

2.7 Typical Characteristics

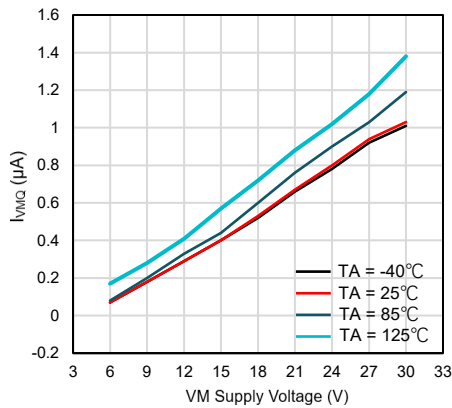


Figure 2-2 VM Sleep Mode Current (I_{VMQ}) vs Supply Voltage (V_{VM})

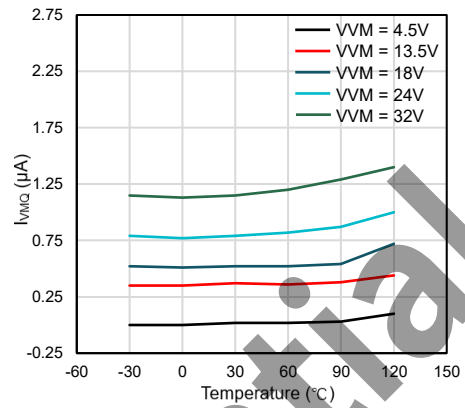


Figure 2-3 VM Sleep Mode Current (I_{VMQ}) vs Ambient Temperature (T_A)

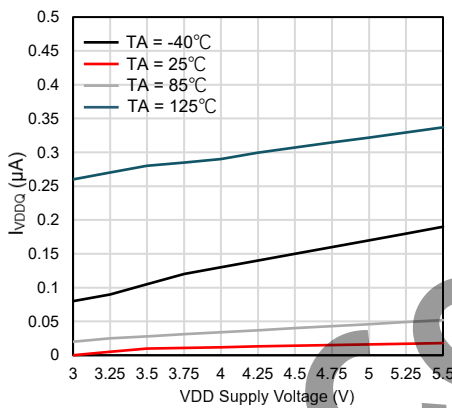


Figure 2-4 VDD Sleep Mode Current (I_{VDDQ}) vs Supply Voltage (V_{VDD})

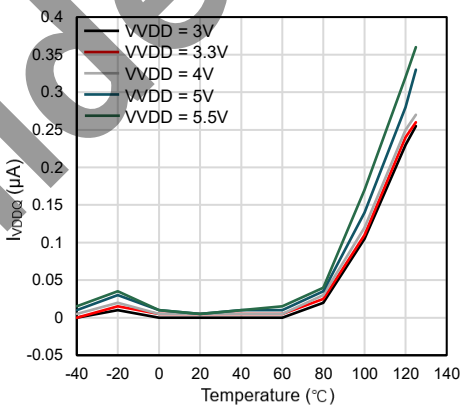


Figure 2-5 VDD Sleep Mode Current (I_{VDDQ}) vs Ambient Temperature (T_A)

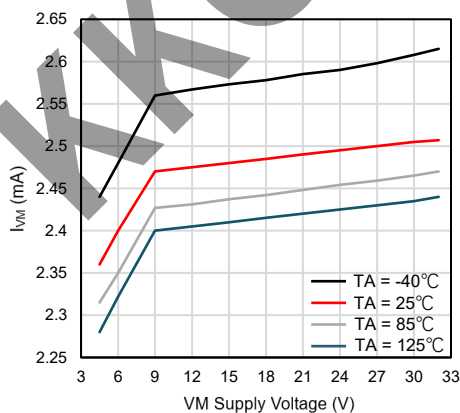


Figure 2-6 VM Operating Mode Current (I_{VM}) vs Supply Voltage (V_{VM})

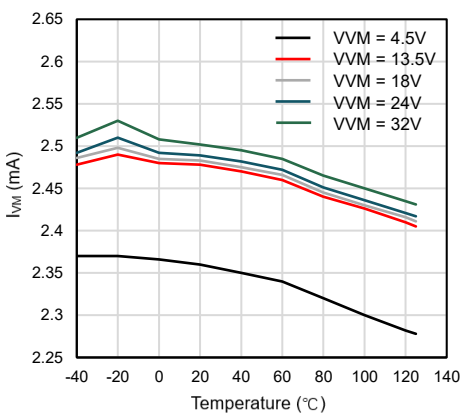


Figure 2-7 VM Operating Mode Current (I_{VM}) vs Ambient Temperature (T_A)

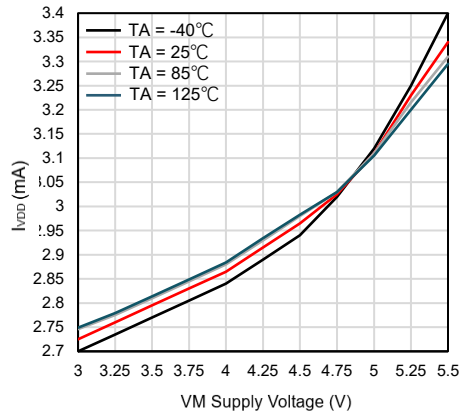


Figure 2-8 VDD Operating Mode Current (I_{VDD}) vs Supply Voltage (V_{VDD})

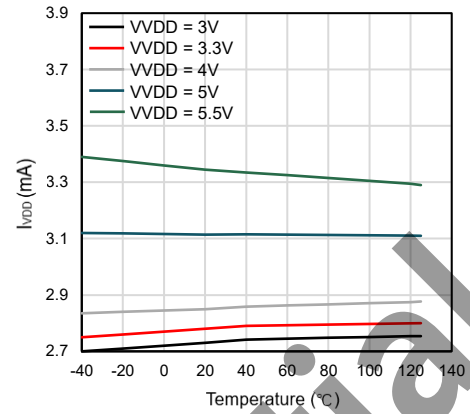


Figure 2-9 VDD Operating Mode Current (I_{VDD}) vs Ambient Temperature (T_A)

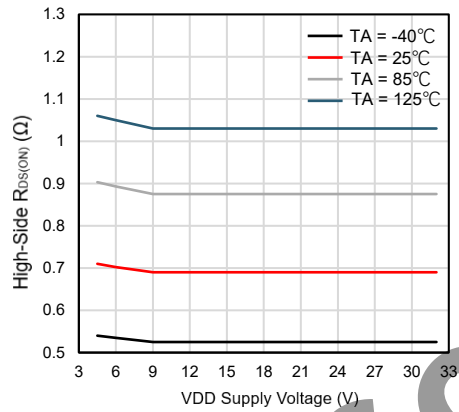


Figure 2-10 High Side On-State Resistance ($R_{DS(ON)}$) vs Supply Voltage (V_{VM})

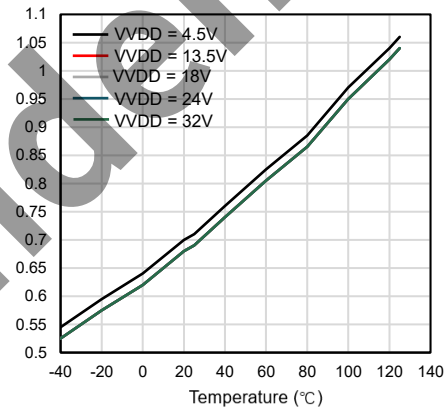


Figure 2-11 High Side On-State Resistance ($R_{DS(ON)}$) vs Ambient Temperature (T_A)

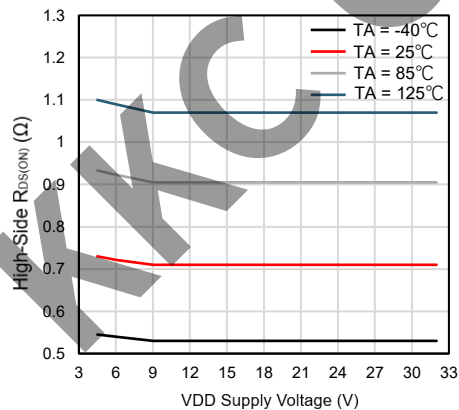


Figure 2-12 Low Side On-State Resistance ($R_{DS(ON)}$) vs Supply Voltage (V_{VM})

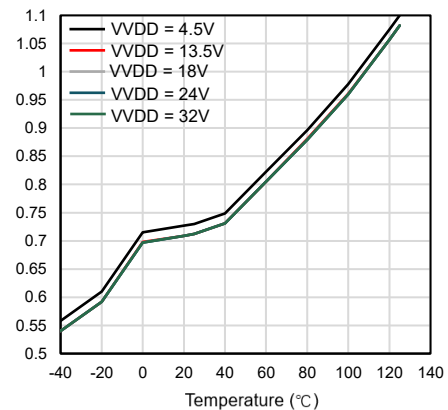


Figure 2-13 Low Side On-State Resistance ($R_{DS(ON)}$) vs Ambient Temperature (T_A)

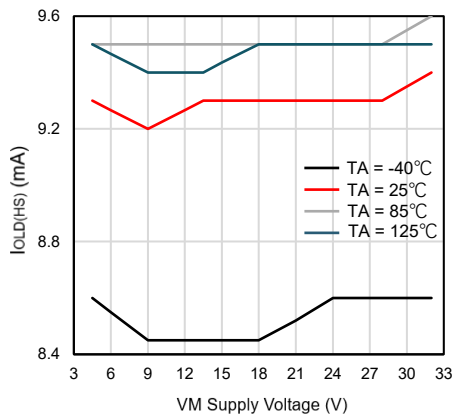


Figure 2-14 High-Side Open-load Detection Current (I_{OLD}) vs Supply Voltage (V_{VM})

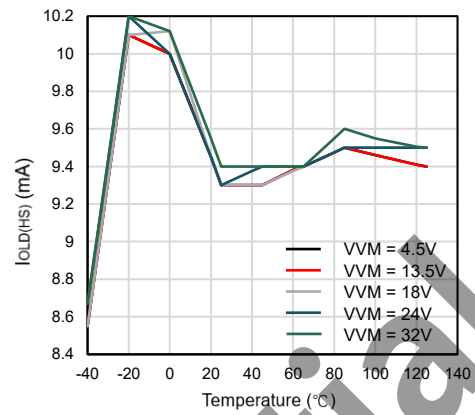


Figure 2-15 High-Side Open-load Detection Current (I_{OLD}) vs Ambient Temperature (T_A)

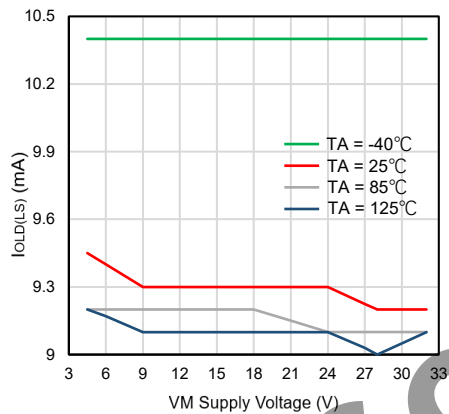


Figure 2-16 Low-Side Open-load Detection Current (I_{OLD}) vs Supply Voltage (V_{VM})

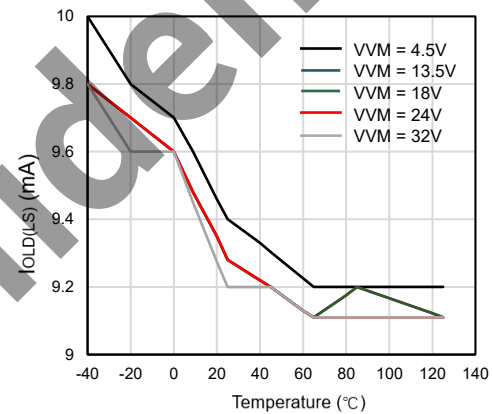


Figure 2-17 Low-Side Open-load Detection Current (I_{OLD}) vs Ambient Temperature (T_A)

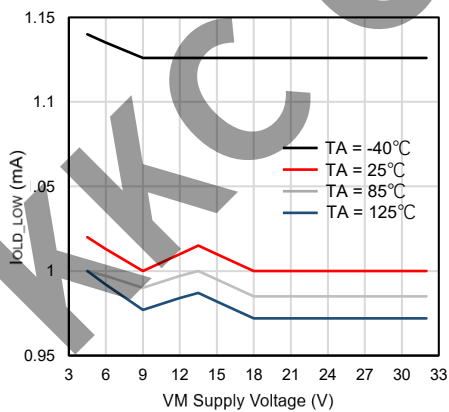


Figure 2-18 Low open-load detection current (I_{OLD_LOW}) vs supply voltage (V_{VM})

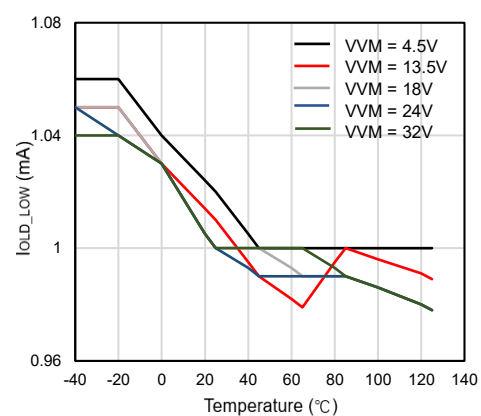


Figure 2-19 Low open-load detection current (I_{OLD_LOW}) vs ambient temperature (T_A)

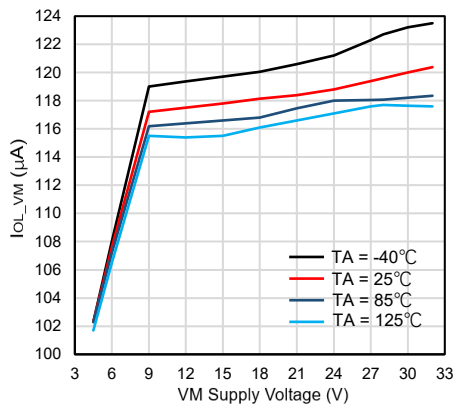


Figure 2-20 Open-load pull-down current (I_{OL_VM}) vs supply voltage (V_{VM})

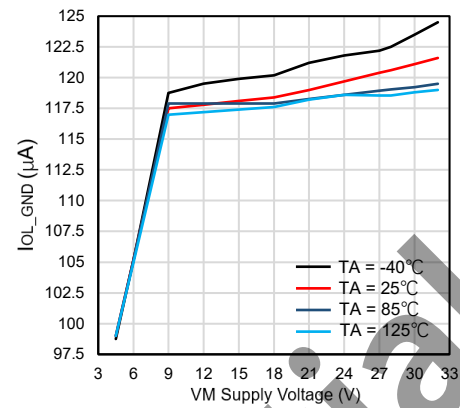


Figure 2-21 Open-load pull-up current (I_{OL_GND}) vs supply voltage (V_{VM})

KKC Confidential

3 Detailed Description

3.1 Overview

The HE99MD90xx family is 4.5V to 32V integrated multi-bridge drivers which supports a maximum voltage of 40V for 400ms in case of load-dump scenario. The half-bridges are designed to support 1A per half-bridge and 6A from the VM/GND pins.

A standard 16bit, 5MHz serial peripheral interface (SPI) provides a simple method for configuring the various device settings and reading fault diagnostic information through an external controller. The device is also equipped with a daisy-chain functionality which allows connecting multiple devices using a single nSCS line and saving on multiple resources.

This device has 4 internal PWM generators which can be mapped to any of the half-bridge through SPI registers. The PWM frequency (4 options) and duty (8bit resolution) for each channel can be selected using the SPI registers. This PWM mode is useful for implementing the current control of motor or dimming control of LEDs.

The device also has numerous integrated protections features which protect the device in case of any abnormal scenario. The over-current protection (OCP) ensures the device protection in any short scenarios like the phase short, phase to ground short and phase to supply short conditions. Undervoltage lockout (UVLO) and overvoltage protection (OVP) ensures the driver operation in fluctuating voltages to support the crank-start and load-dump scenario in automotive applications; In addition to this, the open-load detection (OLD) feature ensures the proper load connection. All devices support active OLD, low-current OLD, and negative-current OLD. Device faults are indicated on the nFAULT pin, and detailed information is available in the device SPI registers.

The device integrates a spread spectrum clocking feature for both the internal digital oscillator and internal charge pump. This feature combined with programmable output slew-rate control minimizes the radiated emissions from the device.

The device is available in a 24-pin ETSSOP package with a thermal pad.

3.2 Feature Description

Table 3-1 lists the recommended values of the external components for the driver.

Table 3-1 HE99MD90xx Driver external components

COMPONENTS	PIN 1	PIN 2	RECOMMENDED
C _{VM1}	VM	GND	X5R or X7R, 0.1μF, VM-rated capacitor
C _{VM2}	VM	GND	≥ 10 μF, VM-rated capacitor
C _{VDD1}	VDD	GND	X5R or X7R, 0.1μF, 6.3V capacitor
C _{VDD2}	VDD	GND	≥ 1 μF, 6.3V capacitor
R _{nFAULT}	VDD	nFAULT	Pullup resistor

3.2.1 Half Bridge Drivers

3.2.1.1 Control Modes

The half-bridge drivers can be programmed to drive loads (motor, solenoids, LEDs) continuously (without PWM) or in chopping mode (with PWM) and in parallel operation for driving high current.

3.2.1.1.1 Continuous Mode (Without PWM)

The half-bridges are configured to operate in the continuous mode without using any PWM switching by default. Any high-side or low-side switch is switched on by individually setting the high-side enable bits (HBx_HS_EN) and low-side enable bits (HBx_LS_EN) in operation control registers (OP_CTRL_1, OP_CTRL_2 and OP_CTRL_3).

Note: If the high-side enable bit (HBx_HS_EN) and low-side enable bit (HBx_LS_EN) of a particular half-bridge is set high (shoot-through configuration), then the particular half-bridge driver will remain in Hi-Z state until the shoot-through condition is cleared

The high-side and low-side enable bits of a particular half-bridge are configured to drive the motor in forward mode, reverse mode, brake mode and coast mode as shown in **Table 3-2**.

Table 3-2 Motor Operation in Continuous Mode (Motor Connected between HB1 and HB2)

nSLEEP	HALF-BRIDGE-1	HALF-BRIDGE-2	OUT1	OUT2	BRIDGE OPERATION (DC MOTOR)
0	HB1_HS_EN = Don't Care HB1_LS_EN = Don't Care	HB2_HS_EN = Don't Care HB2_LS_EN = Don't Care	Z	Z	Sleep Mode
1	HB1_HS_EN = 0 HB1_LS_EN = 0	HB2_HS_EN = 0 HB2_LS_EN = 0	Z	Z	Motor Coast
1	HB1_HS_EN = 1 HB1_LS_EN = 0	HB2_HS_EN = 0 HB2_LS_EN = 1	H	L	Forward Direction
1	HB1_HS_EN = 0 HB1_LS_EN = 1	HB2_HS_EN = 1 HB2_LS_EN = 0	L	H	Reverse Direction
1	HB1_HS_EN = 0 HB1_LS_EN = 1	HB2_HS_EN = 0 HB2_LS_EN = 1	L	L	Motor Brake (Low-Side)
1	HB1_HS_EN = 1 HB1_LS_EN = 0	HB2_HS_EN = 1 HB2_LS_EN = 0	H	H	Motor Brake (High-Side)
1	HB1_HS_EN = 1 HB1_LS_EN = 1	HB2_HS_EN = 1 HB2_LS_EN = 1	Z	Z	Motor Coast

Figure 3-1 shows the bridge configuration for motor operation in forward direction with high-side FET of OUT1 and low-side FET of OUT2 in conducting state with current flowing from OUT1 to OUT2. Similarly, the motor operation in reverse direction is achieved by switching ON the high-side FET of OUT2 and low-side FET of OUT1 such that current flows from OUT2 to OUT1 as shown in **Figure 3-2**.

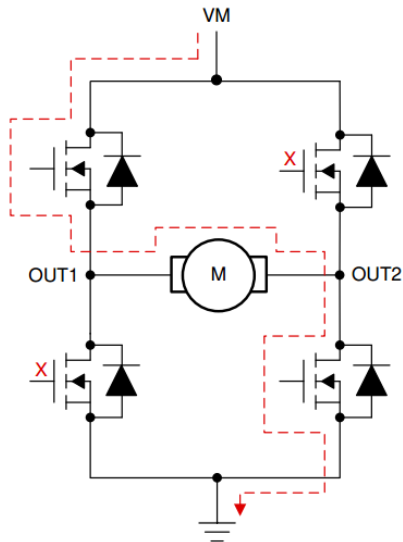


Figure 3-1 Continuous Mode (Forward Direction)

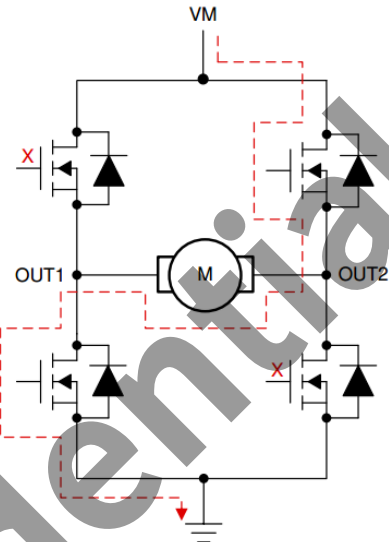


Figure 3-2 Continuous Mode (Reverse Direction)

Figure 3-3 and **Figure 3-4** shows the bridge operation in coast mode with motor initially running in forward and reverse direction respectively. As shown in these figures, due to the energy stored in motor's inductance, the current will continue to flow in motor and take the path flow through the body diodes of FETs.

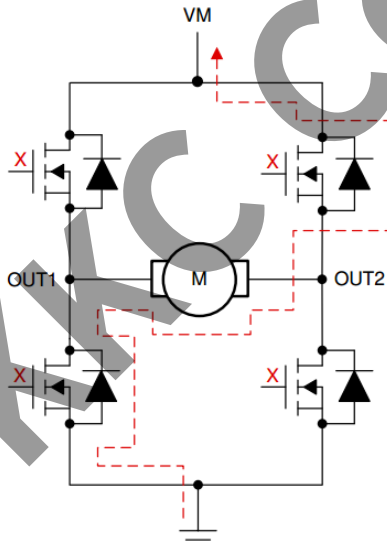


Figure 3-3 Continuous Mode (Coast - From Forward Direction)

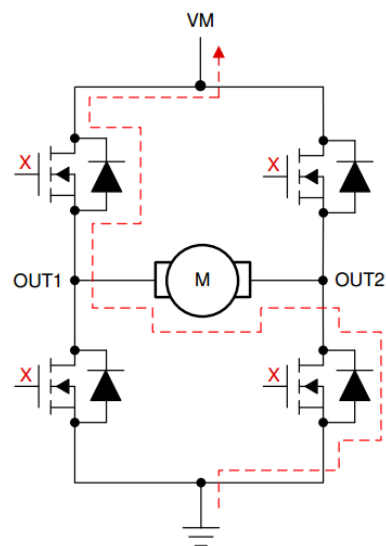


Figure 3-4 Continuous Mode (Coast - From Reverse Direction)

Figure 3-5 shows the low-side braking of the motor when both low-side FET's of the driver are turned ON. In this case, the motor is considered to be operating in forward direction (current flow from OUT1 to OUT2) and then braking is applied. Similarly, for the high-side braking, both high-side FET's of the driver are turned ON as shown in **Figure 3-6**.

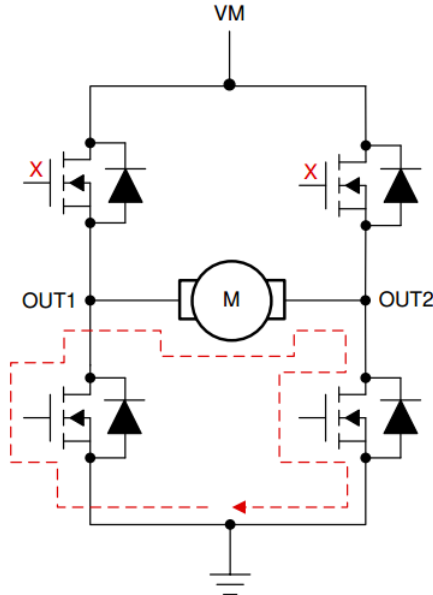


Figure 3-5 Continuous Mode (Brake - Low-Side)

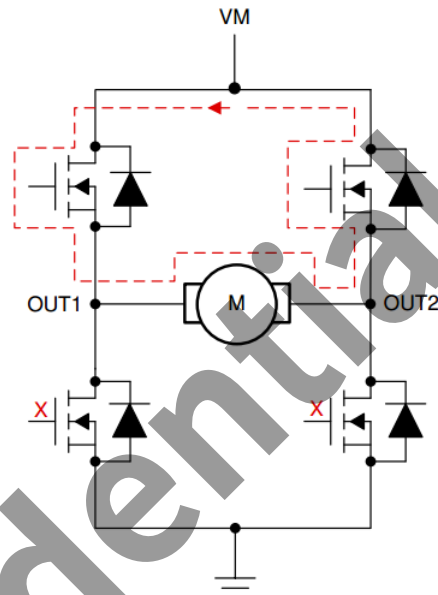


Figure 3-6 Continuous Mode (Brake - High-Side)

3.2.1.1.2 Chopping Mode (With PWM)

The half-bridges can be configured in the chopping mode by enabling the PWM switching on any particular half- bridge or both half-bridges. Each half-bridge can be mapped to any of the 4 PWM channels for which frequency and duty can be controlled independently. User has the flexibility to select the PWM frequency of channels out of 4 settings of 80Hz, 100Hz, 200Hz and 2kHz. Moreover, duty (8-bit resolution) of the 4 PWM generators can be adjusted independently .

The PWM chopping mode operation is done in five steps as follows and explained in detail below.

1. PWM Configuration
2. Free-Wheeling Mode (Synchronous Rectification) Disable / Enable
3. PWM Channels Mapping
4. PWM Channels Configuration (PWM Frequency and PWM Duty)
5. Half-Bridge Enable

3.2.1.1.2.1 PWM Configuration

The operation of selected half-bridge to operate in continuous mode or chopping mode (PWM mode) is selected using the PWM control register (PWM_CTRL_1 and PWM_CTRL_2). The HBx_PWM bit in PWM control register is set to enable the PWM switching in half-bridge.

Note: The default mode of any half-bridge is continuous mode. If the corresponding HBx_PWM bit in PWM_CTRL_x register is not set, then the particular half-bridge will operate in continuous mode.

3.2.1.1.2.2 Free-Wheeling Mode (Synchronous Rectification) Disable / Enable

The synchronous rectification of the half-bridge operating in PWM can be enabled by setting the HBx_FW bit in free-wheeling control registers (FW_CTRL_1 and FW_CTRL_2). **Figure 3-7** shows the operation of the driver when the synchronous rectification mode is disabled. As shown in this figure, during the PWM off time, the high-side diode of the OUT2 conducts to close the current path required for motor.

When synchronous rectification mode is enabled, if either of the low-side or high-side of the half-bridge operates in the PWM switching, the other switch of the same half-bridge operates in complementary fashion.

shows such example of the synchronous rectification, where the high-side FET of OUT2 half-bridge is turned ON when the low-side FET of same half-bridge is turned off in a PWM cycle.

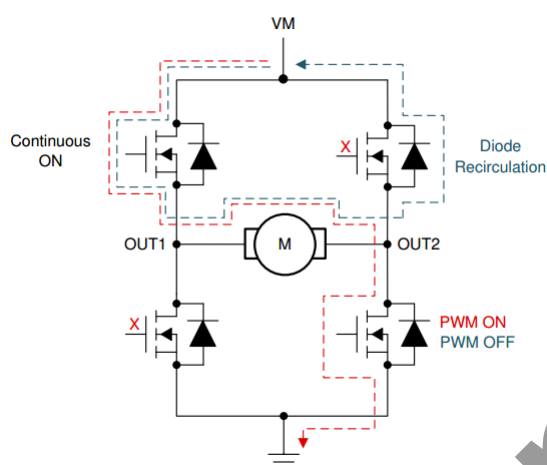


Figure 3-7 PWM Mode (Synchronous Rectification=OFF)

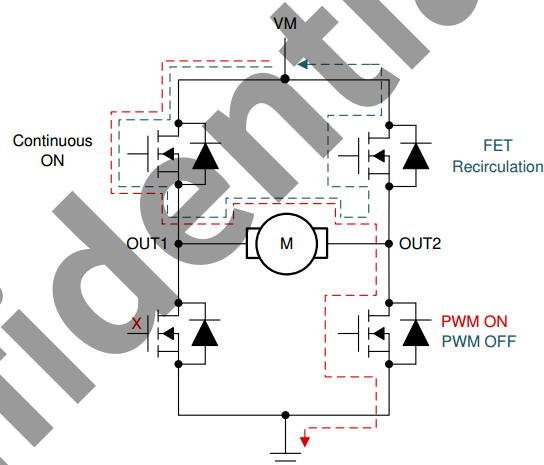


Figure 3-8 PWM Mode (Synchronous Rectification=ON)

Note: The default mode of any half-bridge is asynchronous rectification mode. If the corresponding bit in FW_CTRL_x register is not set, then the particular half-bridge will operate in asynchronous rectification mode.

3.2.1.1.2.3 PWM Channels Mapping

HE99MD90xx devices includes 4 PWM generators which can be mapped to any of the OUTx half bridge outputs using the PWM map control registers. The HBx_PWM_MAP bits in the PWM_MAP_CTRL_x registers are used to map any of the 4 channels to the OUTx outputs as shown in **Table 3-3**.

Table 3-3 PWM Mapping of HE99MD90xx

HBX_PWM MAP BITS	PWM CHANNEL
HBx_PWM_MAP = 00 _b	Channel 1 Selected for OUTx
HBx_PWM_MAP = 01 _b	Channel 2 Selected for OUTx
HBx_PWM_MAP = 10 _b	Channel 3 Selected for OUTx
HBx_PWM_MAP = 11 _b	Channel 4 Selected for OUTx

Note: Any half-bridge is mapped to PWM channel 1 by default.

3.2.1.1.2.4 PWM Channels Configuration (PWM Frequency and PWM Duty)

The frequency and duty of each PWM generator can be controlled independently. The

PWM_CHx_FREQ bits of PWM frequency control register (PWM_FREQ_CTRL) is used to select the frequency of PWM generator as shown in [Table 3-4](#). The PWM duty of each channel is controlled by the PWM duty control register (PWM_DUTY_CTRL_x).

Table 3-4 PWM Frequency

HBX_PWM MAP BITS	PWM CHANNEL
PWM_CHx_FREQ = 00 _b	80 Hz
PWM_CHx_FREQ = 01 _b	100 Hz
PWM_CHx_FREQ = 10 _b	200 Hz
PWM_CHx_FREQ = 11 _b	2000 Hz

3.2.1.1.2.5 Half-bridge Enable

The four steps of PWM mode enable, free-wheeling mode configuration, PWM channel mapping and PWM channels configuration ensure the proper configuration of PWM mode. Once the half-bridge is configured for the PWM generation, the half-bridge is enabled by enabling either of the high-side or low-side switch by individually setting the high-side enable bits (HBx_HS_EN) or low-side enable bits (HBx_LS_EN) in operation control registers (OP_CTRL_1, OP_CTRL_2 and OP_CTRL_3).

Note: The PWM is applicable to either of the high-side or low-side switch depending upon the HBx_HS_EN and HBx_LS_EN bits in OP_CTRL_x registers. In synchronous rectification mode, the opposite side switch will conduct in PWM off time.

3.2.1.1.3 Parallel Mode (Continuous Operation)

Parallel mode in HE99MD90xx device is implemented to support higher current loads which cannot be supported by a single channel. This mode can also be used for reducing the effective on-state resistance ($R_{DS(ON)}$) for achieving a better thermal performance of the device.

The configuration of various mode is very similar to the single half-bridge operation as explained in Continuous Mode (Without PWM) section. Considering six half-bridges for the parallel operation (OUT1, OUT2, OUT3 as group 'X' and OUT4, OUT5, OUT6 as group 'Y'), various modes can be summarized in **Table 3-5**.

Table 3-5 Motor Operation in Parallel Mode (Continuous Operation) (with Motor Connected between OUT1/2/3 and OUT4/5/6)

nSLEEP	HALF-BRIDGE-1 HALF-BRIDGE-2 HALF-BRIDGE-3 (X)	HALF-BRIDGE-4 HALF-BRIDGE-5 HALF-BRIDGE-6 (Y)	OUT1 OUT2 OUT3	OUT4 OUT5 OUT6	BRIDGE OPERATION (DC MOTOR)
0	HBX_HS_EN = Don't Care HBX_LS_EN = Don't Care	HBX_HS_EN = Don't Care HBX_LS_EN = Don't Care	Z	Z	Sleep Mode
1	HBX_HS_EN = 0 HBX_LS_EN = 0	HBX_HS_EN = 0 HBX_LS_EN = 0	Z	Z	Motor Coast
1	HBX_HS_EN = 1 HBX_LS_EN = 0	HBX_HS_EN = 0 HBX_LS_EN = 1	H	L	Forward Direction
1	HBX_HS_EN = 0 HBX_LS_EN = 1	HBX_HS_EN = 1 HBX_LS_EN = 0	L	H	Reverse Direction
1	HBX_HS_EN = 0 HBX_LS_EN = 1	HBX_HS_EN = 0 HBX_LS_EN = 1	L	L	Motor Brake (Low-Side)
1	HBX_HS_EN = 1 HBX_LS_EN = 0	HBX_HS_EN = 1 HBX_LS_EN = 0	H	H	Motor Brake (High-Side)
1	HBX_HS_EN = 1 HBX_LS_EN = 1	HBX_HS_EN = 1 HBX_LS_EN = 1	Z	Z	Motor Coast

Note: For parallel mode operation, the device operation under safe operating area (SOA) is recommended for supply voltage, $V_{VM} \leq 20V$, $HBX_{SR} = HBX_{SR} = 1_b$, $t_{ODP} \leq 10\mu s$ and $PL_MODE_EN = 01_b$.

Figure 3-9 shows three half-bridges (OUT1, OUT2 and OUT3) operating as a parallel high-side switch and other three half-bridges (OUT4, OUT5 and OUT6) are operating as a parallel low-side switch for achieving a forward motor operation. Similarly, the reverse direction of motor is achieved by operation of OUT1, OUT2 and OUT3 as a parallel low-side switch and other three half-bridges (OUT4, OUT5 and OUT6) as parallel high-side switch as shown in **Figure 3-10**.

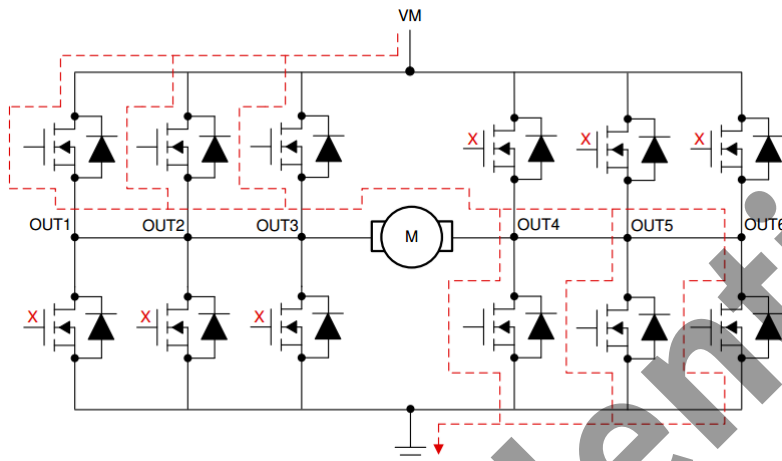


Figure 3-9 Parallel Mode (Forward Direction)

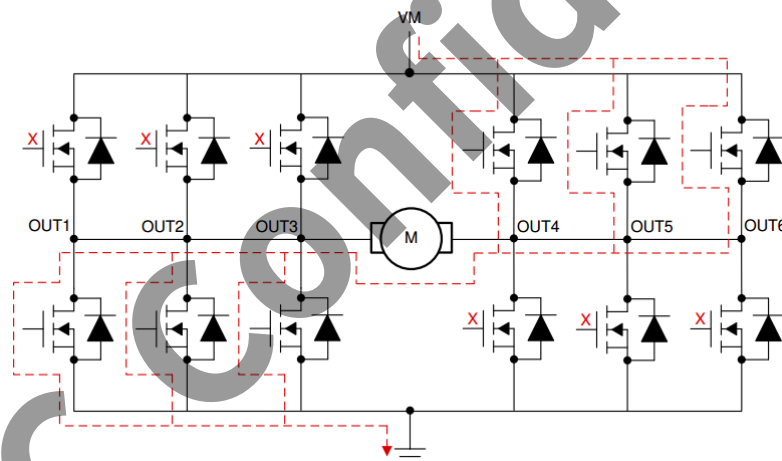


Figure 3-10 Parallel Mode (Reverse Direction)

Figure 3-11 and **Figure 3-12** shows the bridge operation in coast mode with motor initially running in forward and reverse direction respectively. As shown in these figures, the body diodes of the FETs conduct to continue the current flow path due to energy stored in motor's inductance.

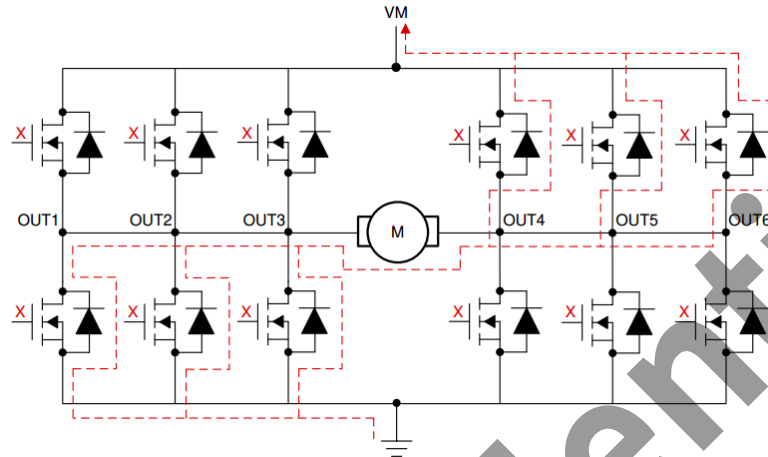


Figure 3-11 Parallel Mode (Coast from Forward Direction)

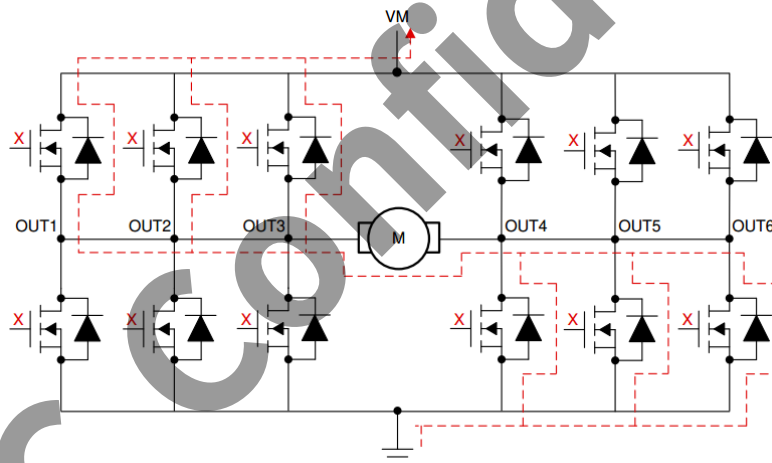


Figure 3-12 Parallel Mode (Coast from Reverse Direction)

The low-side braking of the motor during all the low-side FET's of the driver turning ON is shown in **Figure 3-13**. In this case, the motor is considered to be operating in forward direction (current flow from OUT1/2/3 to OUT4/5/6) and then braking is applied. Similarly, for the high-side braking, all high-side FET's of the driver are turned ON as shown in **Figure 3-14**.

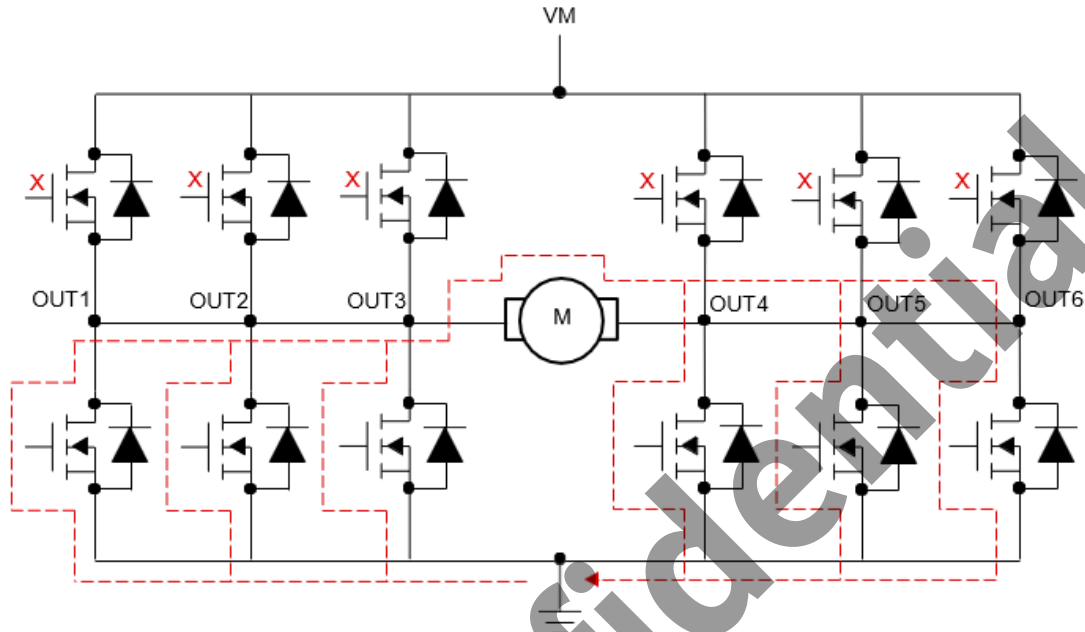


Figure 3-13 Parallel Mode (Brake - Low-Side)

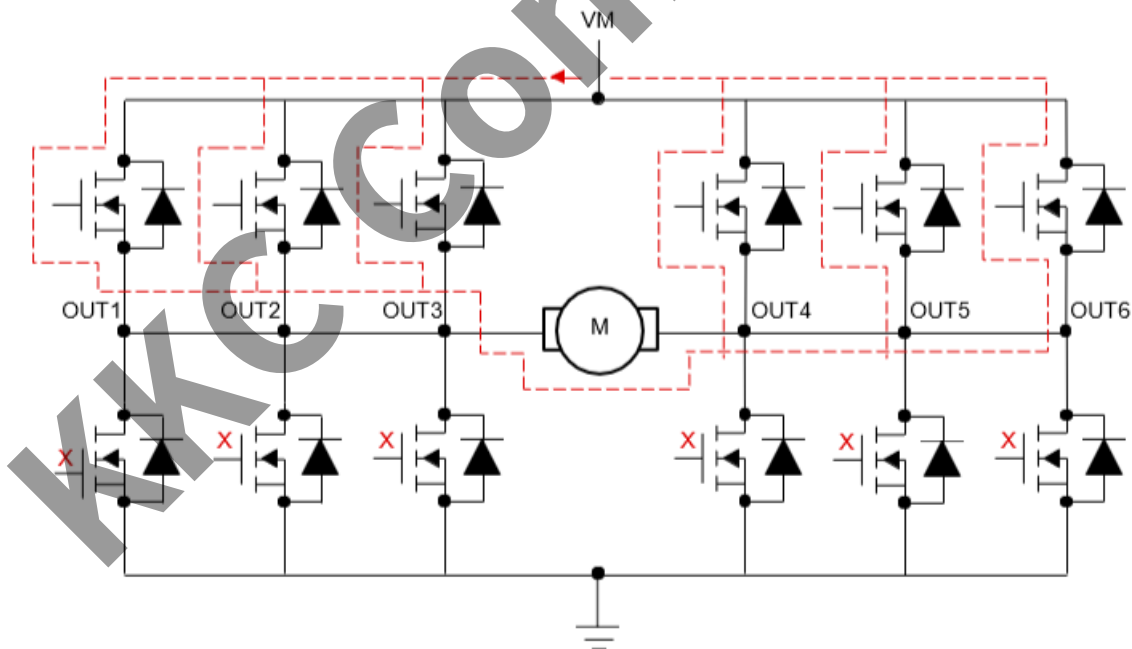


Figure 3-14 Parallel Mode (Brake - High-Side)

3.2.1.1.4 Parallel Mode (PWM Operation)

The half-bridges connected in parallel mode can be configured in the chopping mode by enabling the PWM switching on any particular group of high-side or low-side half-bridges or both group of half-bridges. For PWM operation in parallel mode, all half-bridges are to be mapped to a single PWM channel selected from any of the 4 PWM channels to avoid any delay in the PWM durations which can lead to undesired OCP condition. The user has the flexibility to select the PWM frequency of channels out of 4 settings of 80Hz, 100Hz, 200Hz and 2KHz and the duty adjustment which supports 8-bit resolution. Following steps enable the PWM operation with driver connected for parallel mode and are explained below.

- PWM Configuration
- Free-Wheeling Mode (Synchronous Rectification) Disable / Enable
- PWM Channels Mapping
- PWM Channels Configuration (PWM Frequency and PWM Duty)
- PWM Generators Disable
- Half-Bridge Enable
- PWM Generators Enable

3.2.1.1.4.1 PWM Configuration

The PWM control register (PWM_CTRL_1 and PWM_CTRL_2) are used to select the operation of particular half-bridges in the PWM mode. Considering a case for the motor movement in forward direction as shown in **Figure 3-11**, with low-side FETs of OUT4, OUT5 and OUT6 operating in PWM mode. The HBx_PWM bit in PWM control register is set to enable the PWM switching in selected half-bridges as shown below:

- HB4_PWM = 1_b
- HB5_PWM = 1_b
- HB6_PWM = 1_b

3.2.1.1.4.2 Free-Wheeling Mode (Synchronous Rectification) Disable / Enable

The synchronous rectification of the half bridges operating in PWM mode (OUT4, OUT5 and OUT6) are enabled by setting the corresponding HBx_FW bits in free-wheeling control register (FW_CTRL_1 and FW_CTRL_2). By default, the synchronous rectification mode is disabled.

- HB4_FW = 1_b
- HB5_FW = 1_b
- HB6_FW = 1_b

Figure 3-15 shows the parallel operation of half-bridges in PWM mode with synchronous rectification

disabled. As shown in this figure, during the PWM off time, the high-side diode of the OUT4, OUT5 and OUT6 conducts to close the current path required for motor.

When synchronous rectification mode is enabled, the high-side FETs of OUT4, OUT5 and OUT6 starts conducting during the PWM OFF time to close the motor current path as shown in **Figure 3-16**.

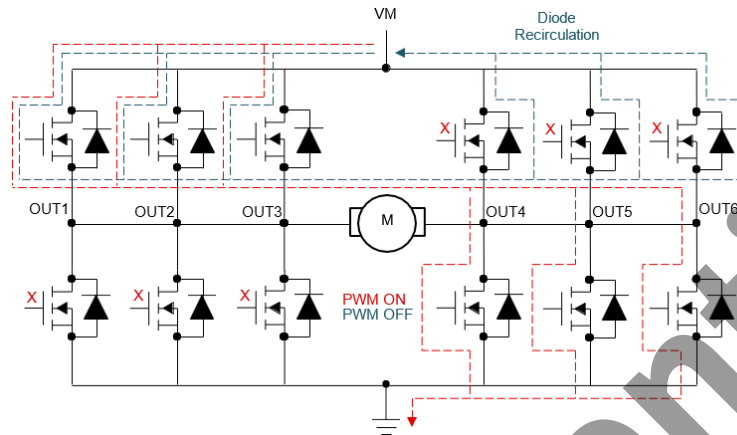


Figure 3-15 Parallel Mode (PWM with Synchronous Rectification = OFF)

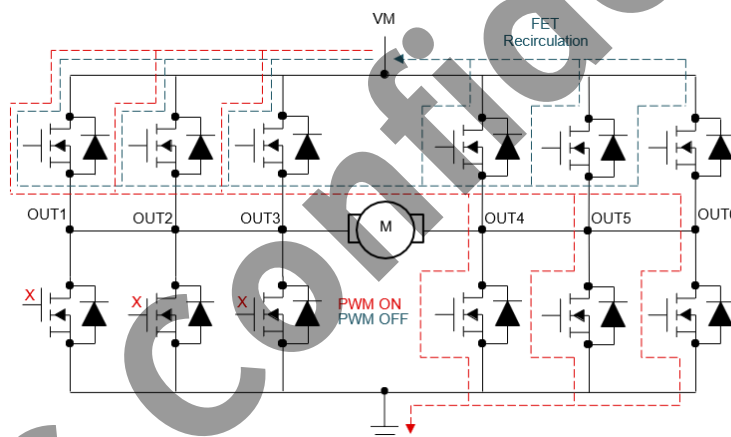


Figure 3-16 Parallel Mode (PWM with Synchronous Rectification = ON)

3.2.1.1.4.3 PWM Channel Mapping

The low-side FET's of half-bridges OUT4, OUT5 and OUT6 are mapped to any of the PWM generator by using the HBx_PWM_MAP bits in PWM mapping control registers. For parallel operation, all the half-bridges operating in PWM mode is mapped to a single PWM generator. Considering that PWM generator-4 is used for the mapping of half-bridges, following bits of the PWM_MAP_CTRL_x registers are affected:

- HB4_PWM_MAP = 11_b
- HB5_PWM_MAP = 11_b
- HB6_PWM_MAP = 11_b

Note: If the PWM of any channel is enabled, then it is mapped to PWM generator-1 by default.

3.2.1.1.4.4 PWM Channels Configuration (PWM Frequency and PWM Duty)

The PWM_CHx_FREQ bits of PWM frequency control registers (PWM_FREQ_CTRL_x) is used to select the frequency of PWM generator. Moreover, the PWM duty of each channel is controlled by the PWM duty control register (PWM_DUTY_CTRL_x). Considering a frequency of 2KHz is selected for the PWM operation (for PWM Generator-4), following frequency control and duty control registers are affected:

- PWM_CH4_FREQ = 11_b
- PWM_CH4_FREQ = '8-bit duty'

3.2.1.1.4.5 PWM Generators Disable

The PWM generators are disabled to ensure that all the half-bridges are turned-on at same time to avoid false OCP conditions for supporting higher current operation. The false OCP condition can arise due to the minimum time required for the SPI delay to switch on various half-bridges available in different registers. This can cause higher current (OCP condition) in one of the paralleled half-bridge while other half-bridge turning ON is delayed to the SPI register write delay and the propagation delay. Therefore, this sequence includes disabling the PWM generators initially, then enabling half-bridges and followed by enabling the PWM generators to avoid such issue. The PWM generator-4 is disabled by using the following command in the PWM_CTRL_x registers:

- PWM_CH4_DIS = 1_b

Note: All PWM generators are enabled by default (Default value of PWM_CTRL_x registers is 00_h).

3.2.1.1.4.6 Half-Bridge Enable

Once the PWM generators are disabled, the high-side and low-side FETs in half-bridges to be paralleled are enabled. High-side switches (connected in parallel) operating in continuous mode are enabled using the following bits in the OP_CTRL_x registers:

- HB1_HS_EN = 1_b
- HB2_HS_EN = 1_b
- HB3_HS_EN = 1_b

Moreover, the low-side switches (connected in parallel) operating in PWM mode are enabled using the following bits in the OP_CTRL_x:

- HB4_LS_EN = 1_b
- HB5_LS_EN = 1_b
- HB6_LS_EN = 1_b

3.2.1.1.4.7 PWM Generators Enable

After the half-bridges are enabled, the PWM generators are also enabled for tuning-on the respective

FETs operating in PWM mode. For this case, the low-side FETs of OUT4, OUT5 and OUT6 are turned ON for PWM operation connected to PWM generator-4. The PWM generator is enabled by the bits in the PWM_CTRL_x registers as shown:

- PWM_CH4_DIS = 0_b

KKC Confidential

3.2.1.2 Half-Bridge Drive Architecture

3.2.1.2.1 Slew Rate

An adjustable gate-drive current control to the MOSFETs of half-bridges is implemented to achieve the slew rate control. The MOSFET VDS slew rates are a critical factor for optimizing radiated emissions, energy and duration of diode recovery spikes and switching voltage transients related to parasitics. These slew rates are predominantly determined by the rate of gate charge to internal MOSFETs as shown in **Figure 3-17**.

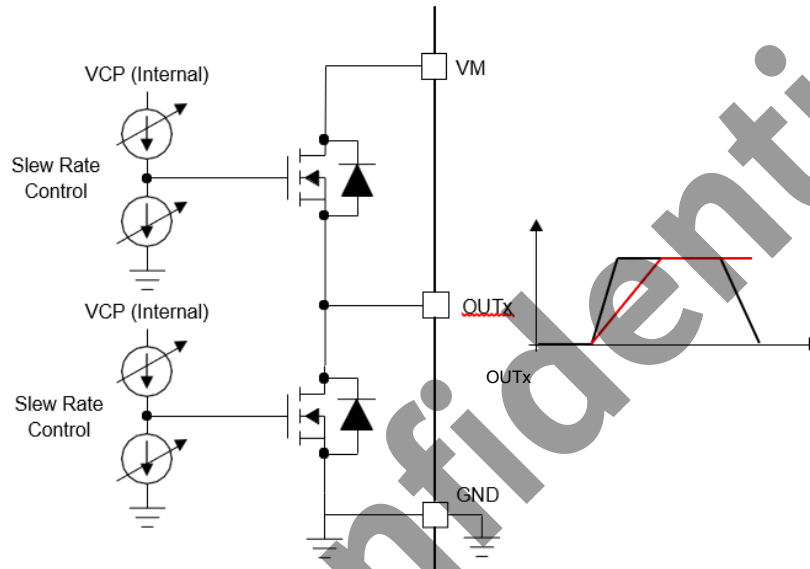


Figure 3-17 Slew Rate Circuit Implementation

The slew rate of each half-bridge can be adjusted by HBx_SR bits in Slew Rate control register (SR_CTRL_1 and SR_CTRL_2). Each half-bridge can be selected to a slew rate of 0.6V/μs or 2.5V/μs. The slew rate is calculated by the rise-time and fall-time of the voltage on OUTx pin as shown in **Figure 3-18**. The slew rate (SR) is calculate as shown in **Figure 3-18**.

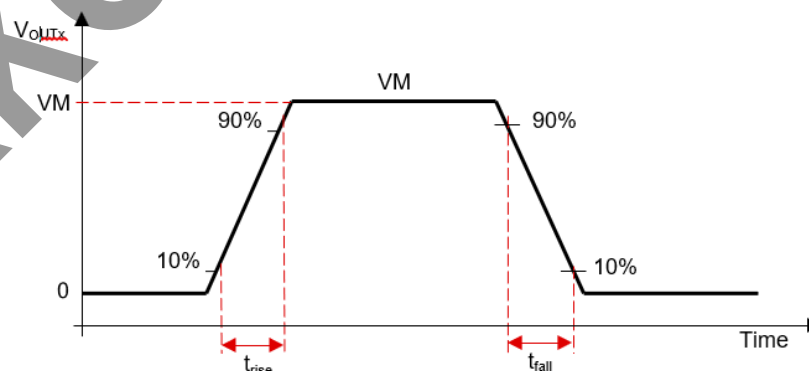


Figure 3-18 Slew Rate Timings

3.2.1.2.2 Cross Conduction (Dead Time)

The device is fully protected for any cross conduction of MOSFETs. In half-bridge configuration, the operation of high-side and low-side MOSFETs are ensured to avoid any shoot through currents by inserting a dead time (t_{dead}). This is implemented by sensing the gate-source voltage (V_{GS}) of the high-side and low-side MOSFETs and ensured that V_{GS} of high-side MOSFET has reached below turn-off levels before switching on the low-side MOSFET of same half-bridge as shown in **Figure 3-19** and **Figure 3-20**.

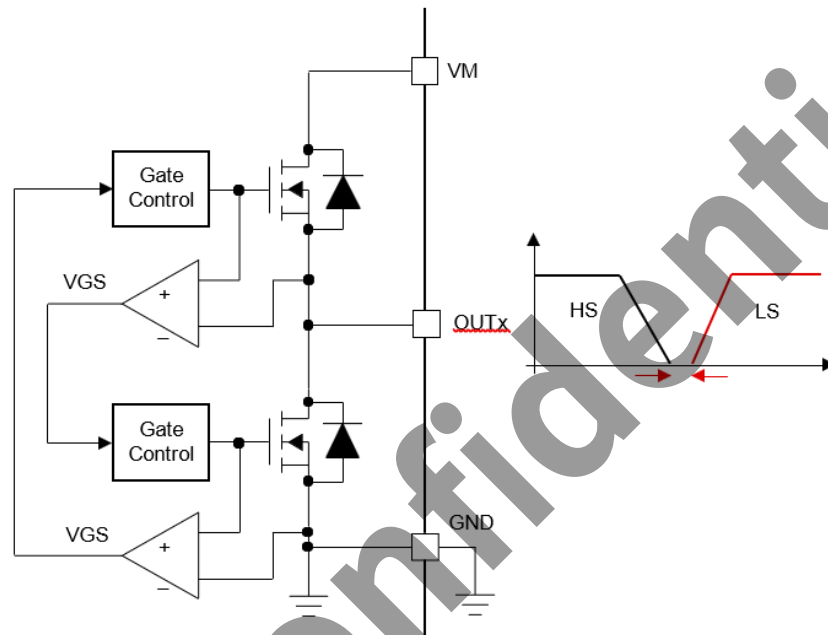


Figure 3-19 Cross Conduction Protection

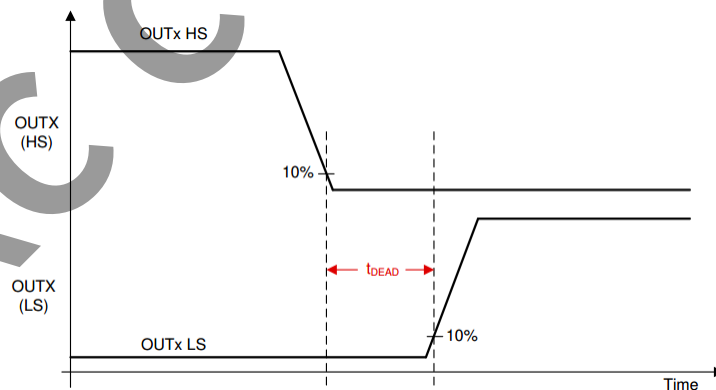


Figure 3-20 Dead Time

3.2.1.2.3 Propagation Delay

Propagation delay refers to the delay time from SPI valid condition to OUTx going high (10% level) as shown in **Figure 3-21**. The propagation constitutes of three major parameters.

1. Digital delay for SPI command decodes.
2. Analog delay for driver switch-on and gate current charging delay.
3. Slew rate delay for OUTx node to reach 10% of the final settling value.

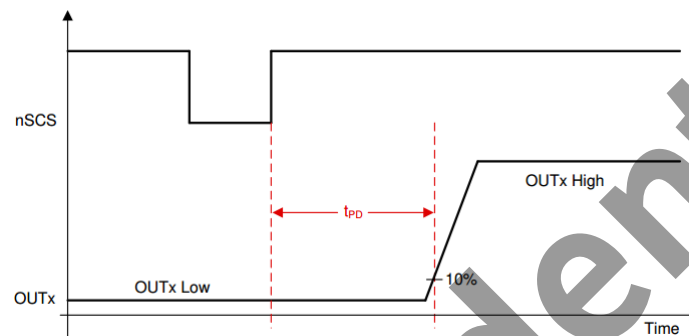


Figure 3-21 Propagation Delay

3.2.2 Pin Diagrams

This section presents the I/O structure of all digital input and output pins.

3.2.2.1 Logic Level Input Pin (nSLEEP, SCLK and SDI)

Figure 3-22 shows the input structure for the logic levels pins, nSLEEP, SCLK and SDI. The input can be with a voltage or external resistor. It is recommended to put SCLK and SDI pin low in device sleep mode to reduce leakage current through internal pull-down resistors.

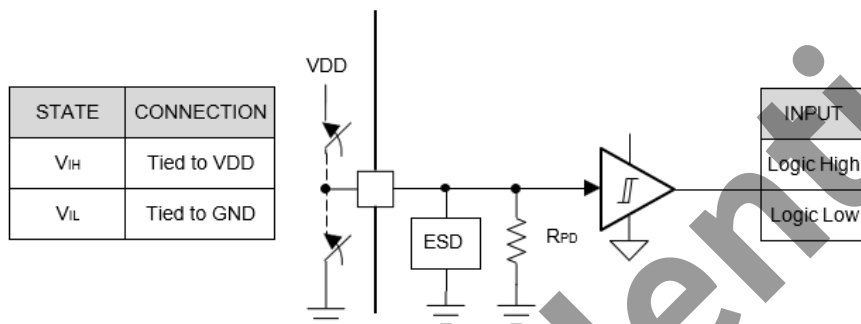


Figure 3-22 Logic Level Input Pin Structure (nSLEEP, SCLK and SDI)

3.2.2.2 Logic Level Input Pin (nSCS)

Figure 3-23 shows the input structure for the logic levels pin, nSCS. The input can be with a voltage or external resistor.

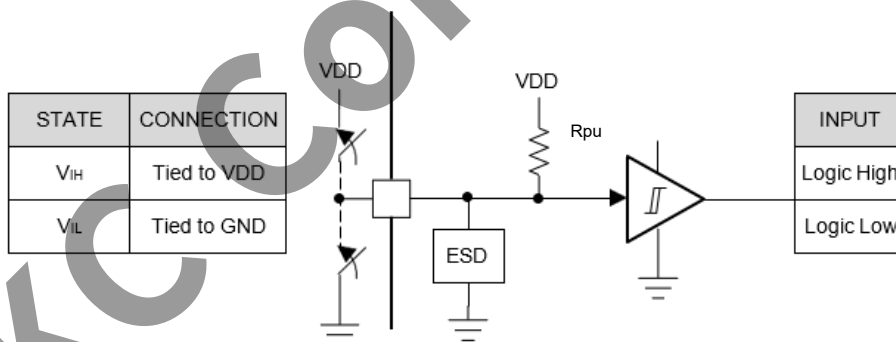


Figure 3-23 Logic Level Input Pin Structure (nSCS)

3.2.2.3 Open Drain Output Pin (nFAULT)

Figure 3-24 shows the structure of the open-drain output pin, nFAULT. The open-drain output requires an external pullup resistor to function properly.

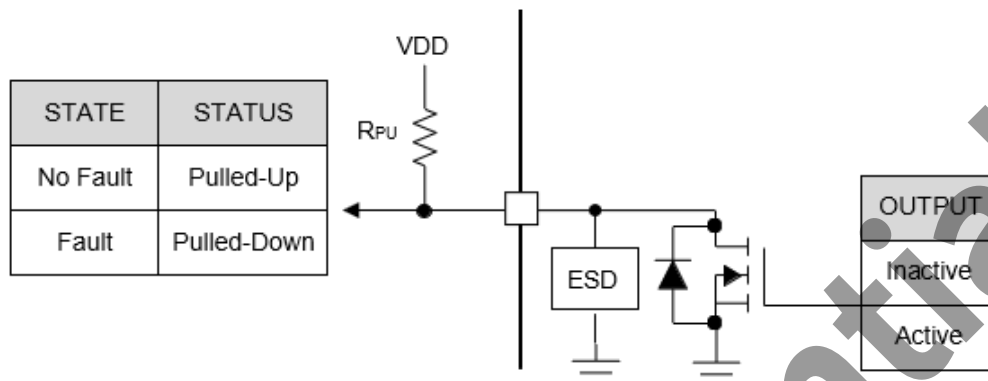


Figure 3-24 Open Drain Output Pin Structure (nFAULT)

3.2.2.4 Push Pull Output Pin (SDO)

Figure 3-25 shows the structure of push-pull pin, SDO.

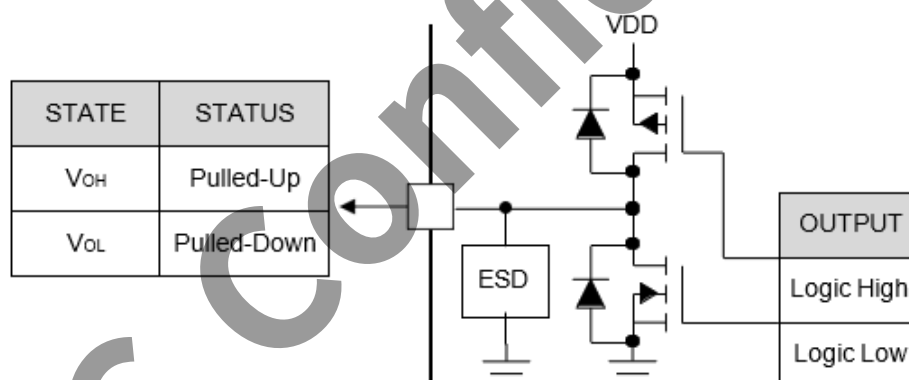


Figure 3-25 Push Pull Output Pin (SDO) Structure

3.2.3 Protection Circuits

The HE99MD90xx device is fully protected against undervoltage, overcurrent, and over-temperature events.

3.2.3.1 VM Supply Undervoltage Lockout (UVLO)

If at any time the input supply voltage on the VM pin falls below the V_{UVLO} threshold, the OVP bit is latched high in the IC status (IC_STAT) register and the operational behavior of the charge pump (CP) subsequent to the VM UVLO event is configurable via the CP_VM_UV_DIS bit in ADD_CONFIG_CTRL register: when CP_VM_UV_DIS bit is set to 0 (default), the charge pump is disabled and the chip is powered down; when set to 1 (high), the charge pump remains operational, all external half-bridges are disabled, and the nFAULT pin is driven low as shown in Figure 3-26. Normal operation resumes (driver operation and the nFAULT pin is released) when the VM undervoltage condition is removed. The UVLO bit remains set until cleared through the CLR_FLT bit.

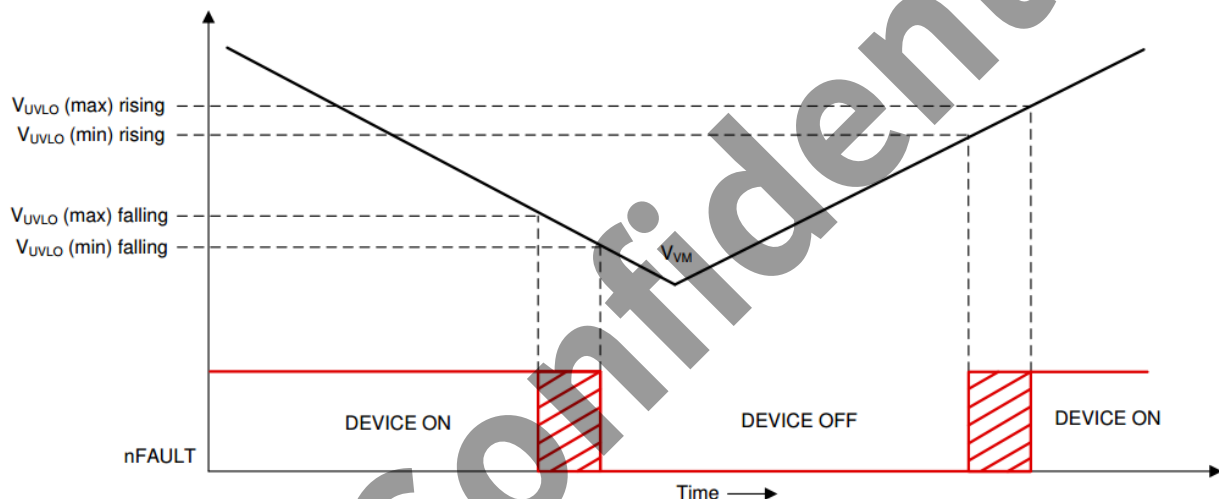


Figure 3-26 VM UVLO Operation

3.2.3.2 VM Supply Overvoltage Protection (OVP)

If at any time the input supply voltage on the VM pin rises above the V_{OVP} threshold, all of the half-bridges are disabled, the charge pump is disabled, and the nFAULT pin is driven low as shown in Figure 3-27. The OVP bit is also latched high in the IC status (IC_STAT) register. Normal operation resumes (driver operation and the nFAULT pin is released) when the VM overvoltage condition is removed. The OVP bit remains set until cleared through the CLR_FLT bit.

An extended overvoltage operation is also supported in this device for higher over-voltage range up to 32V. This operation is enabled by setting the EXT_OVP bit in the configuration (CONFIG_CTRL) register.

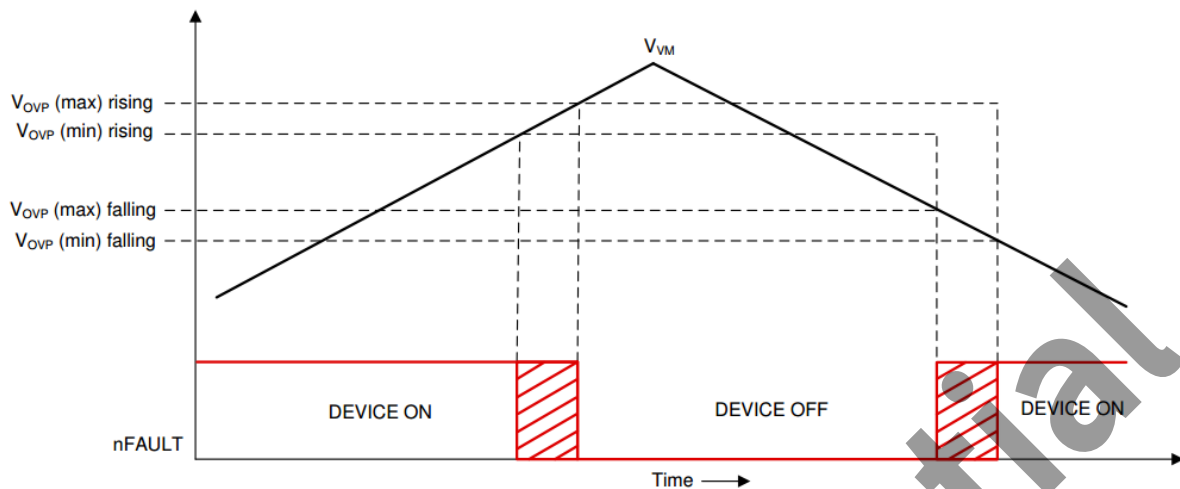


Figure 3-27 Over Voltage Protection

3.2.3.3 Digital Output Supply Power Undervoltage (VDDUV)

If at any time the supply voltage on the VDD pin falls below the V_{VDDUV} threshold, SPI function will be disabled and all input command will be ignored. If VDDUV happens when device is at normal operating mode, channels will be operating at previous state. Channels controlled by SPI will stay in ON/OFF state according to last SPI command. Once recovered, SPI function will be available again, as shown in Figure 3-28.

Note: VDDUV is not reported to nFAULT pin.

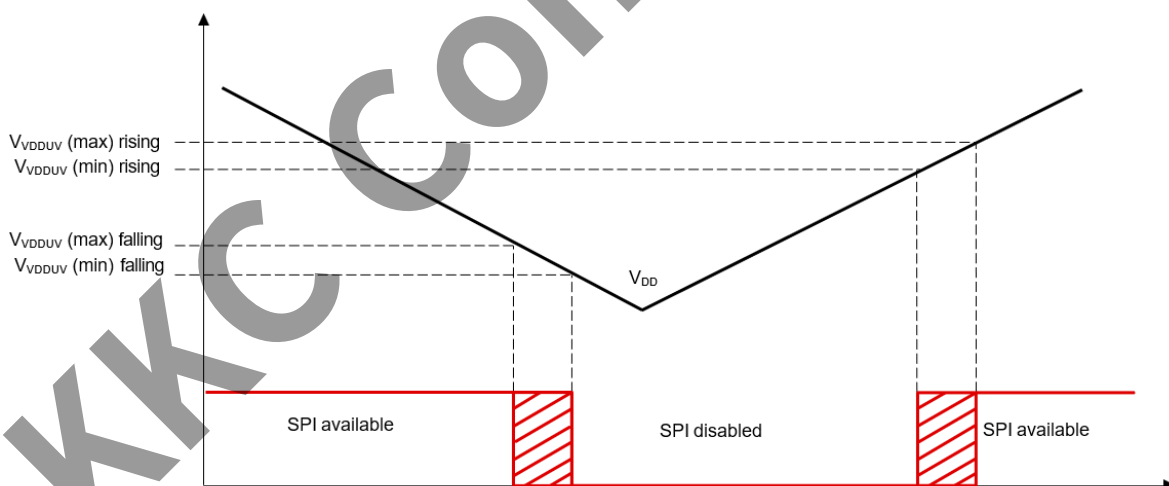


Figure 3-28 VDD UVLO Operation

3.2.3.4 Overcurrent Protection (OCP)

A overcurrent protection circuit is implemented in each driver. If FET current increase higher than I_{OCP} threshold for longer than the t_{OCP} deglitch time, the high-side and the low-side FETs in the corresponding half bridge are disabled and the nFAULT pin is driven low. The OCP bit in the IC status (IC_STAT) register and corresponding bit in overcurrent protection status register (OCP_STAT_x) register is latched high. The charge pump remains active during this condition. The OCP bit in the IC status (IC_STAT) register and corresponding bits (HBx_HS_OCP / HBx_LS_OCP) in overcurrent protection status (OCP_STAT_x) register remains set until cleared through the CLR_FLT bit.

User also has the programmability of disabling the OCP fault on the nFAULT pin by setting the OCP_REP bit in the CONFIG_CTRL register.

The device also provides two slew-rate options for the device turn-off during an OCP event which can be programmed via the PL_MODE_EN bits in OLD_CTRL_2 register. The default option (PL_MODE_EN = 00_b) is the faster slew rate option (typical around 1 μ s) which can be used for the single bridge operation. The slower option (PL_MODE_EN = 01_b) provides a slower slew rate (half-bridge slew rate, HBx_SR) which can be used for the higher current applications in device parallel mode operation.

Table 3-6 Overcurrent Protection

Drive Current	BRIDGE CONFIGURATION	REGISTER SETTINGS	BRIDGE STATE	nFAULT PIN		BITS AFFECTED	RECOVERY
				OCP_REP = 0	OCP_REP = 1		
$I_{LOAD} < I_{OCP}$	OUT1 High-Side ON	HB1_HS_EN = 1	ENABLED	H	H	N/A	N/A
	OUT2 Low-Side ON	HB1_LS_EN = 1	ENABLED	H	H		
$I_{LOAD} < I_{OCP}$	Full Bridge (OUT1/2) Forward Direction	HB1_HS_EN = 1 HB2_LS_EN = 1	ENABLED	H	H		
	Full Bridge (OUT1/2) Reverse Direction	HB1_LS_EN = 1 HB2_HS_EN = 1	ENABLED	H	H		
I_{SHORT} or $I_{LOAD} > I_{OCP}$	OUT1 High-Side ON OUT1 Short to GND	HB1_HS_EN = 1	Hi-Z	L	H	OCP = 1 (IC_STAT) HB1_HS_OCP = 1	OCP Condition Removed CLR_FLT = 1
	OUT1 Low-Side ON OUT1 Short to VM	HB1_LS_EN = 1	Hi-Z	L	H	OCP = 1 (IC_STAT) HB1_LS_OCP = 1	
I_{SHORT} or $I_{LOAD} > I_{OCP}$	Full Bridge (OUT1/2) Forward Direction OUT1 / OUT2 Short	HB1_HS_EN = 1 HB2_LS_EN = 1	Hi-Z	L	H	OCP = 1 (IC_STAT) HB1_HS_OCP = 1 or HB2_LS_OCP = 1 ⁽¹⁾	
	Full Bridge (OUT1/2) Reverse Direction OUT1 / OUT2 Short	HB1_LS_EN = 1 HB2_HS_EN = 1	Hi-Z	L	H	OCP = 1 (IC_STAT) HB1_LS_OCP = 1 or HB2_HS_OCP = 1 ⁽²⁾	

(1) Either of the HB1_HS_OCP or HB2_LS_OCP will set depending upon which half-bridge OCP trigger first.

(2) Either of the HB1_LS_OCP or HB2_HS_OCP will set depending upon which half-bridge OCP trigger first.

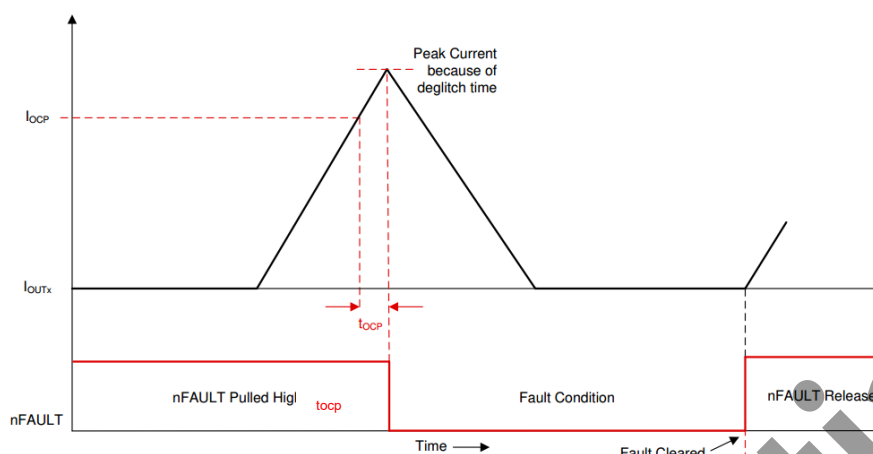


Figure 3-29 Over Current Protection

3.2.3.5 Open-load detection (OLD)

The HE99MD90xx devices implement multiple open-load detection schemes to allow the controller to determine if a load is connected to the OUTx terminals. The OLD schemes available in HE99MD90xx are listed below. [Table 3-7](#) summarizes the use-cases for each OLD scheme.

- Active OLD
- Low-Current Active OLD
- Negative-Current Active OLD

Table 3-7 Summary of OLD features

	ACTIVE OLD	LOW-CURRENT ACTIVE OLD	NEGATIVE-CURRENT ACTIVE OLD
When is it used	Detects the open-load condition while driving a load	Detects the open-load condition while driving a load with small operating current	Detects the open-load condition during current re-circulation (on high-side or low-side) when synchronous rectification is enabled for both half-bridge and full-bridge configurations
State of OUTx for valid OLD	H/L	L	H/L
OLD Trigger Condition	$I_{OUTX} < I_{OLD}$	$I_{OUTX} < I_{OLD_LOW}$	$I_{OUTX} > -I_{OLD_NEG}$ $I_{OUTX} < I_{OLD}$
Tradeoffs	A false flag can occur during current re-circulation if synchronous rectification is ON. Enabling Negative-current OLD will solve this. A false flag can also occur if the operating current for the load is small (below I_{OLD} threshold). See Low-current OLD for solution.	Only applicable for the current flowing in the low-side FETs. The I_{OCP} for the low-side FET is also reduced by 11 times. The $R_{DS(ON)}$ of the low-side FET will increase by 11 times, hence the thermal performance has to be monitored.	Only functional during current re-circulation, however it works in conjunction with active OLD. This feature is not needed if synchronous rectification is disabled.
Can be used with other OLD schemes	Negative-current OLD and low-current OLD	Active OLD and negative-current OLD	Active OLD and low-current OLD

3.2.3.5.1 Active OLD

Active OLD can identify an open-load condition on the OUTx pins while driving a load. As shown in **Figure 3-30**, the HE99MD90xx identifies an open-load fault condition when the current through the MOSFET (I_{OUTx}) is lower than the open-load current threshold (I_{OLD}) for longer than the open-load deglitch time (t_{OLD}). At that point the device takes the following actions.

- OLD bit in the IC status (IC_STAT) register sets to 1
- HBx_HS_OLD or HBx_LS_OLD bit in the open-load status register (OLD_STAT_x) sets to 1 (depending if the fault is on the high-side MOSFET or low-side MOSFET, respectively).
- nFAULT pin is drives low to indicate a fault to the controller.

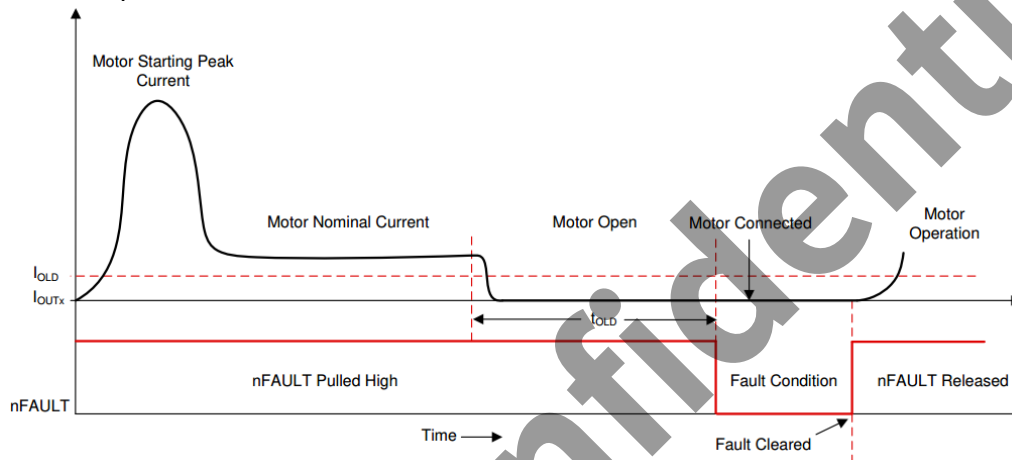


Figure 3-30 Active open-load detection

Normal operation resumes (driver operation resumes, the nFAULT pin goes high, OLD bit is reset to 0) when the open-load condition is removed (the user reconnects the load to the OUTx connection) and the controller writes the CLR_FLT bit to 1.

Note: After the open-load fault condition is removed, the nFAULT pin will be driven high and the fault status are removed when,

- CLR_FLT command is issued after deglitch time (t_{OLD}) while OUTx is set high.
- CLR_FLT command is issued after OUTx is set to Hi-Z.
- CLR_FLT command is issued after HBx_OLD_DIS bit is set. [END]

By default, OLD on the HE99MD90xx device is enabled. The OLD control registers (OLD_CTRL_1 and OLD_CTRL_2) allow the user to disable OLD on the OUTx pins with the HBx_OLD_DIS bits. The OLD_OP bit in the OLD_CTRL_2 register determines the response of the device to an active OLD fault. If OLD_OP = 0, the OUTx pins go to the Hi-Z state to stop driving the outputs. If OLD_OP = 1, the OUTx pins stay in their previous state and do not react to the OLD fault unless the user takes action. Similarly, the OLD_REP bit determines if the OLD fault will report on the nFAULT pin or only in the IC_STAT register. **Table 3-8** summarizes the open-load detection feature and conditions.

Note: By default, the OLD feature is enabled, the outputs disable (go Hi-Z) when the OLD flags, and the nFAULT pin will report the OLD.

Table 3-8 OLD Configuration

LOAD / OPEN	REGISTER SETTINGS	OLD_OP	OLD_REP	OUT1	OUT2	nFAULT	BITS EFFECTED	RECOVERY
Half-Bridge Load Connected	HB1_HS_EN = 1	X	X	H	X	H	N/A	N/A
	HB1_LS_EN = 1	X	X	L	X	H		
Full-Bridge Load Connected	HB1_HS_EN = 1 HB2_LS_EN = 1	X	X	H	L	H		
	HB1_LS_EN = 1 HB2_HS_EN = 1	X	X	L	H	H		
Half-Bridge Open	HB1_HS_EN = 1	0	0	Hi-Z	X	L	OLD = 1 (IC_STAT) HB1_HS_OLD = 1	OLD Condition Removed CLR_FLT = 1
		0	1	Hi-Z	X	H		
		1	0	H	X	L		
		1	1	H	X	H		
	HB1_LS_EN = 1	0	0	Hi-Z	X	L	OLD = 1 (IC_STAT) HB1_LS_OLD = 1	
		0	1	Hi-Z	X	H		
		1	0	L	X	L		
		1	1	L	X	H		
Full-Bridge Open	HB1_HS_EN = 1 HB2_LS_EN = 1	0	0	Hi-Z	Hi-Z	L	OLD = 1 (IC_STAT) HB1_HS_OLD = 1 or HB2_LS_OLD = 1 ⁽¹⁾	
		0	1	Hi-Z	Hi-Z	H		
		1	0	H	L	L		
		1	1	H	L	H		
	HB1_LS_EN = 1 HB2_HS_EN = 1	0	0	Hi-Z	Hi-Z	L	OLD = 1 (IC_STAT) HB1_LS_OLD = 1 or HB2_HS_OLD = 1 ⁽²⁾	
		0	1	Hi-Z	Hi-Z	H		
		1	0	L	H	L		
		1	1	L	H	H		

(1) Either of the HB1_HS_OLD or HB2_LS_OLD will set depending upon which half-bridge OLD triggers first.

(2) Either of the HB1_LS_OLD or HB2_HS_OLD will set depending upon which half-bridge OLD triggers first.

3.2.3.5.2 Negative-current OLD

The HE99MD90xx device also includes a negative-current OLD mode. The negative current can flow either through the body diode of high-side FET or the FET itself depending on whether or not the channel is configured for synchronous rectification.

Figure 3-32 shows the current re-circulation through the body diode of the high-side FET when the synchronous rectification mode is OFF (i.e. HB2_FW = 0). In this case, the active OLD will not falsely report an open-load condition since the OLD circuit only enables when the FET is ON. Negative-current OLD will also work during re-circulation through the low-side FETs.

Figure 3-31 shows the negative current re-circulation through the high-side FET when synchronous rectification is ON (i.e. HB2_FW = 1). In this scenario, for default operation (OLD_NEG_EN = 0), the device can show a false open-load fault since the FET current is lower than the positive OLD threshold. However, when negative-current OLD mode is enabled, the device will only flag an open-load fault if $I_{OUTX} < I_{OLD_NEG}$. This mode is enabled by setting the OLD_NEG_EN bit in OLD_CTRL3 register.

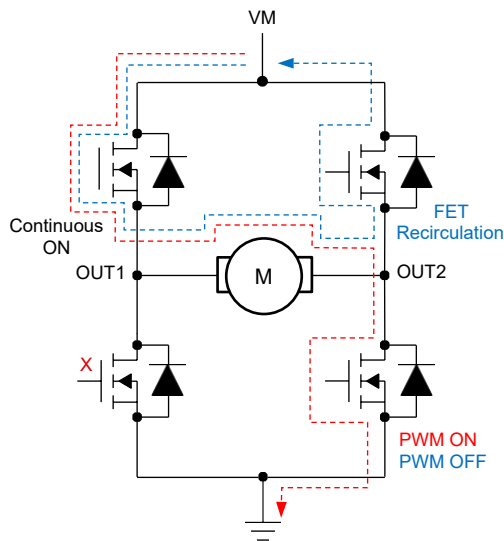


Figure 3-31 Negative Current Flow in High-Side FET of OUT2 Channel

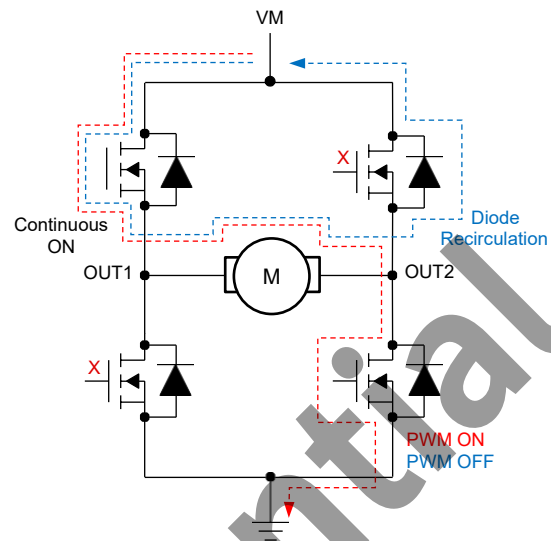


Figure 3-32 Negative Current Flow in OUT2 by Body Diode of High-Side FET

Figure 3-33 shows the waveforms of false open-load detection when the negative-current OLD setting is disabled ($OLD_NEG_EN = 0$). As shown in this figure, the high-side FET of the OUT1 channel is always switched ON and the low-side and high-side FET of the OUT2 channel are operating in complimentary way (i.e. synchronous rectification mode is enabled). In synchronous rectification, the current flows in negative direction from OUT2 to VM (i.e. FET Source to Drain) during the high-side FET conduction. Initially, for the first PWM cycle, the OLD mode is disabled to show the currents in different FETs during the motor operation. When OLD is enabled in second PWM cycle, then the device registers a false open-load detect during the high-side FET conduction as shown in **Figure 3-33**. The nFAULT pin is pulled low and both high-side and low-side FET of OUT2 channels are disabled. The body diode of the high side FET (OUT2) conducts to complete the motor current path.

This false detection of open load is eliminated by enabling the negative-current OLD setting ($OLD_NEG_EN = 1$). As shown in **Figure 3-34**, the negative OLD current setting (I_{OLD_NEG}) is enabled for the high-side FET of OUT2 channel. This setting allows the negative current path (from source to drain) in high-side FET. The nFAULT pin is latched high and OUT2 channel is not disabled when OLD is enabled in second PWM cycle.

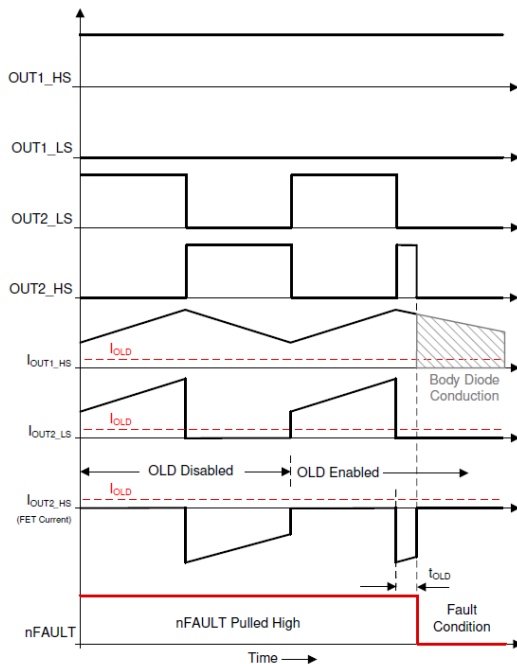


Figure 3-33 Waveforms Showing False OLD With Negative-Current OLD Disabled

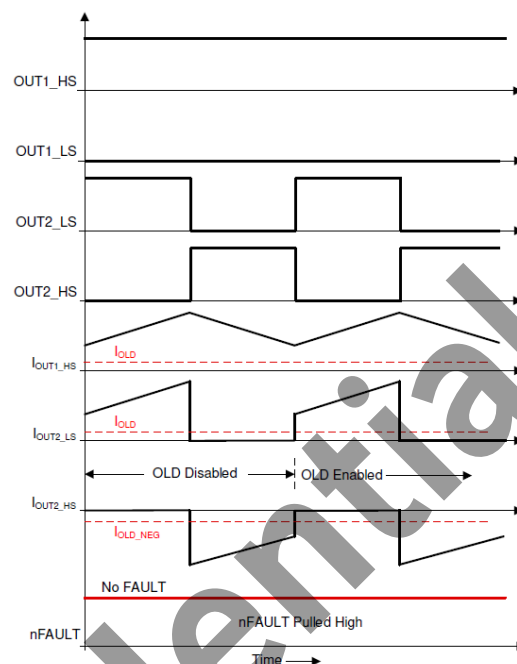


Figure 3-34 Waveforms Showing Operation with Negative-Current OLD Enabled

3.2.3.5.3 Low-current OLD

Low-current open-load detection is another type of active open-load detection in the HE99MD90xx devices. In low-current open-load detection, the current detection threshold is around 10x lower than the active open-load detection scheme. This feature gives the user flexibility to detect a valid open-load condition when driving loads that require low current.

As shown in **Figure 3-35**, if the low-side MOSFET is in operating condition (switched-ON) and the current flowing in the particular MOSFET is lower than the low-current active open-load current threshold (I_{OLD_LOW}) for at least open-load detection deglitch time (t_{OLD}), then an open-load condition is detected. The OLD bit in the IC status (IC_STAT) register is set, the HBx_LS_OLD bit in the open-load status register (OLD_STAT_x) is set and nFAULT pin is driven low during an open-load detect. Normal operation resumes (driver operation and the nFAULT pin is released) when the open-load condition is removed and CLR_FLT command is issued. The OLD bit remains set until cleared through the CLR_FLT bit.

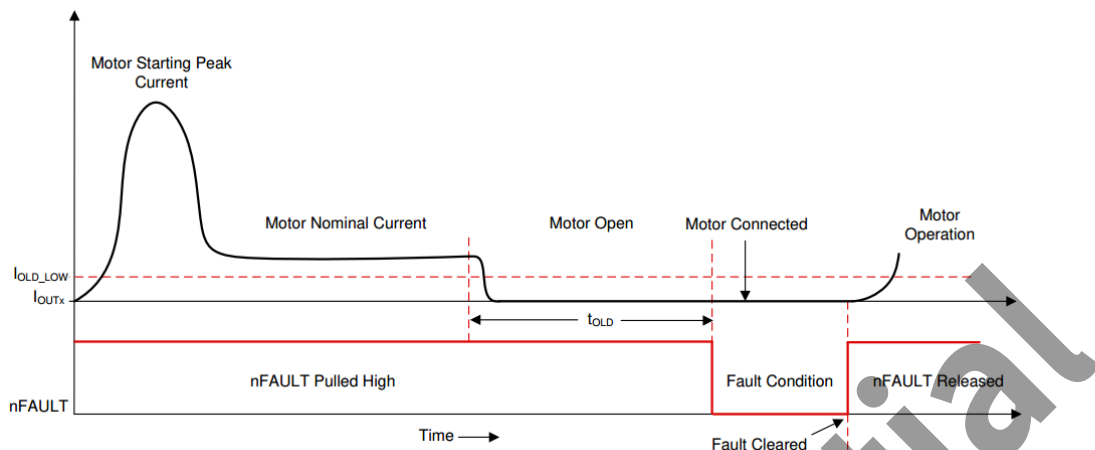


Figure 3-35 Low-Current OLD

Note: The low-current OLD has following limitations

- This feature is only applicable for the current flowing in low-side FET. Once this mode is enabled the corresponding over-current threshold for the low-side FET is also reduced by 11 times (~120 mA min.).
- The $R_{DS(on)}$ of the low side FET will increase by 11 times (~7.5 Ω typical), hence the thermal performance has to be monitored. However, for the lower current the thermal dissipation is limited.

Figure 3-36 shows the flowchart for implementing the low-current active OLD in continuous mode of operation. Following are the steps to configure and detect the low-current active OLD:

- Enable OLD by setting the OLD_OP bit in OLD_CTRL_2 register. This setting will ensure that the OUTx outputs continue to operate when and OLD fault occurs.
- Enable the full-bridge operation by setting the individual HBx_HS_EN/HBx_LS_EN bits in operation control (OP_CTRL_1, OP_CTRL_2 and OP_CTRL_3) registers.
- Check for the nFAULT pin status and the OLD fault in the IC_STAT register.
- If the nFAULT pin is low and the OLD fault is high, then check for the individual HBx_HS_OLD/HBx_LS_OLD bits in OLD status (OLD_STAT_1, OLD_STAT_2 and OLD_STAT_3) registers.
- Disable OLD using the HBx_OLD_DIS bits for the OUTx pins acting as high-side drivers.
- Enable the low-current OLD mode for the half-bridge which low-side is operating by using the HBx_LOLD_EN bit in OLD control (OLD_CTRL_3 and OLD_CTRL_4) registers. This will also disable the high-side OLD for the particular half-bridge.
- Wait for the open-load deglitch time (t_{OLD}).
- Issue the clear fault command (CLR_FLT) to release the nFAULT pin and clear the OLD bits if low-current OLD is not detected.
- If the OLD bit is high and the nFAULT pin is not released (low), then low-current OLD fault is detected.

Note: The low-current OLD is applicable only for low-side FETs. The user has to enable the low- current OLD mode for the corresponding low-side FET.

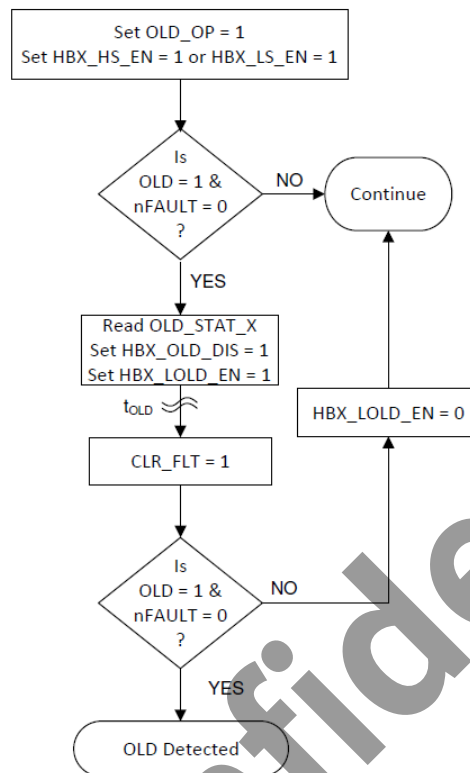


Figure 3-36 Flowchart for Enabling Low-Current OLD (Continuous Mode)

3.2.3.6 Digital Thermal

In order to implement advanced on-chip temperature control, the HE99MD90xx is equipped with a dedicated temperature sensor for each individual channel, and a single comparator is integrated inside the chip. The temperature detection for all channel is performed in a round-robin manner; the detection time interval for each channel can be configured via the (OT_sel) bits, and the filtering time for overtemperature behavior can be configured via the (OT_fit) bits. In case the temperature of an output channel reaches the thermal shutdown threshold, the channel's overtemperature shutdown behavior can be configured via the (TSD_PerHB_EN) bit:

- Standard Mode (default): as soon as any half-bridge reaches thermal threshold the device is switched off.
- Per half-bridge Mode: only the half-bridge that has reached shutdown temperature is switched off.

3.2.3.6.1 Thermal Warning (OTW)

If the die temperature exceeds the trip point of the thermal warning (T_{OTW}), the OTW bit is set in the IC status (IC_STAT) register. If the temperature of the output channel reaches the thermal warning value and is detected by the round-robin monitoring, The OTW_CL1 bit is set in the (ADD_STAT_1) register. The reporting of OTW on the nFAULT pin can be enabled by setting the over-temperature warning reporting (OTW_REP) bit in the configuration control (CONFIG_CTRL) register. The device performs no additional action and continues to function. In this case, the nFAULT pin releases when the die temperature decreases below the hysteresis point of the thermal warning (T_{OTW_HYS}). The OTW bit remains set until cleared through the CLR_FLT bit and the die temperature is lower than thermal warning trip (T_{OTW}).

Note: Over Temperature warning is not reported on nFAULT pin by default.

3.2.3.6.2 Thermal Shutdown (OTSD)

If the temperature of the output channel reaches the thermal shutdown threshold (T_{OTSD}) and is detected by the polling monitoring, then the (OTW_CL1) bit in the (ADD_STAT_1) register will be set, and the OTSD_x bit in either the (ADD_STAT_1) or (TC_STAT_1) register will also be set. In addition, the OTSD bit is latched high in IC status (IC_STAT) register. Normal operation resumes (driver operation and the nFAULT pin is released) when the overtemperature shutdown condition is removed, the die temperature decreases below the hysteresis point of the thermal warning (T_{OTSD_HYS}), and the 12-channel temperature round-robin detection confirms no overtemperature behavior is detected across all channels. The OTSD bit remains latched high indicating that a thermal event occurred until a clear fault command is issued through the CLR_FLT bit. This protection feature cannot be disabled.

3.3 Device Functional Modes

3.3.1 Sleep Mode (nSLEEP = 0)

The nSLEEP pin manages the state of the HE99MD90xx device. When the nSLEEP pin is low, the device enters a low-power sleep mode. In sleep mode, all half-bridge drivers are disabled, the internal charge pump is disabled, the internal regulators are disabled, and the SPI bus is disabled. The t_{SLEEP} time must elapse after a falling edge on the nSLEEP pin before the device enters sleep mode. The device comes out of sleep mode automatically if the nSLEEP pin is pulled high. The t_{WAKE} time must elapse before the device is ready for inputs.

3.3.2 Normal Operating Mode (nSLEEP = 1)

When the nSLEEP pin is high and $V_{VM} > V_{UVLO}$, the device enters operating mode. The t_{WAKE} time must elapse before the device is ready for inputs. In this mode the half bridge drivers, charge pump, internal regulators, and SPI bus are active. [Table 3-9](#) summarizes the different operating modes of HE99MD90xx device.

Table 3-9 Functional Modes

MODE	CONDITION	HALF-BRIDGES	INTERNAL CIRCUITS
Normal Operating	$4.5V < V_{VM} < 24V$ (EXT_OVP = 0 _b) $4.5V < V_{VM} < 32V$ (EXT_OVP = 1 _b) nSLEEP Pin = High	Operating	Operating
Sleep	$4.5V < V_{VM} < 32V$ nSLEEP Pin = Low	Disabled	Disabled
Fault	Any Fault Condition Met	Depend on Fault	Depend on Fault

3.3.3 Fault Mode

The HE99MD90xx is protected against various faults as summarized in [Table 3-10](#).

Table 3-10 Fault Action and Response

FAULT	CONDITION	CONFIGURATION	REPORT	HALF-BRIDGE	LOGIC	RECOVERY
VM Undervoltage (UVLO)	$V_{VM} < V_{UVLO}$ (Max. 4.5V)	—	nFAULT Pin IC_STAT Register	Hi-Z	Active	Automatic: $V_{VM} > V_{UVLO}$
VDD Undervoltage (UVLO)	$V_{VDD} < V_{UVLO}$ (Max 3V)	—	IC_STAT Register	Hi-Z	Reset	Automatic: $V_{VDD} > V_{UVLO}$
VM Overvoltage (OVP)	$V_{VM} > V_{OVP}$ (Min. 24V)	EXT_OVP = 0	nFAULT Pin IC_STAT Register	Hi-Z	Active	Automatic: $V_{VM} < V_{OVP}$
	$V_{VM} > V_{OVP}$ (Min. 32V)	EXT_OVP = 1				
Over Current Protection (OCP)	$I_{OUT} > I_{OCP}$ (Min. 1.3A)	OCP_REP = 0	IC_STAT Register	Hi-Z	Active	CLR_FLT = 1 & $I_{OUT} < I_{OCP}$
		OCP_REP = 1	nFAULT Pin IC_STAT Register	Hi-Z	Active	
Open-Load Detect (OLD)	$I_{OUT} < I_{OLD}$ (Max. 15mA)	OLD_OP = 0 OLD_REP = 0	nFAULT Pin IC_STAT Register	Hi-Z	Active	CLR_FLT = 1 & $I_{LOAD} > I_{OLD}$
		OLD_OP = 0 OLD_REP = 1	IC_STAT Register	Hi-Z	Active	
		OLD_OP = 1 OLD_REP = 0	nFAULT Pin IC_STAT Register	Operating	Active	
		OLD_OP = 1 OLD_REP = 1	IC_STAT Register	Operating	Active	
	$R_{LOAD} > R_{OLD}$ (Max. 100KΩ)	OLD_REP = 0	nFAULT Pin IC_STAT Register	N/A	Active	CLR_FLT = 1 & OLD Sequenced & $R_{LOAD} < R_{OLD}$
		OLD_REP = 1	IC_STAT Register	N/A	Active	
Over-Temperature Warning (OTW)	$T_J > T_{OTW}$ (Min. 120°C)	OTW_REP = 0	IC_STAT Register	Operating	Active	No Action
		OTW_REP = 1	nFAULT Pin IC_STAT Register	Operating	Active	Automatic: $T_J < T_{OTW} - T_{OTW_HYS}$
Over-Temperature Shutdown (OTSD)	$T_J > T_{OTSD}$ (Min. 150°C)	—	nFAULT Pin IC_STAT Register	Hi-Z	Active	Automatic: $T_J < T_{OTSD} - T_{OTSD_HYS}$

3.4 Programming

3.4.1 SPI

SPI bus is used to set device configurations, operating parameters, and read out diagnostic information on the HE99MD90xx device. The SPI operates in slave mode and connects to a master controller. The SPI input data (SDI) word consists of a 16-bit word, with an 8-bit command and 8 bits of data. The SPI output data (SDO) word consists of 8-bit register data and the first 8 bits make up the Status Register with Fault Status indication. The data sequence between the MCU and the SPI slave driver is shown in **Figure 3-37**.

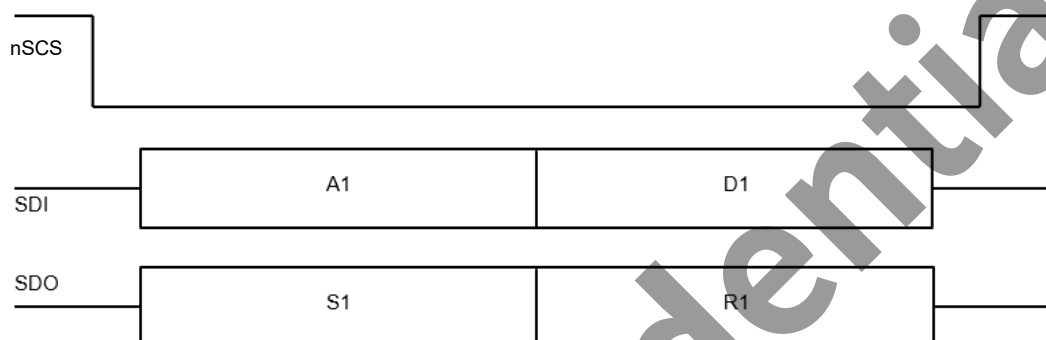


Figure 3-37 SPI Data Frame

A valid frame must meet the following conditions:

The SCLK pin should be low when the nSCS pin transitions from high to low and from low to high.

- The nSCS pin should be pulled high for at least 400ns between words. When the nSCS pin is pulled high, any signals at the SCLK and SDI pins are ignored and the SDO pin is placed in the Hi-Z state.

Data is captured on the falling edge of SCLK and data is propagated on the rising edge of SCLK.

The most significant bit (MSB) is shifted in and out first.

A full 16 SCLK cycles must occur for transaction to be valid.

If the data word sent to the SDI pin is less than or more than 16 bits, a frame error occurs, and the data word is ignored.

For a write command, the existing data in the register being written to is shifted out on the SDO pin following the 8-bit command data.

3.4.2 SPI Format

The SDI input data word is 16 bits long and consists of the following format:

- 1 read or write bit, W (bit B14)
- 6 address bits, A (bits B13 through B8)
- 8 data bits, D (bits B7 through B0)

The SDO output data word is 16 bits long and the first 8 bits makes up the IC status register. The report word is the content of the register being accessed.

For a write command ($W0 = 0$), the response word on the SDO pin is the data currently in the register being written to.

For a read command ($W0 = 1$), the response word is the data currently in the register being read.

Table 3-11 SDI Input Data Word Format

		R/W	Address						Data							
Bit	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
Data	0	W0	A5	A4	A3	A2	A1	A0	D7	D6	D5	D4	D3	D2	D1	D0

Table 3-12 SDO Output Data Word Format

	IC Status								Report							
Bit	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
Data	1	1	OT	OLD	OCF	UVLO	OVP	NVDDUV	D7	D6	D5	D4	D3	D2	D1	D0

3.4.3 SPI Interface for Multiple Slaves

Multiple HE99MD90xx devices can be connected to the master controller with and without the daisy chain. For connecting a 'n' number of HE99MD90xx to a master controller without using a daisy chain, 'n' number of I/O resources from master controller has to be utilized for nSCS pins as shown **Figure 3-38**. Whereas, if the daisy chain configuration is used, then a single nSCS line can be used for connecting multiple HE99MD90xx devices as shown in **Figure 3-39**.

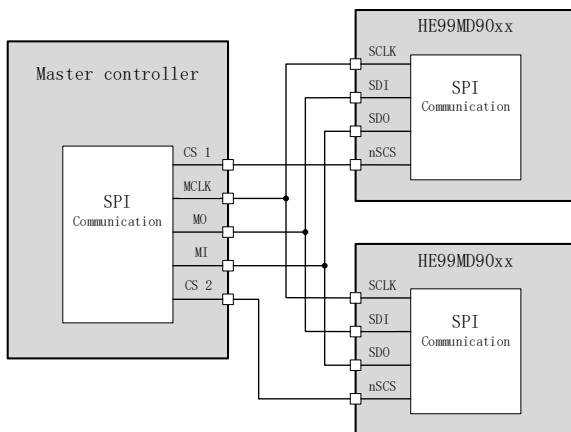


Figure 3-38 SPI Operation Without Daisy Chain

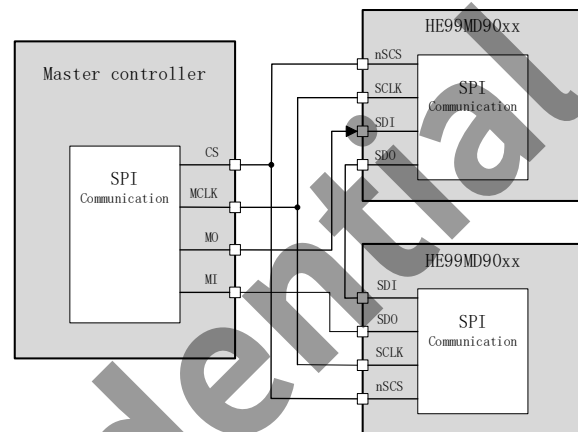


Figure 3-39 SPI Operation With Daisy Chain

3.4.3.1 SPI Interface for Multiple Slaves in Daisy Chain

The HE99MD90xx device can be connected in a daisy chain configuration to save GPIO ports when multiple devices are communicating to the same MCU. **Figure 3-40** shows the topology when 3 devices are connected in series with waveforms.

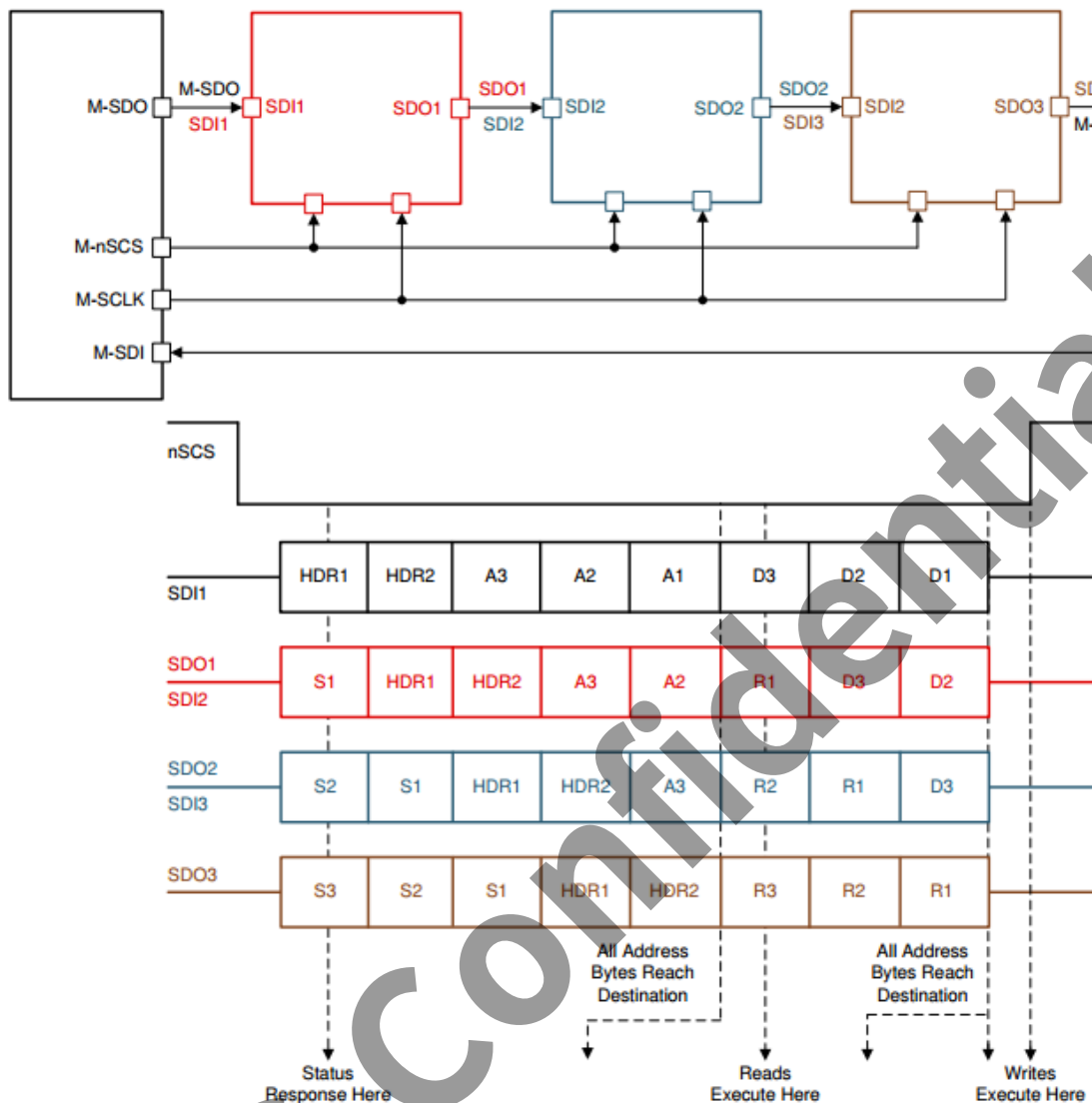


Figure 3-40 Daisy Chain SPI Operation

The first device in the chain shown above receives data from the master controller in the following format. See SDI1 in [Figure 3-40](#)

- 2 bytes of Header
- 3 bytes of Address
- 3 bytes of Data

After the data has been transmitted through the chain, the master controller receives it in the following format. See SDO3 in [Figure 3-40](#)

- 3 bytes of Status
- 2 bytes of Header (should be identical to the information controller sent)

- 3 bytes of Report

The Header bytes contain information of the number of devices connected in the chain, and a global clear fault command that will clear the fault registers of all the devices on the rising edge of the chip select (nSCS) signal. N5 through N0 are 6 bits dedicated to show the number of devices in the chain as shown in **Figure 3-41**. Up to 63 devices can be connected in series per daisy chain connection.

The 5 LSBs of the HDR2 register are don't care bits that can be used by the MCU to determine integrity of the daisy chain connection. Header bytes must start with 1 and 0 for the two MSBs.

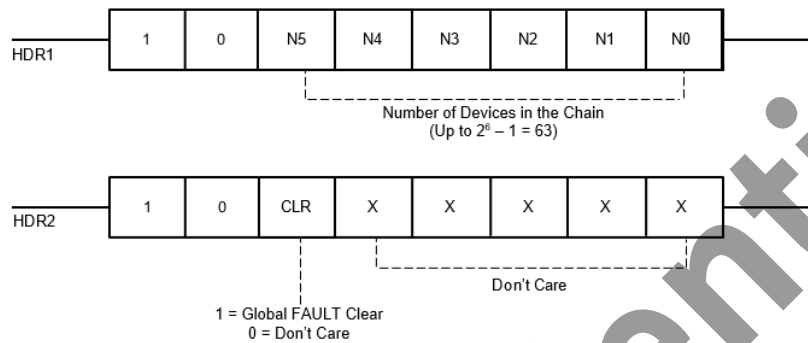


Figure 3-41 Header Bits

The Status byte provides information about the fault status register for each device in the daisy chain as shown in **Figure 3-42**. That way the master controller does not have to initiate a read command to read the fault status from any particular device. This saves the controller additional read commands and makes the system more efficient to determine fault conditions flagged in a device.

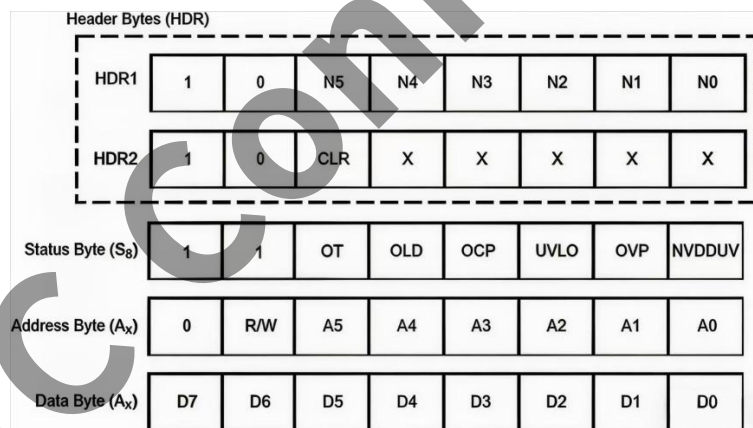


Figure 3-42 Daisy Chain Read Registers

When data passes through a device, it determines the position of itself in the chain by counting the number of Status bytes it receives following by the first Header byte. For example, in this 3-device configuration, device 2 in the chain will receive two Status bytes before receiving HDR1 byte, followed by HDR2 byte.

From the two Status bytes it knows that its position is second in the chain, and from HDR2 byte it knows how many devices are connected in the chain. That way it only loads the relevant address and data byte in its buffer and bypasses the other bits. This protocol allows for faster communication without adding latency to the system for up to 63 devices in the chain.

The address and data bytes remain the same with respect to a single device connection. The Report

bytes (R1 through R3), as shown in the figure above, is the content of the register being accessed.

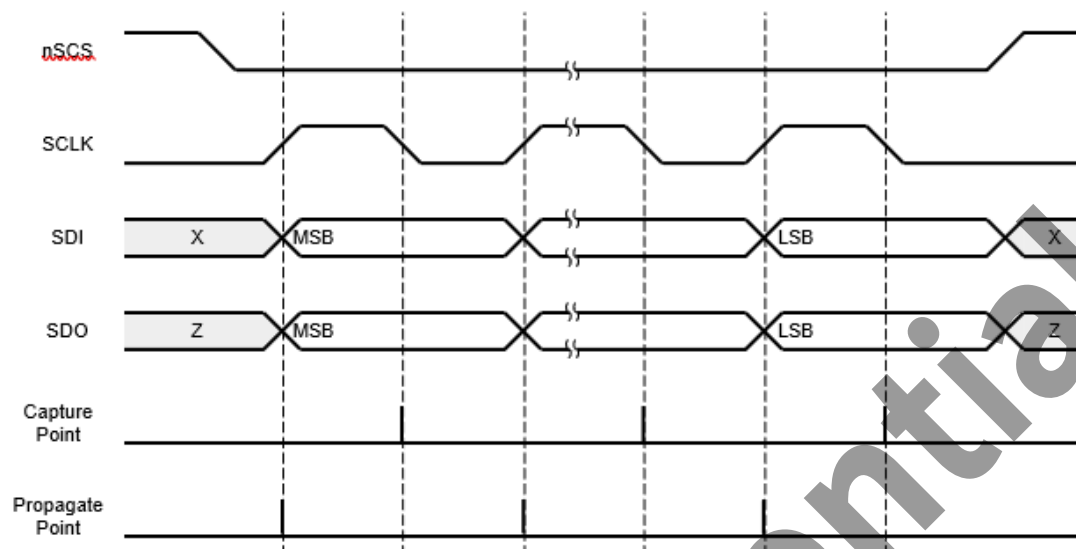


Figure 3-43 SPI Slave Timing Diagram

3.5 Register Map

This section contains the register maps and bit descriptions for all of the HE99MD90xx devices. [Table 3-13](#) summarizes the differences among the part numbers in the HE99MD90xx family.

Table 3-13 Summary of HE99MD90xx Device Family

DEVICE	NUMBER OF HALF-BRIDGES	NUMBER OF PWM GENERATORS	OPEN-LOAD DETECTION SCHEMES	LINK TO REGISTER MAP
HE99MD9012	12	4	Active OLD, Low-Current Active OLD, Negative-Current Active OLD	Table 3-15
HE99MD9010	10			Table 3-16
HE99MD9008N	8			Table 3-17
HE99MD9006	4			Table 3-18

Complex bit access types are encoded to fit into small table cells. [Table 3-14](#) shows the codes that are used for access types in this section.

Table 3-14 Control Registers Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
Write Type		
W	W	Write
Reset or Default Value		
-n		Value after reset or the default value

3.5.1 HE99MD90xx Register Maps

Table 3-15 HE99MD9012 Register Map

Name	7	6	5	4	3	2	1	0	Type	Address
IC_STAT	1'b1	OTSD	OTW	OLD	OCP	UVLO	OVP	nVDDUV	R	00h
OCP_STAT_1	HB4_HS_OCP	HB4_LS_OCP	HB3_HS_OCP	HB3_LS_OCP	HB2_HS_OCP	HB2_LS_OCP	HB1_HS_OCP	HB1_LS_OCP	R	01h
OCP_STAT_2	HB8_HS_OCP	HB8_LS_OCP	HB7_HS_OCP	HB7_LS_OCP	HB6_HS_OCP	HB6_LS_OCP	HB5_HS_OCP	HB5_LS_OCP	R	02h
OCP_STAT_3	HB12_HS_OCP	HB12_LS_OCP	HB11_HS_OCP	HB11_LS_OCP	HB10_HS_OCP	HB10_LS_OCP	HB9_HS_OCP	HB9_LS_OCP	R	03h
OLD_STAT_1	HB4_HS_OLD	HB4_LS_OLD	HB3_HS_OLD	HB3_LS_OLD	HB2_HS_OLD	HB2_LS_OLD	HB1_HS_OLD	HB1_LS_OLD	R	04h
OLD_STAT_2	HB8_HS_OLD	HB8_LS_OLD	HB7_HS_OLD	HB7_LS_OLD	HB6_HS_OLD	HB6_LS_OLD	HB5_HS_OLD	HB5_LS_OLD	R	05h
OLD_STAT_3	HB12_HS_OLD	HB12_LS_OLD	HB11_HS_OLD	HB11_LS_OLD	HB10_HS_OLD	HB10_LS_OLD	HB9_HS_OLD	HB9_LS_OLD	R	06h
CONFIG_CTRL	Reserved				OCP_REP	OTW_REP	EXT_OVP	CLR_FLT	RW	07h
OP_CTRL_1	HB4_HS_EN	HB4_LS_EN	HB3_HS_EN	HB3_LS_EN	HB2_HS_EN	HB2_LS_EN	HB1_HS_EN	HB1_LS_EN	RW	08h
OP_CTRL_2	HB8_HS_EN	HB8_LS_EN	HB7_HS_EN	HB7_LS_EN	HB6_HS_EN	HB6_LS_EN	HB5_HS_EN	HB5_LS_EN	RW	09h
OP_CTRL_3	HB12_HS_EN	HB12_LS_EN	HB11_HS_EN	HB11_LS_EN	HB10_HS_EN	HB10_LS_EN	HB9_HS_EN	HB9_LS_EN	RW	0Ah
PWM_CTRL_1	HB8_PWM	HB7_PWM	HB6_PWM	HB5_PWM	HB4_PWM	HB3_PWM	HB2_PWM	HB1_PWM	RW	0Bh
PWM_CTRL_2	PWM_CH4_DIS	PWM_CH3_DIS	PWM_CH2_DIS	PWM_CH1_DIS	HB12_PWM	HB11_PWM	HB10_PWM	HB9_PWM	RW	0Ch
FW_CTRL_1	HB8_FW	HB7_FW	HB6_FW	HB5_FW	HB4_FW	HB3_FW	HB2_FW	HB1_FW	RW	0Dh
FW_CTRL_2	Reserved				HB12_FW	HB11_FW	HB10_FW	HB9_FW	RW	0Eh
PWM_MAP_CTRL_1	HB4_PWM_MAP		HB3_PWM_MAP		HB2_PWM_MAP		HB1_PWM_MAP		RW	0Fh
PWM_MAP_CTRL_2	HB8_PWM_MAP		HB7_PWM_MAP		HB6_PWM_MAP		HB5_PWM_MAP		RW	10h
PWM_MAP_CTRL_3	HB12_PWM_MAP		HB11_PWM_MAP		HB10_PWM_MAP		HB9_PWM_MAP		RW	11h
PWM_FREQ_CTRL	PWM_CH4_FREQ		PWM_CH3_FREQ		PWM_CH2_FREQ		PWM_CH1_FREQ		RW	12h
PWM_DUTY_CTRL_1	PWM_DUTY_CH1								RW	13h
PWM_DUTY_CTRL_2	PWM_DUTY_CH2								RW	14h
PWM_DUTY_CTRL_3	PWM_DUTY_CH3								RW	15h
PWM_DUTY_CTRL	PWM_DUTY_CH4								RW	16h

Name	7	6	5	4	3	2	1	0	Type	Address
L_4										
SR_CTRL_1	HB8_SR	HB7_SR	HB6_SR	HB5_SR	HB4_SR	HB3_SR	HB2_SR	HB1_SR	RW	17h
SR_CTRL_2	Reserved				HB12_SR	HB11_SR	HB10_SR	HB9_SR	RW	18h
OLD_CTRL_1	HB8_OLD_DIS	HB7_OLD_DIS	HB6_OLD_DIS	HB5_OLD_DIS	HB4_OLD_DIS	HB3_OLD_DIS	HB2_OLD_DIS	HB1_OLD_DIS	RW	19h
OLD_CTRL_2	OLD_REP	OLD_OP	PL_MODE_EN		HB12_OLD_DIS	HB11_OLD_DIS	HB10_OLD_DIS	HB9_OLD_DIS	RW	1Ah
OLD_CTRL_3	OCP_DEG			OLD_NEG_EN	HB12_LOLD_EN	HB11_LOLD_EN	HB10_LOLD_EN	HB9_LOLD_EN	RW	1Bh
OLD_CTRL_4	HB8_LOLD_EN	HB7_LOLD_EN	HB6_LOLD_EN	HB5_LOLD_EN	HB4_LOLD_EN	HB3_LOLD_EN	HB2_LOLD_EN	HB1_LOLD_EN	RW	24h
ADD_CONFIG_CTRL	Reserved				CP_OTSD_DIS	CP_VM_UV_DIS	TSD_PerHB_EN	OVP_OP	RW	28h
ADD_STAT_1	OTSD<11:7>					Reserved	OTSD_CL1	OTW_CL1	R	30h
TC_STAT_1	CP_UV	OTSD<6:0>							R	31h
ADD_CTRL_EX	Reserved		OT_SEL_WIDTH		OT_FLT		OSC_MODU_DIS	OSC_MODU_SEL	RW	32h
ADD_OLD_CTRL_1	HS8_NEG_OLD_EN	HS7_NEG_OLD_EN	HS6_NEG_OLD_EN	HS5_NEG_OLD_EN	HS4_NEG_OLD_EN	HS3_NEG_OLD_EN	HS2_NEG_OLD_EN	HS1_NEG_OLD_EN	RW	33h
ADD_OLD_CTRL_2	HS12_NEG_OLD_EN	HS11_NEG_OLD_EN	HS10_NEG_OLD_EN	HS9_NEG_OLD_EN	LS12_NEG_OLD_EN	LS11_NEG_OLD_EN	LS10_NEG_OLD_EN	LS9_NEG_OLD_EN	RW	34h
ADD_OLD_CTRL_3	LS8_NEG_OLD_EN	LS7_NEG_OLD_EN	LS6_NEG_OLD_EN	LS5_NEG_OLD_EN	LS4_NEG_OLD_EN	LS3_NEG_OLD_EN	LS2_NEG_OLD_EN	LS1_NEG_OLD_EN	RW	35h

Table 3-16 HE99MD9010 Register Map

Name	7	6	5	4	3	2	1	0	Type	Address
IC_STAT	1'b1	OTSD	OTW	OLD	OCP	UVLO	OVP	nVDDUV	R	00h
OCP_STAT_1	HB4_HS_OCP	HB4_LS_OCP	HB3_HS_OCP	HB3_LS_OCP	HB2_HS_OCP	HB2_LS_OCP	HB1_HS_OCP	HB1_LS_OCP	R	01h
OCP_STAT_2	HB8_HS_OCP	HB8_LS_OCP	HB7_HS_OCP	HB7_LS_OCP	HB6_HS_OCP	HB6_LS_OCP	HB5_HS_OCP	HB5_LS_OCP	R	02h
OCP_STAT_3	Reserved				HB10_HS_OCP	HB10_LS_OCP	HB9_HS_OCP	HB9_LS_OCP	R	03h
OLD_STAT_1	HB4_HS_OLD	HB4_LS_OLD	HB3_HS_OLD	HB3_LS_OLD	HB2_HS_OLD	HB2_LS_OLD	HB1_HS_OLD	HB1_LS_OLD	R	04h
OLD_STAT_2	HB8_HS_OLD	HB8_LS_OLD	HB7_HS_OLD	HB7_LS_OLD	HB6_HS_OLD	HB6_LS_OLD	HB5_HS_OLD	HB5_LS_OLD	R	05h
OLD_STAT_3	Reserved				HB10_HS_OLD	HB10_LS_OLD	HB9_HS_OLD	HB9_LS_OLD	R	06h
CONFIG_CTRL	Reserved				OCP_REP	OTW_REP	EXT_OVP	CLR_FLT	RW	07h
OP_CTRL_1	HB4_HS_EN	HB4_LS_EN	HB3_HS_EN	HB3_LS_EN	HB2_HS_EN	HB2_LS_EN	HB1_HS_EN	HB1_LS_EN	RW	08h
OP_CTRL_2	HB8_HS_EN	HB8_LS_EN	HB7_HS_EN	HB7_LS_EN	HB6_HS_EN	HB6_LS_EN	HB5_HS_EN	HB5_LS_EN	RW	09h
OP_CTRL_3	Reserved				HB10_HS_EN	HB10_LS_EN	HB9_HS_EN	HB9_LS_EN	RW	0Ah
PWM_CTRL_1	HB8_PWM	HB7_PWM	HB6_PWM	HB5_PWM	HB4_PWM	HB3_PWM	HB2_PWM	HB1_PWM	RW	0Bh
PWM_CTRL_2	PWM_CH4_DIS	PWM_CH3_DIS	PWM_CH2_DIS	PWM_CH1_DIS	Reserved		HB10_PWM	HB9_PWM	RW	0Ch
FW_CTRL_1	HB8_FW	HB7_FW	HB6_FW	HB5_FW	HB4_FW	HB3_FW	HB2_FW	HB1_FW	RW	0Dh
FW_CTRL_2	Reserved						HB10_FW	HB9_FW	RW	0Eh
PWM_MAP_CTRL_1	HB4_PWM_MAP		HB3_PWM_MAP		HB2_PWM_MAP		HB1_PWM_MAP		RW	0Fh
PWM_MAP_CTRL_2	HB8_PWM_MAP		HB7_PWM_MAP		HB6_PWM_MAP		HB5_PWM_MAP		RW	10h
PWM_MAP_CTRL_3	Reserved				HB10_PWM_MAP		HB9_PWM_MAP		RW	11h
PWM_FREQ_CTRL	PWM_CH4_FREQ		PWM_CH3_FREQ		PWM_CH2_FREQ		PWM_CH1_FREQ		RW	12h
PWM_DUTY_CTRL_1	PWM_DUTY_CH1								RW	13h
PWM_DUTY_CTRL_2	PWM_DUTY_CH2								RW	14h
PWM_DUTY_CTRL_3	PWM_DUTY_CH3								RW	15h
PWM_DUTY_CTRL_4	PWM_DUTY_CH4								RW	16h
SR_CTRL_1	HB8_SR	HB7_SR	HB6_SR	HB5_SR	HB4_SR	HB3_SR	HB2_SR	HB1_SR	RW	17h

Name	7	6	5	4	3	2	1	0	Type	Address
SR_CTRL_2	Reserved						HB10_SR	HB9_SR	RW	18h
OLD_CTRL_1	HB8_OLD_DIS	HB7_OLD_DIS	HB6_OLD_DIS	HB5_OLD_DIS	HB4_OLD_DIS	HB3_OLD_DIS	HB2_OLD_DIS	HB1_OLD_DIS	RW	19h
OLD_CTRL_2	OLD_REP	OLD_OP	PL_MODE_EN		Reserved		HB10_OLD_DIS	HB9_OLD_DIS	RW	1Ah
OLD_CTRL_3	OCP_DEG			OLD_NEG_EN	Reserved		HB10_LOLD_EN	HB9_LOLD_EN	RW	1Bh
OLD_CTRL_4	HB8_LOLD_EN	HB7_LOLD_EN	HB6_LOLD_EN	HB5_LOLD_EN	HB4_LOLD_EN	HB3_LOLD_EN	HB2_LOLD_EN	HB1_LOLD_EN	RW	24h
ADD_CONFIG_CTRL	Reserved				CP_OTSD_DIS	CP_VM_UV_DIS	TSD_PerHB_EN	OVP_OP	RW	28h
ADD_STAT_1	Reserved		OTSD<9:7>			Reserved	OTSD_CL1	OTW_CL1	R	30h
TC_STAT_1	CP_UV	OTSD<6:0>							R	31h
ADD_CTRL_EX	Reserved		OT_SEL_WIDTH		OT_FLT		OSC_MODU_DIS	OSC_MODU_SEL	RW	32h
ADD_OLD_CTRL_1	HS8_NEG_OLD_EN	HS7_NEG_OLD_EN	HS6_NEG_OLD_EN	HS5_NEG_OLD_EN	HS4_NEG_OLD_EN	HS3_NEG_OLD_EN	HS2_NEG_OLD_EN	HS1_NEG_OLD_EN	RW	33h
ADD_OLD_CTRL_2	Reserved		HS10_NEG_OLD_EN	HS9_NEG_OLD_EN	Reserved		LS10_NEG_OLD_EN	LS9_NEG_OLD_EN	RW	34h
ADD_OLD_CTRL_3	LS8_NEG_OLD_EN	LS7_NEG_OLD_EN	LS6_NEG_OLD_EN	LS5_NEG_OLD_EN	LS4_NEG_OLD_EN	LS3_NEG_OLD_EN	LS2_NEG_OLD_EN	LS1_NEG_OLD_EN	RW	35h

Table 3-17 HE99MD9008N Register Map

Name	7	6	5	4	3	2	1	0	Typ e	Address
IC_STAT	1'b1	OTSD	OTW	OLD	OCP	UVLO	OVP	nVDDUV	R	00h
OCP_STAT_1	HB4_HS_OCP	HB4_LS_OCP	HB3_HS_OCP	HB3_LS_OCP	HB2_HS_OCP	HB2_LS_OCP	HB1_HS_OCP	HB1_LS_OCP	R	01h
OCP_STAT_2	HB8_HS_OCP	HB8_LS_OCP	HB7_HS_OCP	HB7_LS_OCP	HB6_HS_OCP	HB6_LS_OCP	HB5_HS_OCP	HB5_LS_OCP	R	02h
OCP_STAT_3	Reserved								R	03h
OLD_STAT_1	HB4_HS_OLD	HB4_LS_OLD	HB3_HS_OLD	HB3_LS_OLD	HB2_HS_OLD	HB2_LS_OLD	HB1_HS_OLD	HB1_LS_OLD	R	04h
OLD_STAT_2	HB8_HS_OLD	HB8_LS_OLD	HB7_HS_OLD	HB7_LS_OLD	HB6_HS_OLD	HB6_LS_OLD	HB5_HS_OLD	HB5_LS_OLD	R	05h
OLD_STAT_3	Reserved								R	06h
CONFIG_CTRL	Reserved				OCP_REP	OTW_REP	EXT_OVP	CLR_FLT	RW	07h
OP_CTRL_1	HB4_HS_EN	HB4_LS_EN	HB3_HS_EN	HB3_LS_EN	HB2_HS_EN	HB2_LS_EN	HB1_HS_EN	HB1_LS_EN	RW	08h
OP_CTRL_2	HB8_HS_EN	HB8_LS_EN	HB7_HS_EN	HB7_LS_EN	HB6_HS_EN	HB6_LS_EN	HB5_HS_EN	HB5_LS_EN	RW	09h
OP_CTRL_3	Reserved								RW	0Ah
PWM_CTRL_1	HB8_PWM	HB7_PWM	HB6_PWM	HB5_PWM	HB4_PWM	HB3_PWM	HB2_PWM	HB1_PWM	RW	0Bh
PWM_CTRL_2	PWM_CH4_DIS	PWM_CH3_DIS	PWM_CH2_DIS	PWM_CH1_DIS	Reserved				RW	0Ch
FW_CTRL_1	HB8_FW	HB7_FW	HB6_FW	HB5_FW	HB4_FW	HB3_FW	HB2_FW	HB1_FW	RW	0Dh
FW_CTRL_2	Reserved								RW	0Eh
PWM_MAP_CTRL_1	HB4_PWM_MAP		HB3_PWM_MAP		HB2_PWM_MAP		HB1_PWM_MAP		RW	0Fh
PWM_MAP_CTRL_2	HB8_PWM_MAP		HB7_PWM_MAP		HB6_PWM_MAP		HB5_PWM_MAP		RW	10h
PWM_MAP_CTRL_3	Reserved								RW	11h
PWM_FREQ_CTRL	PWM_CH4_FREQ		PWM_CH3_FREQ		PWM_CH2_FREQ		PWM_CH1_FREQ		RW	12h
PWM_DUTY_CTRL_1	PWM_DUTY_CH1								RW	13h
PWM_DUTY_CTRL_2	PWM_DUTY_CH2								RW	14h
PWM_DUTY_CTRL_3	PWM_DUTY_CH3								RW	15h
PWM_DUTY_CTRL_4	PWM_DUTY_CH4								RW	16h
SR_CTRL_1	HB8_SR	HB7_SR	HB6_SR	HB5_SR	HB4_SR	HB3_SR	HB2_SR	HB1_SR	RW	17h

Name	7	6	5	4	3	2	1	0	Type	Address
SR_CTRL_2	Reserved								RW	18h
OLD_CTRL_1	HB8_OLD_DIS	HB7_OLD_DIS	HB6_OLD_DIS	HB5_OLD_DIS	HB4_OLD_DIS	HB3_OLD_DIS	HB2_OLD_DIS	HB1_OLD_DIS	RW	19h
OLD_CTRL_2	OLD_REP	OLD_OP	PL_MODE_EN		Reserved				RW	1Ah
OLD_CTRL_3	OCP_DEG			OLD_NEG_EN	Reserved				RW	1Bh
OLD_CTRL_4	HB8_LOLD_EN	HB7_LOLD_EN	HB6_LOLD_EN	HB5_LOLD_EN	HB4_LOLD_EN	HB3_LOLD_EN	HB2_LOLD_EN	HB1_LOLD_EN	RW	24h
ADD_CONFIG_CTRL	Reserved				CP_OTSD_DIS	CP_VM_UV_DIS	TSD_PerHB_EN	OVP_OP	RW	28h
ADD_STAT_1	Reserved				OTSD<7>	Reserved	OTSD_CL1	OTW_CL1	R	30h
TC_STAT_1	CP_UV	OTSD<6:0>							R	31h
ADD_CTRL_EX	Reserved		OT_SEL_WIDTH		OT_FLT		OSC_MODU_DS	OSC_MODU_SEL	RW	32h
ADD_OLD_CTRL_1	HS8_NEG_OLD_EN	HS7_NEG_OLD_EN	HS6_NEG_OLD_EN	HS5_NEG_OLD_EN	HS4_NEG_OLD_EN	HS3_NEG_OLD_EN	HS2_NEG_OLD_EN	HS1_NEG_OLD_EN	RW	33h
ADD_OLD_CTRL_2	Reserved								RW	34h
ADD_OLD_CTRL_3	LS8_NEG_OLD_EN	LS7_NEG_OLD_EN	LS6_NEG_OLD_EN	LS5_NEG_OLD_EN	LS4_NEG_OLD_EN	LS3_NEG_OLD_EN	LS2_NEG_OLD_EN	LS1_NEG_OLD_EN	RW	35h

Table 3-18 HE99MD9006 Register Map

Name	7	6	5	4	3	2	1	0	Typ e	Address s
IC_STAT	1'b1	OTSD	OTW	OLD	OCP	UVLO	OVP	nVDDUV	R	00h
OCP_STAT_1	HB4_HS_OC P	HB4_LS_OCP	HB3_HS_OCP	HB3_LS_OCP	HB2_HS_OCP	HB2_LS_OCP	HB1_HS_OCP	HB1_LS_OCP	R	01h
OCP_STAT_2	Reserved				HB6_HS_OCP	HB6_LS_OCP	HB5_HS_OCP	HB5_LS_OCP	R	02h
OCP_STAT_3	Reserved								R	03h
OLD_STAT_1	HB4_HS_OLD	HB4_LS_OLD	HB3_HS_OLD	HB3_LS_OLD	HB2_HS_OLD	HB2_LS_OLD	HB1_HS_OLD	HB1_LS_OLD	R	04h
OLD_STAT_2	Reserved				HB6_HS_OLD	HB6_LS_OLD	HB5_HS_OLD	HB5_LS_OLD	R	05h
OLD_STAT_3	Reserved								R	06h
CONFIG_CTRL	Reserved				OCP_REP	OTW_REP	EXT_OVP	CLR_FLT	RW	07h
OP_CTRL_1	HB4_HS_EN	HB4_LS_EN	HB3_HS_EN	HB3_LS_EN	HB2_HS_EN	HB2_LS_EN	HB1_HS_EN	HB1_LS_EN	RW	08h
OP_CTRL_2	Reserved				HB6_HS_EN	HB6_LS_EN	HB5_HS_EN	HB5_LS_EN	RW	09h
OP_CTRL_3	Reserved								RW	0Ah
PWM_CTRL_1	Reserved		HB6_PWM	HB5_PWM	HB4_PWM	HB3_PWM	HB2_PWM	HB1_PWM	RW	0Bh
PWM_CTRL_2	PWM_CH4_D IS	PWM_CH3_D IS	PWM_CH2_DIS	PWM_CH1_DIS	Reserved				RW	0Ch
FW_CTRL_1	Reserved		HB6_FW	HB5_FW	HB4_FW	HB3_FW	HB2_FW	HB1_FW	RW	0Dh
FW_CTRL_2	Reserved								RW	0Eh
PWM_MAP_CTRL_1	HB4_PWM_MAP		HB3_PWM_MAP		HB2_PWM_MAP		HB1_PWM_MAP		RW	0Fh
PWM_MAP_CTRL_2					HB6_PWM_MAP		HB5_PWM_MAP		RW	10h
PWM_MAP_CTRL_3	Reserved								RW	11h
PWM_FREQ_CTRL	PWM_CH4_FREQ		PWM_CH3_FREQ		PWM_CH2_FREQ		PWM_CH1_FREQ		RW	12h
PWM_DUTY_CTRL_1	PWM_DUTY_CH1								RW	13h
PWM_DUTY_CTRL_2	PWM_DUTY_CH2								RW	14h
PWM_DUTY_CTRL_3	PWM_DUTY_CH3								RW	15h
PWM_DUTY_CTRL_4	PWM_DUTY_CH4								RW	16h
SR_CTRL_1	Reserved		HB6_SR	HB5_SR	HB4_SR	HB3_SR	HB2_SR	HB1_SR	RW	17h

Name	7	6	5	4	3	2	1	0	Type	Address
SR_CTRL_2	Reserved								RW	18h
OLD_CTRL_1	Reserved		HB6_OLD_DIS	HB5_OLD_DIS	HB4_OLD_DIS	HB3_OLD_DIS	HB2_OLD_DIS	HB1_OLD_DIS	RW	19h
OLD_CTRL_2	OLD_REP	OLD_OP	PL_MODE_EN		Reserved				RW	1Ah
OLD_CTRL_3	OCP_DEG			OLD_NEG_EN	Reserved				RW	1Bh
OLD_CTRL_4	Reserved		HB6_LOLD_EN	HB5_LOLD_EN	HB4_LOLD_EN	HB3_LOLD_EN	HB2_LOLD_EN	HB1_LOLD_EN	RW	24h
ADD_CONFIG_CTL	Reserved				CP_OTSD_DIS	CP_VM_UV_DIS	TSD_PerHB_EN	OVP_OP	RW	28h
ADD_STAT_1	Reserved						OTSD_CL1	OTW_CL1	R	30h
TC_STAT_1	CP_UV	OTSD<6:0>							R	31h
ADD_CTRL_EX	Reserved		OT_SEL_WIDTH		OT_FLT		OSC_MODU_DIS	OSC_MODU_SEL	RW	32h
ADD_OLD_CTRL_1	Reserved		HS6_NEG_OLD_EN	HS5_NEG_OLD_EN	HS4_NEG_OLD_EN	HS3_NEG_OLD_EN	HS2_NEG_OLD_EN	HS1_NEG_OLD_EN	RW	33h
ADD_OLD_CTRL_2	Reserved								RW	34h
ADD_OLD_CTRL_3	Reserved		LS6_NEG_OLD_EN	LS5_NEG_OLD_EN	LS4_NEG_OLD_EN	LS3_NEG_OLD_EN	LS2_NEG_OLD_EN	LS1_NEG_OLD_EN	RW	35h

(1) After this register address, the functions are similar between HE99MD9012, HE99MD9010, HE99MD9008N and HE99MD9006.

3.5.1.1 Status Registers

The status registers are used to report warning and fault conditions. The status registers are read-only registers.

Table 3-19 lists the memory-mapped registers for the status registers. All register offset addresses not listed in **Table 3-19** should be considered as reserved locations and the register contents should not be modified.

Table 3-19 Status Registers Summary Table

Address	Register Name	Section
0x00	IC Status	3.5.1.1.1
0x01	Overcurrent Protection (OCP) Status 1	3.5.1.1.2
0x02	Overcurrent Protection (OCP) Status 2	3.5.1.1.3
0x03	Overcurrent Protection (OCP) Status 3	3.5.1.1.4
0x04	Open-Load Detect (OLD) Status 1	3.5.1.1.5
0x05	Open-Load Detect (OLD) Status 2	3.5.1.1.6
0x06	Open-Load Detect (OLD) Status 3	3.5.1.1.7
0x30	Additional Status (ADD_STAT_1) Register	3.5.1.1.8
0x31	Digital Thermal Status (TC_STAT_1) Register	3.5.1.1.9

3.5.1.1.1 IC Status (IC_STAT) Register (Address = 0x00) [reset = 0x81]

The IC status (IC_STAT) register is shown in **Table 3-20** and described in **Table 3-21**.

Register access type: Read only

Table 3-20 IC Status Register

7	6	5	4	3	2	1	0
Reserved	OTSD	OTW	OLD	OCP	UVLO	OVP	nVDDUV
R-1 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-1 _b

Table 3-21 IC Status Register Field Descriptions

Bit	Field	Type	Reset	Description
7	Reserved	R	1 _b	Reserved
6	OTSD	R	0 _b	0 _b = No overtemperature shutdown is detected 1 _b = Overcurrent condition is detected
5	OTW	R	0 _b	0 _b = No overtemperature warning is detected 1 _b = Overcurrent condition is detected
4	OLD	R	0 _b	0 _b = No open-load condition is detected 1 _b = Open-load condition is detected
3	OCP	R	0 _b	0 _b = No overcurrent condition is detected 1 _b = Overcurrent condition is detected
2	UVLO	R	0 _b	0 _b = No undervoltage lock-out condition is detected 1 _b = Under-voltage lock-out condition is detected
1	OVP	R	0 _b	0 _b = No overvoltage condition is detected 1 _b = Overvoltage condition is detected
0	nVDDUV	R	1 _b	0 _b = VDD undervoltage condition is detected 1 _b = NO VDD undervoltage condition is detected

3.5.1.1.2 Overcurrent Protection (OCP) Status 1 (OCP_STAT_1) Register (Address = 0x01) [reset = 0x00]

The overcurrent protection (OCP) status 1 register is shown in [Table 3-22](#) and described in [Table 3-23](#).

Register access type: Read only

Table 3-22 Overcurrent Protection (OCP) Status 1 Register

7	6	5	4	3	2	1	0
HB4_HS_OCP	HB4_LS_OCP	HB3_HS_OCP	HB3_LS_OCP	HB2_HS_OCP	HB2_LS_OCP	HB1_HS_OCP	HB1_LS_OCP
R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b

Table 3-23 Overcurrent Protection (OCP) Status 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB4_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 4 1 _b = Overcurrent detected on high-side switch of half-bridge 4
6	HB4_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 4 1 _b = Overcurrent detected on low-side switch of half-bridge 4
5	HB3_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 3 1 _b = Overcurrent detected on high-side switch of half-bridge 3
4	HB3_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 3 1 _b = Overcurrent detected on low-side switch of half-bridge 3
3	HB2_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 2 1 _b = Overcurrent detected on high-side switch of half-bridge 2
2	HB2_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 2 1 _b = Overcurrent detected on low-side switch of half-bridge 2
1	HB1_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 1 1 _b = Overcurrent detected on high-side switch of half-bridge 1
0	HB1_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 1 1 _b = Overcurrent detected on low-side switch of half-bridge 1

3.5.1.1.3 Overcurrent Protection (OCP) Status 2 (OCP_STAT_2) Register (Address = 0x02) [reset = 0x00]

The overcurrent protection (OCP) status 2 register is shown in [Table 3-24](#) and described in [Table 3-25](#).

Register access type: Read only

Table 3-24 Overcurrent Protection (OCP) Status 2 Register

7	6	5	4	3	2	1	0
HB8_HS_OCP	HB8_LS_OCP	HB7_HS_OCP	HB7_LS_OCP	HB6_HS_OCP	HB6_LS_OCP	HB5_HS_OCP	HB5_LS_OCP
R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b

Table 3-25 Overcurrent Protection (OCP) Status 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 8 1 _b = Overcurrent detected on high-side switch of half-bridge 8
6	HB8_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 8 1 _b = Overcurrent detected on low-side switch of half-bridge 8
5	HB7_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 7 1 _b = Overcurrent detected on high-side switch of half-bridge 7
4	HB7_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 7 1 _b = Overcurrent detected on low-side switch of half-bridge 7
3	HB6_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 6 1 _b = Overcurrent detected on high-side switch of half-bridge 6
2	HB6_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 6 1 _b = Overcurrent detected on low-side switch of half-bridge 6
1	HB5_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 5 1 _b = Overcurrent detected on high-side switch of half-bridge 5
0	HB5_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 5 1 _b = Overcurrent detected on low-side switch of half-bridge 5

3.5.1.1.4 Overcurrent Protection (OCP) Status 3 (OCP_STAT_3) Register (Address = 0x03) [reset = 0x00]

The overcurrent protection (OCP) status 3 register is shown in [Table 3-26](#) and described in [Table 3-27](#).

Register access type: Read only

Table 3-26 Overcurrent Protection (OCP) Status 3 Register

7	6	5	4	3	2	1	0
HB12_HS_OCP	HB12_LS_OCP	HB11_HS_OCP	HB11_LS_OCP	HB10_HS_OCP	HB10_LS_OCP	HB9_HS_OCP	HB9_LS_OCP
R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b

Table 3-27 Overcurrent Protection (OCP) Status 3 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB12_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 12 1 _b = Overcurrent detected on high-side switch of half-bridge 12
6	HB12_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 12 1 _b = Overcurrent detected on low-side switch of half-bridge 12
5	HB11_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 11 1 _b = Overcurrent detected on high-side switch of half-bridge 11
4	HB11_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 11 1 _b = Overcurrent detected on low-side switch of half-bridge 11
3	HB10_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 10 1 _b = Overcurrent detected on high-side switch of half-bridge 10
2	HB10_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 10 1 _b = Overcurrent detected on low-side switch of half-bridge 10
1	HB9_HS_OCP	R	0 _b	0 _b = No overcurrent detected on high-side switch of half-bridge 9 1 _b = Overcurrent detected on high-side switch of half-bridge 9
0	HB9_LS_OCP	R	0 _b	0 _b = No overcurrent detected on low-side switch of half-bridge 9 1 _b = Overcurrent detected on low-side switch of half-bridge 9

3.5.1.1.5 Open-Load Detect (OLD) Status 1 (OLD_STAT_1) Register (Address = 0x04) [reset = 0x00]

The open-load detect (OLD) status 1 register is shown in [Table 3-28](#) and described in [Table 3-29](#).

Register access type: Read only

Table 3-28 Open-Load Detect (OLD) Status 1 Register

7	6	5	4	3	2	1	0
HB4_HS_OLD	HB4_LS_OLD	HB3_HS_OLD	HB3_LS_OLD	HB2_HS_OLD	HB2_LS_OLD	HB1_HS_OLD	HB1_LS_OLD
R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b

Table 3-29 Open-Load Detect (OLD) Status 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB4_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 4 1 _b = Open load detected on high-side switch of half-bridge 4
6	HB4_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 4 1 _b = Open load detected on low-side switch of half-bridge 4
5	HB3_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 3 1 _b = Open load detected on high-side switch of half-bridge 3
4	HB3_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 3 1 _b = Open load detected on low-side switch of half-bridge 3
3	HB2_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 2 1 _b = Open load detected on high-side switch of half-bridge 2
2	HB2_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 2 1 _b = Open load detected on low-side switch of half-bridge 2
1	HB1_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 1 1 _b = Open load detected on high-side switch of half-bridge 1
0	HB1_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 1 1 _b = Open load detected on low-side switch of half-bridge 1

3.5.1.1.6 Open-Load Detect (OLD) Status 2 (OLD_STAT_2) Register (Address = 0x05) [reset = 0x00]

The open-load detect (OLD) status 2 register is shown in [Table 3-30](#) and described in [Table 3-31](#).

Register access type: Read only

Table 3-30 Open-Load Detect (OLD) Status 2 Register

7	6	5	4	3	2	1	0
HB8_HS_OLD	HB8_LS_OLD	HB7_HS_OLD	HB7_LS_OLD	HB6_HS_OLD	HB6_LS_OLD	HB5_HS_OLD	HB5_LS_OLD
R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b

Table 3-31 Open-Load Detect (OLD) Status 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 8 1 _b = Open load detected on high-side switch of half-bridge 8
6	HB8_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 8 1 _b = Open load detected on low-side switch of half-bridge 8
5	HB7_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 7 1 _b = Open load detected on high-side switch of half-bridge 7
4	HB7_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 7 1 _b = Open load detected on low-side switch of half-bridge 7
3	HB6_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 6 1 _b = Open load detected on high-side switch of half-bridge 6
2	HB6_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 6 1 _b = Open load detected on low-side switch of half-bridge 6
1	HB5_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 5 1 _b = Open load detected on high-side switch of half-bridge 5
0	HB5_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 5 1 _b = Open load detected on low-side switch of half-bridge 5

3.5.1.1.7 Open-Load Detect (OLD) Status 3 (OLD_STAT_3) Register (Address = 0x06) [reset = 0x00]

The open-load detect (OLD) status 3 register is shown in [Table 3-32](#) and described in [Table 3-33](#).

Register access type: Read only

Table 3-32 Open-Load Detect (OLD) Status 3 Register

7	6	5	4	3	2	1	0
HB12_HS_OLD	HB12_LS_OLD	HB11_HS_OLD	HB11_LS_OLD	HB10_HS_OLD	HB10_LS_OLD	HB9_HS_OLD	HB9_LS_OLD
R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b	R-0 _b

Table 3-33 Open-Load Detect (OLD) Status 3 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB12_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 12 1 _b = Open load detected on high-side switch of half-bridge 12
6	HB12_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 12 1 _b = Open load detected on low-side switch of half-bridge 12
5	HB11_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 11 1 _b = Open load detected on high-side switch of half-bridge 11
4	HB11_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 11 1 _b = Open load detected on low-side switch of half-bridge 11
3	HB10_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 10 1 _b = Open load detected on high-side switch of half-bridge 10
2	HB10_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 10 1 _b = Open load detected on low-side switch of half-bridge 10
1	HB9_HS_OLD	R	0 _b	0 _b = No open load detected on high-side switch of half-bridge 9 1 _b = Open load detected on high-side switch of half-bridge 9
0	HB9_LS_OLD	R	0 _b	0 _b = No open load detected on low-side switch of half-bridge 9 1 _b = Open load detected on low-side switch of half-bridge 9

3.5.1.1.8 Additional Status (ADD_STAT_1) Register (Address = 0x30) [reset = 0x00]

The additional configuration (ADD_STAT_1) register is shown in [Table 3-34](#) and described in [Table 3-35](#).

Register access type: Read/Write

Table 3-34 Additional Configuration (ADD_STAT_1) Register

7	6	5	4	3	2	1	0
OTSD<11:7>					Reserved	OTSD_CL1	OTW_CL1
R/W-00000 _b					R -0 _b	R -0 _b	R -0 _b

Table 3-35 Additional Configuration (ADD_STAT_1) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	OTSD_11	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-12 is detected 1 _b = Overtemperature shutdown on half-bridge-12 is detected
6	OTSD_10	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-11 is detected 1 _b = Overtemperature shutdown on half-bridge-11 is detected
5	OTSD_9	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-10 is detected 1 _b = Overtemperature shutdown on half-bridge-10 is detected
4	OTSD_8	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-9 is detected 1 _b = Overtemperature shutdown on half-bridge-9 is detected
3	OTSD_7	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-8 is detected 1 _b = Overtemperature shutdown on half-bridge-8 is detected
2	Reserved	R	0 _b	Reserved
1	OTSD_CL1	R	0 _b	0 _b = No overtemperature shutdown on HE99MD9012 is detected 1 _b = Overtemperature shutdown on HE99MD9012 is detected
0	OTW_CL1	R	0 _b	0 _b = No overtemperature warning on HE99MD9012 is detected 1 _b = Overtemperature warning on HE99MD9012 is detected

3.5.1.1.9 Digital Thermal Status (TC_STAT_1) Register (Address = 0x31) [reset = 0x00]

Digital Thermal Status (TC_STAT_1) register is shown in [Table 3-36](#) and described in [Table 3-37](#).

Register access type: Read

Table 3-36 Digital Thermal Status (TC_STAT_1) Register

7	6	5	4	3	2	1	0
CP_UV	OTSD						
R-0 _b	R-0000000 _b						

Table 3-37 Digital Thermal Status (TC_STAT_1) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	CP_UV	R	0 _b	0 _b = No Charge pump undervoltage condition is detected 1 _b = Charge pump undervoltage condition is detected

6	OTSD_6	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-7 is detected 1 _b = Overtemperature shutdown on half-bridge-7 is detected
5	OTSD_5	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-6 is detected 1 _b = Overtemperature shutdown on half-bridge-6 is detected
4	OTSD_4	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-5 is detected 1 _b = Overtemperature shutdown on half-bridge-5 is detected
3	OTSD_3	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-4 is detected 1 _b = Overtemperature shutdown on half-bridge-4 is detected
2	OTSD_2	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-3 is detected 1 _b = Overtemperature shutdown on half-bridge-3 is detected
1	OTSD_1	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-2 is detected 1 _b = Overtemperature shutdown on half-bridge-2 is detected
0	OTSD_0	R	0 _b	0 _b = No Overtemperature shutdown on half-bridge-1 is detected 1 _b = Overtemperature shutdown on half-bridge-1 is detected

3.5.1.2 Control Registers

The control registers are used to configure the device. The control registers are read and write capable.

Table 3-38 lists the memory-mapped registers for the control registers. All register offset addresses not listed in **Table 3-38** should be considered as reserved locations and the register contents should not be modified.

Table 3-38 Control Registers Summary Table

Address	Register Name	Section
0x07	Configuration Register	3.5.1.2.1
0x08	Operation Control 1 Register	3.5.1.2.2
0x09	Operation Control 2 Register	3.5.1.2.3
0x0A	Operation Control 3 Register	3.5.1.2.4
0x0B	PWM Control 1 Register	3.5.1.2.5
0x0C	PWM Control 2 Register	3.5.1.2.6
0x0D	Free-Wheeling Control 1 Register	3.5.1.2.7
0x0E	Free-Wheeling Control 2 Register	3.5.1.2.8
0x0F	PWM Map Control 1 Register	3.5.1.2.9
0x10	PWM Map Control 2 Register	3.5.1.2.10
0x11	PWM Map Control 3 Register	3.5.1.2.11
0x12	PWM Frequency Control Register	3.5.1.2.12
0x13	PWM Duty Control Channel 1 Register	3.5.1.2.13
0x14	PWM Duty Control Channel 2 Register	3.5.1.2.14
0x15	PWM Duty Control Channel 3 Register	3.5.1.2.15
0x16	PWM Duty Control Channel 4 Register	3.5.1.2.16

Address	Register Name	Section
0x17	Slew Rate Control 1 Register	3.5.1.2.17
0x18	Slew Rate Control 2 Register	3.5.1.2.18
0x19	Open-Load Detect (OLD) Control 1 Register	3.5.1.2.19
0x1A	Open-Load Detect (OLD) Control 2 Register	3.5.1.2.20
0x1B	Open-Load Detect (OLD) Control 3 Register	3.5.1.2.21
0x24	Open-Load Detect (OLD) Control 4 Register	3.5.1.2.22
0x28	Additional Configuration Register	3.5.1.2.23
0x32	Additional Control External Register	3.5.1.2.24
0x33	Additional Open-Load Detect Control 1 Register	3.5.1.2.25
0x34	Additional Open-Load Detect Control 2 Register	3.5.1.2.26
0x35	Additional Open-Load Detect Control 3 Register	3.5.1.2.27

3.5.1.2.1 Configuration (CONFIG_CTRL) Register (Address = 0x07) [reset = 0x00]

The configuration register is shown in [Table 3-39](#) and described in [Table 3-40](#).

Register access type: Read/Write

Table 3-39 Configuration Register

7	6	5	4	3	2	1	0
Reserved	IC_ID			OCP_REP	OTW_REP	EXT_OVP	CLR_FLT
R/W-0 _b	R-XXX _b			R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-40 Configuration Register Field Descriptions

Bit	Field	Type	Reset	Description
7	Reserved	R/W	0 _b	Reserved
6-4	IC_ID	R	XXX _b	000 _b = Device connected is HE99MD9012 (12 Channel Device) Others = Reserved
3	OCP_REP	R/W	0 _b	0 _b = Overcurrent condition is reported in nFAULT pin 1 _b = Overcurrent condition warning is not reported on the nFAULT pin
2	OTW_REP	R/W	0 _b	0 _b = Overtemperature warning is not reported in nFAULT pin 1 _b = Overtemperature warning is reported on the nFAULT pin
1	EXT_OVP	R/W	0 _b	0 _b = Overvoltage protection threshold is at 23V (typ) 1 _b = Overvoltage protection threshold is at 33V (typ)
0	CLR_FLT	R/W	0 _b	0 _b = Faults not cleared 1 _b = Clear all faults

Note: CLR_FLT bit is an auto-clear bit and will always read 0_b.

3.5.1.2.2 Operation Control 1 (OP_CTRL_1) Register (Address = 0x08) [reset = 0x00]

The operation control 1 register is shown in [Table 3-41](#) and described in [Table 3-42](#).

Register access type: Read/Write

Table 3-41 Operation Control 1 Register

7	6	5	4	3	2	1	0
HB4_HS_EN	HB4_LS_EN	HB3_HS_EN	HB3_LS_EN	HB2_HS_EN	HB2_LS_EN	HB1_HS_EN	HB1_LS_EN
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-42 Operation Control 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB4_HS_EN	R/W	0 _b	0 _b = Half-bridge 4 high-side switch is disabled 1 _b = Half-bridge 4 high-side switch is enabled
6	HB4_LS_EN	R/W	0 _b	0 _b = Half-bridge 4 low-side switch is disabled 1 _b = Half-bridge 4 low-side switch is enabled
5	HB3_HS_EN	R/W	0 _b	0 _b = Half-bridge 3 high-side switch is disabled 1 _b = Half-bridge 3 high-side switch is enabled
4	HB3_LS_EN	R/W	0 _b	0 _b = Half-bridge 3 low-side switch is disabled 1 _b = Half-bridge 3 low-side switch is enabled
3	HB2_HS_EN	R/W	0 _b	0 _b = Half-bridge 2 high-side switch is disabled 1 _b = Half-bridge 2 high-side switch is enabled
2	HB2_LS_EN	R/W	0 _b	0 _b = Half-bridge 2 low-side switch is disabled 1 _b = Half-bridge 2 low-side switch is enabled
1	HB1_HS_EN	R/W	0 _b	0 _b = Half-bridge 1 high-side switch is disabled 1 _b = Half-bridge 1 high-side switch is enabled
0	HB1_LS_EN	R/W	0 _b	0 _b = Half-bridge 1 low-side switch is disabled 1 _b = Half-bridge 1 low-side switch is enabled

3.5.1.2.3 Operation Control 2 (OP_CTRL_2) Register (Address = 0x09) [reset = 0x00]

The operation control 2 register is shown in [Table 3-43](#) and described in [Table 3-44](#).

Register access type: Read/Write

Table 3-43 Operation Control 2 Register

7	6	5	4	3	2	1	0
HB8_HS_EN	HB8_LS_EN	HB7_HS_EN	HB7_LS_EN	HB6_HS_EN	HB6_LS_EN	HB5_HS_EN	HB5_LS_EN
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-44 Operation Control 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_HS_EN	R/W	0 _b	0 _b = Half-bridge 8 high-side switch is disabled 1 _b = Half-bridge 8 high-side switch is enabled
6	HB8_LS_EN	R/W	0 _b	0 _b = Half-bridge 8 low-side switch is disabled 1 _b = Half-bridge 8 low-side switch is enabled
5	HB7_HS_EN	R/W	0 _b	0 _b = Half-bridge 7 high-side switch is disabled 1 _b = Half-bridge 7 high-side switch is enabled
4	HB7_LS_EN	R/W	0 _b	0 _b = Half-bridge 7 low-side switch is disabled 1 _b = Half-bridge 7 low-side switch is enabled
3	HB6_HS_EN	R/W	0 _b	0 _b = Half-bridge 6 high-side switch is disabled 1 _b = Half-bridge 6 high-side switch is enabled
2	HB6_LS_EN	R/W	0 _b	0 _b = Half-bridge 6 low-side switch is disabled 1 _b = Half-bridge 6 low-side switch is enabled
1	HB5_HS_EN	R/W	0 _b	0 _b = Half-bridge 5 high-side switch is disabled 1 _b = Half-bridge 5 high-side switch is enabled
0	HB5_LS_EN	R/W	0 _b	0 _b = Half-bridge 5 low-side switch is disabled 1 _b = Half-bridge 5 low-side switch is enabled

3.5.1.2.4 Operation Control 3 (OP_CTRL_3) Register (Address = 0x0A) [reset = 0x00]

The operation control 3 register is shown in [Table 3-45](#) and described in [Table 3-46](#).

Register access type: Read/Write

Table 3-45 Operation Control 3 Register

7	6	5	4	3	2	1	0
HB12_HS_EN	HB12_LS_EN	HB11_HS_EN	HB11_LS_EN	HB10_HS_EN	HB10_LS_EN	HB9_HS_EN	HB9_LS_EN
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-46 Operation Control 3 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB12_HS_EN	R/W	0 _b	0 _b = Half-bridge 12 high-side switch is disabled 1 _b = Half-bridge 12 high-side switch is enabled
6	HB12_LS_EN	R/W	0 _b	0 _b = Half-bridge 12 low-side switch is disabled 1 _b = Half-bridge 12 low-side switch is enabled
5	HB11_HS_EN	R/W	0 _b	0 _b = Half-bridge 11 high-side switch is disabled 1 _b = Half-bridge 11 high-side switch is enabled
4	HB11_LS_EN	R/W	0 _b	0 _b = Half-bridge 11 low-side switch is disabled 1 _b = Half-bridge 11 low-side switch is enabled
3	HB10_HS_EN	R/W	0 _b	0 _b = Half-bridge 10 high-side switch is disabled 1 _b = Half-bridge 10 high-side switch is enabled
2	HB10_LS_EN	R/W	0 _b	0 _b = Half-bridge 10 low-side switch is disabled 1 _b = Half-bridge 10 low-side switch is enabled
1	HB9_HS_EN	R/W	0 _b	0 _b = Half-bridge 9 high-side switch is disabled 1 _b = Half-bridge 9 high-side switch is enabled
0	HB9_LS_EN	R/W	0 _b	0 _b = Half-bridge 9 low-side switch is disabled 1 _b = Half-bridge 9 low-side switch is enabled

3.5.1.2.5 PWM Control 1 (PWM_CTRL_1) Register (Address = 0x0B) [reset = 0x00]

The PWM control 1 register is shown in [Table 3-47](#) and described in [Table 3-48](#).

Register access type: Read/Write

Table 3-47 PWM Control 1 Register

7	6	5	4	3	2	1	0
HB8_PWM	HB7_PWM	HB6_PWM	HB5_PWM	HB4_PWM	HB3_PWM	HB2_PWM	HB1_PWM
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-48 PWM Control 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_PWM	R/W	0 _b	0 _b = Half-bridge 8 is operating in continuous mode 1 _b = Half-bridge 8 is operating in PWM mode
6	HB7_PWM	R/W	0 _b	0 _b = Half-bridge 7 is operating in continuous mode 1 _b = Half-bridge 7 is operating in PWM mode
5	HB6_PWM	R/W	0 _b	0 _b = Half-bridge 6 is operating in continuous mode 1 _b = Half-bridge 6 is operating in PWM mode
4	HB5_PWM	R/W	0 _b	0 _b = Half-bridge 5 is operating in continuous mode 1 _b = Half-bridge 5 is operating in PWM mode
3	HB4_PWM	R/W	0 _b	0 _b = Half-bridge 4 is operating in continuous mode 1 _b = Half-bridge 4 is operating in PWM mode
2	HB3_PWM	R/W	0 _b	0 _b = Half-bridge 3 is operating in continuous mode 1 _b = Half-bridge 3 is operating in PWM mode
1	HB2_PWM	R/W	0 _b	0 _b = Half-bridge 2 is operating in continuous mode 1 _b = Half-bridge 2 is operating in PWM mode
0	HB1_PWM	R/W	0 _b	0 _b = Half-bridge 1 is operating in continuous mode 1 _b = Half-bridge 1 is operating in PWM mode

3.5.1.2.6 PWM Control 2 (PWM_CTRL_2) Register (Address = 0x0C) [reset = 0x00]

The PWM control 1 register is shown in [Table 3-49](#) and described in [Table 3-50](#).

Register access type: Read/Write

Table 3-49 PWM Control 2 Register

7	6	5	4	3	2	1	0
PWM_CH4_DIS	PWM_CH3_DIS	PWM_CH2_DIS	PWM_CH1_DIS	HB12_PWM	HB11_PWM	HB10_PWM	HB9_PWM
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-50 PWM Control 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	PWM_CH4_DIS	R/W	0 _b	0 _b = PWM Generator-4 is enabled 1 _b = PWM Generator-4 is disabled
6	PWM_CH3_DIS	R/W	0 _b	0 _b = PWM Generator-3 is enabled 1 _b = PWM Generator-3 is disabled
5	PWM_CH2_DIS	R/W	0 _b	0 _b = PWM Generator-2 is enabled 1 _b = PWM Generator-2 is disabled
4	PWM_CH1_DIS	R/W	0 _b	0 _b = PWM Generator-1 is enabled 1 _b = PWM Generator-1 is disabled
3	HB12_PWM	R/W	0 _b	0 _b = Half-bridge 12 is operating in continuous mode 1 _b = Half-bridge 12 is operating in PWM mode
2	HB11_PWM	R/W	0 _b	0 _b = Half-bridge 11 is operating in continuous mode 1 _b = Half-bridge 11 is operating in PWM mode
1	HB10_PWM	R/W	0 _b	0 _b = Half-bridge 10 is operating in continuous mode 1 _b = Half-bridge 10 is operating in PWM mode
0	HB9_PWM	R/W	0 _b	0 _b = Half-bridge 9 is operating in continuous mode 1 _b = Half-bridge 9 is operating in PWM mode

3.5.1.2.7 Free-Wheeling Control 1 (FW_CTRL_1) Register (Address = 0x0D) [reset = 0x00]

The Free-Wheeling Control 1 register is shown in [Table 3-51](#) and described in [Table 3-52](#).

Register access type: Read/Write

Table 3-51 Free-Wheeling Control 1 Register

7	6	5	4	3	2	1	0
HB8_FW	HB7_FW	HB6_FW	HB5_FW	HB4_FW	HB3_FW	HB2_FW	HB1_FW
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-52 Free-Wheeling Control 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 8 is enabled 1 _b = Active free-wheeling on half-bridge 8 is enabled
6	HB7_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 7 is enabled 1 _b = Active free-wheeling on half-bridge 7 is enabled
5	HB6_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 6 is enabled 1 _b = Active free-wheeling on half-bridge 6 is enabled
4	HB5_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 5 is enabled 1 _b = Active free-wheeling on half-bridge 5 is enabled
3	HB4_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 4 is enabled 1 _b = Active free-wheeling on half-bridge 4 is enabled
2	HB3_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 3 is enabled 1 _b = Active free-wheeling on half-bridge 3 is enabled
1	HB2_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 2 is enabled 1 _b = Active free-wheeling on half-bridge 2 is enabled
0	HB1_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 1 is enabled 1 _b = Active free-wheeling on half-bridge 1 is enabled

3.5.1.2.8 Free-Wheeling Control 2 (FW_CTRL_2) Register (Address = 0x0E) [reset = 0x00]

The Free-Wheeling Control 2 register is shown in [Table 3-53](#) and described in [Table 3-54](#).

Register access type: Read/Write

Table 3-53 Free-Wheeling Control 2 Register

7	6	5	4	3	2	1	0
Reserved				HB4_FW	HB3_FW	HB2_FW	HB1_FW
R/W-0000 _b				R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-54 Free-Wheeling Control 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	Reserved	R/W	0000 _b	Reserved
3	HB12_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 12 is enabled 1 _b = Active free-wheeling on half-bridge 12 is enabled
2	HB11_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 11 is enabled 1 _b = Active free-wheeling on half-bridge 11 is enabled
1	HB10_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 10 is enabled 1 _b = Active free-wheeling on half-bridge 10 is enabled
0	HB9_FW	R/W	0 _b	0 _b = Passive free-wheeling on half-bridge 9 is enabled 1 _b = Active free-wheeling on half-bridge 9 is enabled

3.5.1.2.9 PWM Map Control 1 (PWM_MAP_CTRL_1) Register (Address = 0x0F) [reset = 0x00]

The PWM Map Control 1 register is shown in [Table 3-55](#) and described in [Table 3-56](#).

Register access type: Read/Write

Table 3-55 PWM Map Control 1 Register

7	6	5	4	3	2	1	0
HB4_PWM_MAP		HB3_PWM_MAP		HB2_PWM_MAP		HB1_PWM_MAP	
R/W-00 _b		R/W-00 _b		R/W-00 _b		R/W-00 _b	

Table 3-56 PWM Map Control 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	HB4_PWM_MAP	R/W	00 _b	00 _b = HB4 mapped to PWM channel 1 01 _b = HB4 mapped to PWM channel 2 10 _b = HB4 mapped to PWM channel 3 11 _b = HB4 mapped to PWM channel 4
5-4	HB3_PWM_MAP	R/W	00 _b	00 _b = HB3 mapped to PWM channel 1 01 _b = HB3 mapped to PWM channel 2 10 _b = HB3 mapped to PWM channel 3 11 _b = HB3 mapped to PWM channel 4
3-2	HB2_PWM_MAP	R/W	00 _b	00 _b = HB2 mapped to PWM channel 1 01 _b = HB2 mapped to PWM channel 2 10 _b = HB2 mapped to PWM channel 3 11 _b = HB2 mapped to PWM channel 4
1-0	HB1_PWM_MAP	R/W	00 _b	00 _b = HB1 mapped to PWM channel 1 01 _b = HB1 mapped to PWM channel 2 10 _b = HB1 mapped to PWM channel 3 11 _b = HB1 mapped to PWM channel 4

3.5.1.2.10 PWM Map Control 2 (PWM_MAP_CTRL_2) Register (Address = 0x10) [reset = 0x00]

The PWM Map Control 2 register is shown in [Table 3-57](#) and described in [Table 3-58](#).

Register access type: Read/Write

Table 3-57 PWM Map Control 2 Register

7	6	5	4	3	2	1	0
HB8_PWM_MAP		HB7_PWM_MAP		HB6_PWM_MAP		HB5_PWM_MAP	
R/W-00 _b		R/W-00 _b		R/W-00 _b		R/W-00 _b	

Table 3-58 PWM Map Control 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	HB8_PWM_MAP	R/W	00 _b	00 _b = HB8 mapped to PWM channel 1 01 _b = HB8 mapped to PWM channel 2 10 _b = HB8 mapped to PWM channel 3 11 _b = HB8 mapped to PWM channel 4
5-4	HB7_PWM_MAP	R/W	00 _b	00 _b = HB7 mapped to PWM channel 1 01 _b = HB7 mapped to PWM channel 2 10 _b = HB7 mapped to PWM channel 3 11 _b = HB7 mapped to PWM channel 4
3-2	HB6_PWM_MAP	R/W	00 _b	00 _b = HB6 mapped to PWM channel 1 01 _b = HB6 mapped to PWM channel 2 10 _b = HB6 mapped to PWM channel 3 11 _b = HB6 mapped to PWM channel 4
1-0	HB5_PWM_MAP	R/W	00 _b	00 _b = HB5 mapped to PWM channel 1 01 _b = HB5 mapped to PWM channel 2 10 _b = HB5 mapped to PWM channel 3 11 _b = HB5 mapped to PWM channel 4

3.5.1.2.11 PWM Map Control 3 (PWM_MAP_CTRL_3) Register (Address = 0x11) [reset = 0x00]

The PWM Map Control 3 register is shown in [Table 3-59](#) and described in [Table 3-60](#).

Register access type: Read/Write

Table 3-59 PWM Map Control 3 Register

7	6	5	4	3	2	1	0
HB12_PWM_MAP		HB11_PWM_MAP		HB10_PWM_MAP		HB9_PWM_MAP	
R/W-00 _b		R/W-00 _b		R/W-00 _b		R/W-00 _b	

Table 3-60 PWM Map Control 3 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	HB12_PWM_MAP	R/W	00 _b	00 _b = HB12 mapped to PWM channel 1 01 _b = HB12 mapped to PWM channel 2 10 _b = HB12 mapped to PWM channel 3 11 _b = HB12 mapped to PWM channel 4
5-4	HB11_PWM_MAP	R/W	00 _b	00 _b = HB11 mapped to PWM channel 1 01 _b = HB11 mapped to PWM channel 2 10 _b = HB11 mapped to PWM channel 3 11 _b = HB11 mapped to PWM channel 4
3-2	HB10_PWM_MAP	R/W	00 _b	00 _b = HB10 mapped to PWM channel 1 01 _b = HB10 mapped to PWM channel 2 10 _b = HB10 mapped to PWM channel 3 11 _b = HB10 mapped to PWM channel 4
1-0	HB9_PWM_MAP	R/W	00 _b	00 _b = HB9 mapped to PWM channel 1 01 _b = HB9 mapped to PWM channel 2 10 _b = HB9 mapped to PWM channel 3 11 _b = HB9 mapped to PWM channel 4

3.5.1.2.12 PWM Frequency Control (PWM_FREQ_CTRL) Register (Address = 0x12) [reset = 0x00]

The PWM Frequency Control register is shown in [Table 3-61](#) and described in [Table 3-62](#).

Register access type: Read/Write

Table 3-61 PWM Frequency Control Register

7	6	5	4	3	2	1	0
PWM_CH4_FREQ		PWM_CH3_FREQ		PWM_CH2_FREQ		PWM_CH1_FREQ	
R/W-00 _b		R/W-00 _b		R/W-00 _b		R/W-00 _b	

Table 3-62 PWM Frequency Control Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	PWM_CH4_FREQ	R/W	00 _b	00 _b = PWM frequency is 80 Hz 01 _b = PWM frequency is 100 Hz 10 _b = PWM frequency is 200 Hz 11 _b = PWM frequency is 2000 Hz
5-4	PWM_CH3_FREQ	R/W	00 _b	00 _b = PWM frequency is 80 Hz 01 _b = PWM frequency is 100 Hz 10 _b = PWM frequency is 200 Hz 11 _b = PWM frequency is 2000 Hz
3-2	PWM_CH2_FREQ	R/W	00 _b	00 _b = PWM frequency is 80 Hz 01 _b = PWM frequency is 100 Hz 10 _b = PWM frequency is 200 Hz 11 _b = PWM frequency is 2000 Hz
1-0	PWM_CH1_FREQ	R/W	00 _b	00 _b = PWM frequency is 80 Hz 01 _b = PWM frequency is 100 Hz 10 _b = PWM frequency is 200 Hz 11 _b = PWM frequency is 2000 Hz

3.5.1.2.13 PWM Duty Control Channel 1 (PWM_DUTY_CTRL_1) Register (Address = 0x13) [reset = 0x00]

The channel 1 PWM duty cycle control register is shown in [Table 3-63](#) and described in [Table 3-64](#).

Register access type: Read/Write

Table 3-63 PWM Duty Control 1 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH1							
R/W-00000000 _b							

Table 3-64 PWM Duty Control 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	PWM_DUTY_CH1	R/W	00000000 _b	00000000 _b = 0% PWM Duty 11111111 _b = 100% PWM Duty Calculate duty as decimal (xxxxxxx _b) × 1/255

3.5.1.2.14 PWM Duty Control Channel 2 (PWM_DUTY_CTRL_2) Register (Address = 0x14) [reset = 0x00]

The channel 2 PWM duty cycle control register is shown in [Table 3-65](#) and described in [Table 3-66](#).

Register access type: Read/Write

Table 3-65 PWM Duty Control 2 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH2							
R/W-00000000 _b							

Table 3-66 PWM Duty Control 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	PWM_DUTY_CH2	R/W	00000000 _b	00000000 _b = 0% PWM Duty 11111111 _b = 100% PWM Duty Calculate duty as decimal (xxxxxxx _b) × 1/255

3.5.1.2.15 PWM Duty Control Channel 3 (PWM_DUTY_CTRL_3) Register (Address = 0x15) [reset = 0x00]

The channel 3 PWM duty cycle control register is shown in [Table 3-67](#) and described in [Table 3-68](#).

Register access type: Read/Write

Table 3-67 PWM Duty Control 3 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH3							
R/W-00000000 _b							

Table 3-68 PWM Duty Control 3 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	PWM_DUTY_CH3	R/W	00000000 _b	00000000 _b = 0% PWM Duty 11111111 _b = 100% PWM Duty Calculate duty as decimal (xxxxxxx _b) × 1/255

3.5.1.2.16 PWM Duty Control Channel 4 (PWM_DUTY_CTRL_4) Register (Address = 0x16) [reset = 0x00]

The channel 4 PWM duty cycle control register is shown in [Table 3-69](#) and described in [Table 3-70](#).

Register access type: Read/Write

Table 3-69 PWM Duty Control 4 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH4							
R/W-00000000 _b							

Table 3-70 PWM Duty Control 4 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-0	PWM_DUTY_CH4	R/W	00000000 _b	00000000 _b = 0% PWM Duty 11111111 _b = 100% PWM Duty Calculate duty as decimal (xxxxxxx _b) × 1/255

3.5.1.2.17 Slew Rate Control 1 (SR_CTRL_1) Register (Address = 0x17) [reset = 0x00]

The Slew Rate Control 1 register is shown in [Table 3-71](#) and described in [Table 3-72](#).

Register access type: Read/Write

Table 3-71 Slew Rate Control 1 Register

7	6	5	4	3	2	1	0
HB8_SR	HB7_SR	HB6_SR	HB5_SR	HB4_SR	HB3_SR	HB2_SR	HB1_SR
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-72 Slew Rate Control 1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
6	HB7_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
5	HB6_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
4	HB5_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
3	HB4_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
2	HB3_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
1	HB2_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
0	HB1_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs

3.5.1.2.18 Slew Rate Control 2 (SR_CTRL_2) Register (Address = 0x18) [reset = 0x00]

The Slew Rate Control 2 register is shown in [Table 3-73](#) and described in [Table 3-74](#).

Register access type: Read/Write

Table 3-73 Slew Rate Control 2 Register

7	6	5	4	3	2	1	0
Reserved				HB12_SR	HB11_SR	HB10_SR	HB9_SR
R/W-0000 _b				R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-74 Slew Rate Control 2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	Reserved	R/W	0000 _b	Reserved
3	HB12_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
2	HB11_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
1	HB10_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs
0	HB9_SR	R/W	0 _b	0 _b = 0.6V/μs 1 _b = 2.5V/μs

3.5.1.2.19 Open-Load Detect (OLD) Control 1 (OLD_CTRL_1) Register (Address = 0x19) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_1) Register-1 is shown in [Table 3-75](#) and described in [Table 3-76](#).

Register access type: Read/Write

Table 3-75 Open-Load Detect (OLD) Control (OLD_CTRL_1) Register

7	6	5	4	3	2	1	0
HB8_OLD_DIS	HB7_OLD_DIS	HB6_OLD_DIS	HB5_OLD_DIS	HB4_OLD_DIS	HB3_OLD_DIS	HB2_OLD_DIS	HB1_OLD_DIS
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-76 Open-Load Detect (OLD) Control (OLD_CTRL_1) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 8 is enabled 1 _b = Open-load on half-bridge 8 is disabled
6	HB7_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 7 is enabled 1 _b = Open-load on half-bridge 7 is disabled
5	HB6_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 6 is enabled 1 _b = Open-load on half-bridge 6 is disabled
4	HB5_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 5 is enabled 1 _b = Open-load on half-bridge 5 is disabled
3	HB4_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 4 is enabled 1 _b = Open-load on half-bridge 4 is disabled
2	HB3_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 3 is enabled 1 _b = Open-load on half-bridge 3 is disabled
1	HB2_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 2 is enabled 1 _b = Open-load on half-bridge 2 is disabled
0	HB1_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 1 is enabled 1 _b = Open-load on half-bridge 1 is disabled

3.5.1.2.20 Open-Load Detect (OLD) Control 2 (OLD_CTRL_2) Register (Address = 0x1A) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_2) Register-2 is shown in [Table 3-77](#) and described in [Table 3-78](#).

Register access type: Read/Write

Table 3-77 Open-Load Detect (OLD) Control (OLD_CTRL_2) Register

7	6	5	4	3	2	1	0
OLD_REP	OLD_OP	PL_MODE_EN		HB12_OLD_DIS	HB11_OLD_DIS	HB2_OLD_DIS	HB1_OLD_DIS
R/W-0 _b	R/W-0 _b	W-00 _b		R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-78 Open-Load Detect (OLD) Control (OLD_CTRL_2) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	OLD_REP	R/W	0 _b	0 _b = Report on nFAULT pin during OLD condition 1 _b = No report on nFAULT pin during OLD condition
6	OLD_OP	R/W	0 _b	0 _b = Half bridges are not active after OLD condition detect 1 _b = Half bridges are active after OLD condition detect
5-4	PL_MODE_EN	W	00 _b	00 _b = Parallel mode OCP fast turn-off slew is enabled 01 _b = Parallel mode OCP slow turn-off slew is enabled 10 _b = Invalid Setting 11 _b = Invalid Setting
3	HB12_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 12 is enabled 1 _b = Open-load on half-bridge 12 is disabled
2	HB11_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 11 is enabled 1 _b = Open-load on half-bridge 11 is disabled
1	HB10_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 10 is enabled 1 _b = Open-load on half-bridge 10 is disabled
0	HB9_OLD_DIS	R/W	0 _b	0 _b = Open-load detection on half-bridge 9 is enabled 1 _b = Open-load on half-bridge 9 is disabled

Note: The PL_MODE_EN is write only bits and always read 00_b.

3.5.1.2.21 Open-Load Detect (OLD) Control 3 (OLD_CTRL_3) Register (Address = 0x1B) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_3) Register-3 is shown in [Table 3-79](#) and described in [Table 3-80](#).

Register access type: Read/Write

Table 3-79 Open-Load Detect (OLD) Control (OLD_CTRL_3) Register

7	6	5	4	3	2	1	0
OCP_DEG			OLD_NEG_EN	HB12_LOLD_EN	HB11_LOLD_EN	HB10_LOLD_EN	HB9_LOLD_EN
R/W-000 _b			R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-80 Open-Load Detect (OLD) Control (OLD_CTRL_3) Register Field Descriptions

Bit	Field	Type	Reset	Description
7-5	OCP_DEG	R/W	000 _b	000 _b = OCP deglitch time is 10μs 001 _b = OCP deglitch time is 5μs 010 _b = OCP deglitch time is 2.5μs 011 _b = OCP deglitch time is 1μs 100 _b = OCP deglitch time is 60μs 101 _b = OCP deglitch time is 40μs 110 _b = OCP deglitch time is 30μs 111 _b = OCP deglitch time is 20μs
4	OLD_NEG_EN	R/W	0 _b	0 _b = Negative-current OLD mode is disabled 1 _b = Negative-current OLD mode is enabled
3	HB12_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 12 is disabled 1 _b = Low-current OLD on half-bridge 12 is enabled
2	HB11_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 11 is disabled 1 _b = Low-current OLD on half-bridge 11 is enabled
1	HB10_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 10 is disabled 1 _b = Low-current OLD on half-bridge 10 is enabled
0	HB9_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 9 is disabled 1 _b = Low-current OLD on half-bridge 9 is enabled

3.5.1.2.22 Open Load Detect (OLD) Control 4 (OLD_CTRL_4) Register (Address = 0x24) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_4) Register-4 is shown in [Table 3-81](#) and described in [Table 3-82](#).

Register access type: Read/Write

Table 3-81 Open-Load Detect (OLD) Control (OLD_CTRL_4) Register

7	6	5	4	3	2	1	0
HB8_LOLD_EN	HB7_LOLD_EN	HB6_LOLD_EN	HB5_LOLD_EN	HB4_LOLD_EN	HB3_LOLD_EN	HB2_LOLD_EN	HB1_LOLD_EN
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-82 Open-Load Detect (OLD) Control (OLD_CTRL_4) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HB8_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 8 is disabled 1 _b = Low-current OLD on half-bridge 8 is enabled
6	HB7_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 7 is disabled 1 _b = Low-current OLD on half-bridge 7 is enabled
5	HB6_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 6 is disabled 1 _b = Low-current OLD on half-bridge 6 is enabled
4	HB5_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 5 is disabled 1 _b = Low-current OLD on half-bridge 5 is enabled
3	HB4_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 4 is disabled 1 _b = Low-current OLD on half-bridge 4 is enabled
2	HB3_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 3 is disabled 1 _b = Low-current OLD on half-bridge 3 is enabled
1	HB2_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 2 is disabled 1 _b = Low-current OLD on half-bridge 2 is enabled
0	HB1_LOLD_EN	R/W	0 _b	0 _b = Low-current OLD on half-bridge 1 is disabled 1 _b = Low-current OLD on half-bridge 1 is enabled

3.5.1.2.23 Additional Configuration (ADD_CONFIG_CTRL) Register (Address = 0x28) [reset = 0x00]

The additional configuration (ADD_CONFIG_CTRL) register is shown in [Table 3-83](#) and described in [Table 3-84](#).

Register access type: Read/Write

Table 3-83 Additional Configuration (ADD_CONFIG_CTRL) Register

7	6	5	4	3	2	1	0
Reserved				CP_OTSD_DIS	CP_VM_UV_DIS	TSD_PerHB_EN	OVP_OP
R/W-0000 _b				R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-84 Additional Configuration (ADD_CONFIG_CTRL) Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	Reserved	R/W	0000 _b	Reserved
3	CP_OTSD_DIS	R/W	0 _b	0 _b = When the OTSD occurs, the charge pump will be turned off. 1 _b = When the OTSD occurs, the charge pump will not turn off
2	CP_VM_UV_DIS	R/W	0 _b	0 _b = When the VM_UV occurs, the charge pump will be turned off. 1 _b = When the VM_UV occurs, the charge pump will not turn off
1	TSD_PerHB_EN	R/W	0 _b	0 _b = Standard Mode: as soon as any half-bridge reaches thermal threshold the device is switched off 1 _b = Per half-bridge Mode: only the half-bridge that has reached shutdown temperature is switched off
0	OVP_OP	R/W	0 _b	0 _b = Half bridges are not active after OVP condition detect 1 _b = Half bridges are active after OVP condition detect

3.5.1.2.24 Additional Control External (ADD_CTRL_EX) Register (Address = 0x32) [reset = 0x00]

Additional Control External (ADD_CTRL_EX) register is shown in [Table 3-85](#) and described in [Table 3-86](#).

Register access type: Read/Write

Table 3-85 Additional Control External (ADD_CTRL_EX) Register

7	6	5	4	3	2	1	0
Reserved		OT_SEL_WIDTH		OT_FLT		OSC_MODU_DIS	OSC_MODU_SEL
R/W-00 _b		R/W-00 _b		R/W-00 _b		R/W-0 _b	R/W-0 _b

Table 3-86 Additional Control External (ADD_CTRL_EX) Register Field Descriptions

Bit	Field	Type	Reset	Description
-----	-------	------	-------	-------------

7-6	Reserved	R/W	00 _b	Reserved
5-4	OT_SEL_WIDTH	R/W	01 _b	Single output half-bridge monitoring time 00 _b = pulse width 32us 01 _b = pulse width 16us 10 _b = pulse width 48us 11 _b = pulse width 64us
3-2	OT_FLT	R/W	10 _b	OT/OTW filter time 00 _b = filter time 20us 01 _b = filter time 5us 10 _b = filter time 10us 11 _b = filter time 40us
1	OSC_MODU_DIS	R/W	0 _b	0 _b = spread spectrum mode enable 1 _b = spread spectrum mode disable
0	OSC_MODU_SEL	R/W	0 _b	0 _b = spread spectrum range $\pm 4\%$ 1 _b = spread spectrum range $\pm 12\%$

Note: HE99MD9012 have 12 output channel , which are monitored in turn. Each output channel monitoring time can be selected by OT_SEL_WIDTH.

3.5.1.2.25 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register (Address = 0x33) [reset = 0x00]

Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register is shown in [Table 3-87](#) and described in [Table 3-88](#).

Register access type: Read/Write

Table 3-87 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register

7	6	5	4	3	2	1	0
HS8_NEG_OLD_EN	HS7_NEG_OLD_EN	HS6_NEG_OLD_EN	HS5_NEG_OLD_EN	HS4_NEG_OLD_EN	HS3_NEG_OLD_EN	HS2_NEG_OLD_EN	HS1_NEG_OLD_EN
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-88 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HS8_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 8 is disabled 1 _b = Negative-current OLD on high-side 8 is enabled
6	HS7_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 7 is disabled 1 _b = Negative-current OLD on high-side 7 is enabled
5	HS6_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 6 is disabled 1 _b = Negative-current OLD on high-side 6 is enabled
4	HS5_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 5 is disabled 1 _b = Negative-current OLD on high-side 5 is enabled
3	HS4_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 4 is disabled 1 _b = Negative-current OLD on high-side 4 is enabled
2	HS3_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 3 is disabled 1 _b = Negative-current OLD on high-side 3 is enabled
1	HS2_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 2 is disabled 1 _b = Negative-current OLD on high-side 2 is enabled
0	HS1_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 1 is disabled 1 _b = Negative-current OLD on high-side 1 is enabled

3.5.1.2.26 Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register (Address = 0x34) [reset = 0x00]

Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register is shown in [Table 3-89](#) and described in [Table 3-90](#).

Table 3-89 Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register

7	6	5	4	3	2	1	0
HS12_NEG_OLD_EN	HS11_NEG_OLD_EN	HS10_NEG_OLD_EN	HS9_NEG_OLD_EN	LS12_NEG_OLD_EN	LS11_NEG_OLD_EN	LS10_NEG_OLD_EN	LS9_NEG_OLD_EN
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-90 Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	HS12_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 12 is disabled 1 _b = Negative-current OLD on high-side 12 is enabled
6	HS11_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 11 is disabled 1 _b = Negative-current OLD on high-side 11 is enabled
5	HS10_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 10 is disabled 1 _b = Negative-current OLD on high-side 10 is enabled
4	HS9_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on high-side 9 is disabled 1 _b = Negative-current OLD on high-side 9 is enabled
3	LS12_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 12 is disabled 1 _b = Negative-current OLD on low-side 12 is enabled
2	LS11_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 11 is disabled 1 _b = Negative-current OLD on low-side 11 is enabled
1	LS10_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 10 is disabled 1 _b = Negative-current OLD on low-side 10 is enabled
0	LS9_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 9 is disabled 1 _b = Negative-current OLD on low-side 9 is enabled

3.5.1.2.27 Additional Open-Load Detect Control (ADD_OLD_CTRL_3) Register (Address = 0x35) [reset = 0x00]

Additional Open-Load Detect Control (ADD_OLD_CTRL_3) Register is shown in [Table 3-91](#) and described in [Table 3-92](#).

Register access type: Read/Write

Table 3-91 Additional Open-Load Detect Control (ADD_OLD_CTRL_3) Register

7	6	5	4	3	2	1	0
LS8_NEG_O LD_EN	LS7_NEG_O LD_EN	LS6_NEG_O LD_EN	LS5_NEG_O LD_EN	LS4_NEG_O LD_EN	LS3_NEG_O LD_EN	LS2_NEG_O LD_EN	LS1_NEG_O LD_EN
R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b	R/W-0 _b

Table 3-92 Additional Open-Load Detect Control (ADD_OLD_CTRL_3) Register Field Descriptions

Bit	Field	Type	Reset	Description
7	LS8_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 8 is disabled 1 _b = Negative-current OLD on low-side 8 is enabled
6	LS7_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 7 is disabled 1 _b = Negative-current OLD on low-side 7 is enabled
5	LS6_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 6 is disabled 1 _b = Negative-current OLD on low-side 6 is enabled
4	LS5_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 5 is disabled 1 _b = Negative-current OLD on low-side 5 is enabled
3	LS4_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 4 is disabled 1 _b = Negative-current OLD on low-side 4 is enabled
2	LS3_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 3 is disabled 1 _b = Negative-current OLD on low-side 3 is enabled
1	LS2_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 2 is disabled 1 _b = Negative-current OLD on low-side 2 is enabled
0	LS1_NEG_OLD_EN	R/W	0 _b	0 _b =Negative-current OLD on low-side 1 is disabled 1 _b = Negative-current OLD on low-side 1 is enabled

4 Typical Application

4.1 Primary Application

The HE99MD90xx device is primarily used for the control of multiple brushed DC motors which can be connected in independent-type, sequential-type or the parallel-type motor connection as shown in **Figure 4-1**.

An automotive battery powers the device to power supply pin (VM). A 3.3V regulated power supply is generated for the supplying power to the SPI (VDD) of the device. A micro-controller is connected to the HE99MD90xx device via the SPI interface (4-lines) for control, configuration and diagnostics. The device operating or sleep state is controlled by the nSLEEP pin and nFAULT pins is used as an additional hardware diagnostic.

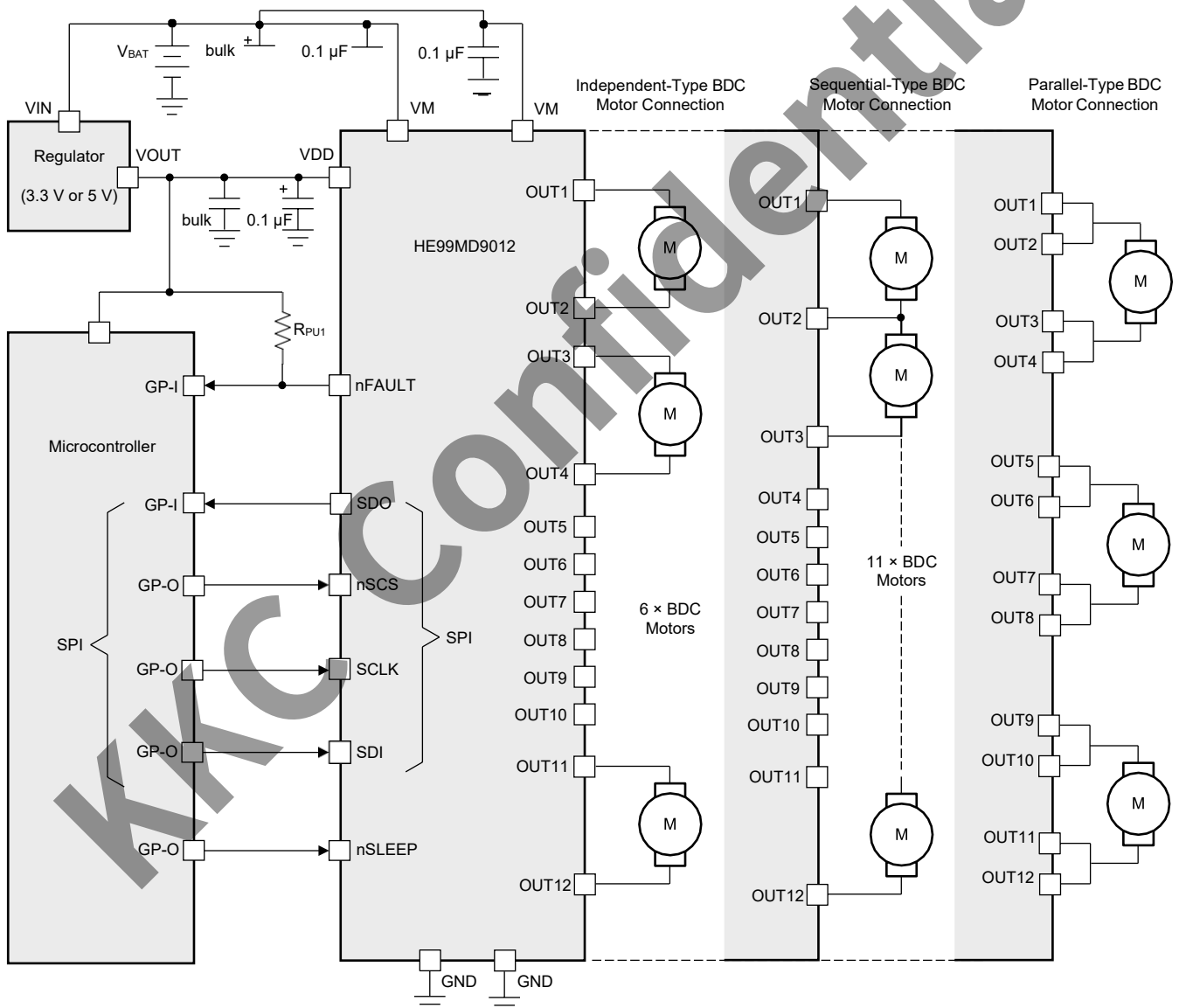


Figure 4-1 Primary Application Schematic (Automotive HVAC Application)

4.1.1 Design Requirements

Table 4-1 lists example input parameters for the system design.

Table 4-1 Design Parameters

DESIGN PARAMETERS	REFERENCE	EXAMPLE VALUE
Supply voltage	V_{VM}	13.5-V
Supply digital voltage	V_{VDD}	3.3-V
Number of motor connected	N	6 motors
Number of motor operating in normal operation	N_F	4 motors
Number of motor operating in stall condition	N_S	2 motors
Motor RMS current	I_{RMS}	200mA
Motor peak current	I_{PEAK}	800mA
Motor resistance	R_{MOTOR}	16.9 Ω
Motor inductance	L_{MOTOR}	10mH
PWM Frequency	f_{PWM}	2kHz (Internal)
Rise and fall time for continuous mode (SR = 0)	$t_{RISE_CONT}, t_{FALL_CONT}$	22.5 μ s
Rise and fall time for PWM mode (SR = 1)	$t_{RISE_PWM}, t_{FALL_PWM}$	5.4 μ s

4.1.2 Detailed Design Procedure

The design procedure includes the selection of motor current rating the power dissipation to meet the desired thermal performance.

4.1.2.1 Motor Current Rating

Motor specification selection is the most importance criteria for the design. Each half-bridge (OUTx) of the HE99MD90xx device is designed to handle RMS current of 1A and the peak current is limited by the minimum over-current (OCP) limit of 1.3A. Therefore, a motor with peak starting current higher than 1.3A is expected to hit OCP limit. For higher peak current motors (starting current higher than 1.3A), following methods can be implemented:

- **Current Chopping:** During starting, if supply voltage is connected directly to the motor, then due to low back-emf (when speed is zero or low), a huge peak current is demanded by the motor. This peak current is only limited by the motor's winding resistance (R_{MOTOR}). This peak current of motor can be limited by starting the motor with low-duty PWM switching operation and then gradually increasing (duty-ramping) the duty with speed to 100% PWM operation (equivalent to motor operating in continuous mode). This duty-ramping provides enough time to ramp motor speed and build sufficient back-emf which limits the peak current. The HE99MD90xx device implements a 2kHz PWM switching operation which is suitable for the HVAC damper motors.
- **OCP Deglitch Time Adjustments:** This method is applicable if the motor inertia is low and the motor can quickly pick up the speed. For this method, the motor starting current should settle to lower than minimum over-current limit (I_{OCP}) before OCP deglitch time (t_{OCP}) is over. The device provides multiple (8 settings) OCP deglitch time settings with a default deglitch time of 10 μ s and can be increased to a maximum value of 60 μ s.

Note: For multiple motor connection, it has to be ensured that the total device current should be lower than the maximum current-carrying capability of the power-supply (VM/GND) pins i.e. 6A (maximum).

4.2 Alternative Application

The HE99MD90xx device can alternatively be used for the mirrors targeting the mirror-fold, mirror x-y direction control and side indicator LED's as shown in **Figure 4-2**.

The half-bridges are connected in parallel to support the higher current requirement of the mirror fold application. Whereas single half-bridges can be used for driving the low-current motors used for the mirror X and Y positioning. Moreover, the LED's used inside indicators, puddle lamp is lower current which can be easily driven by single half-bridges.

The driver is powered by the automotive battery with a 5V regulated power supply generated for the supplying power to the digital pin (VDD). A micro-controller is connected to the HE99MD90xx device with the SPI interface (4-lines) for control, configuration and diagnostics. The device operating or sleep state is controlled by the nSLEEP pin and nFAULT pins is used as an additional hardware diagnostic.

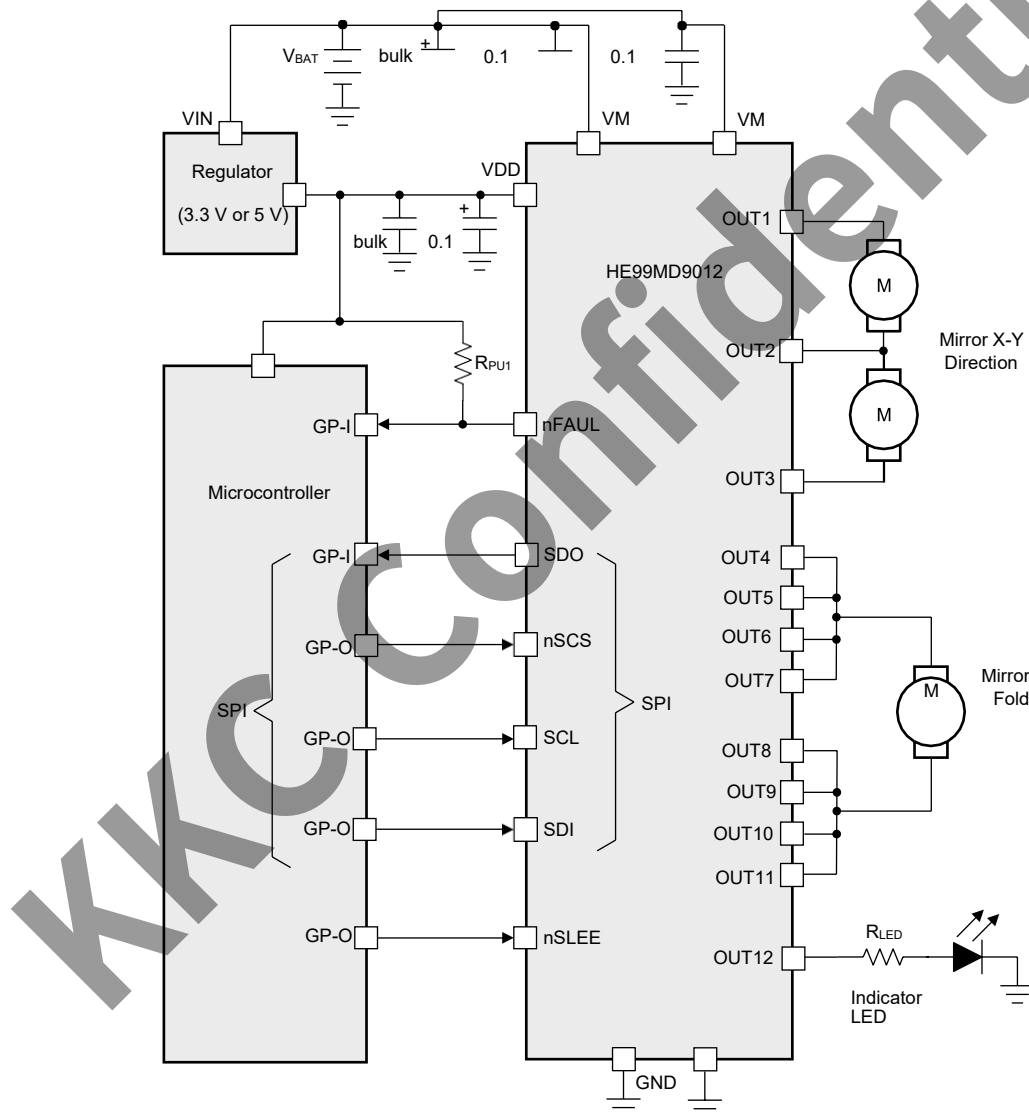


Figure 4-2 Alternative Application Schematic (Automotive Side-Mirror Application)

5 Package and information

5.1 ETSSOP mechanical data

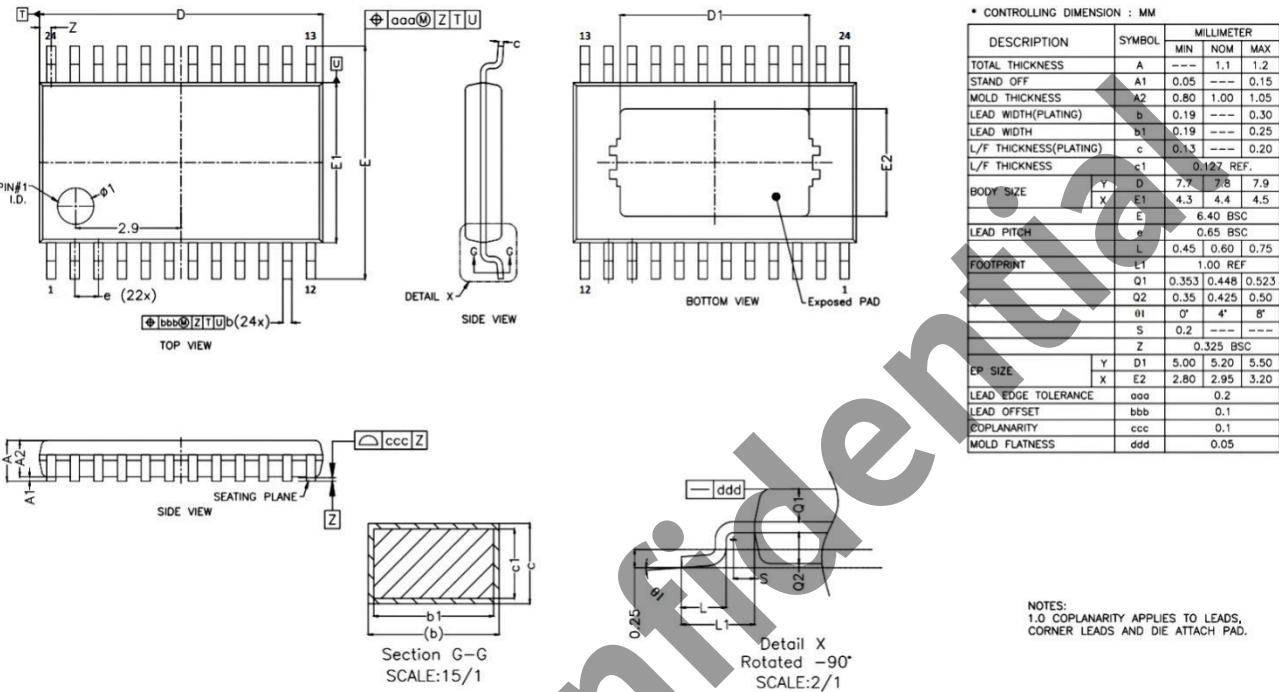
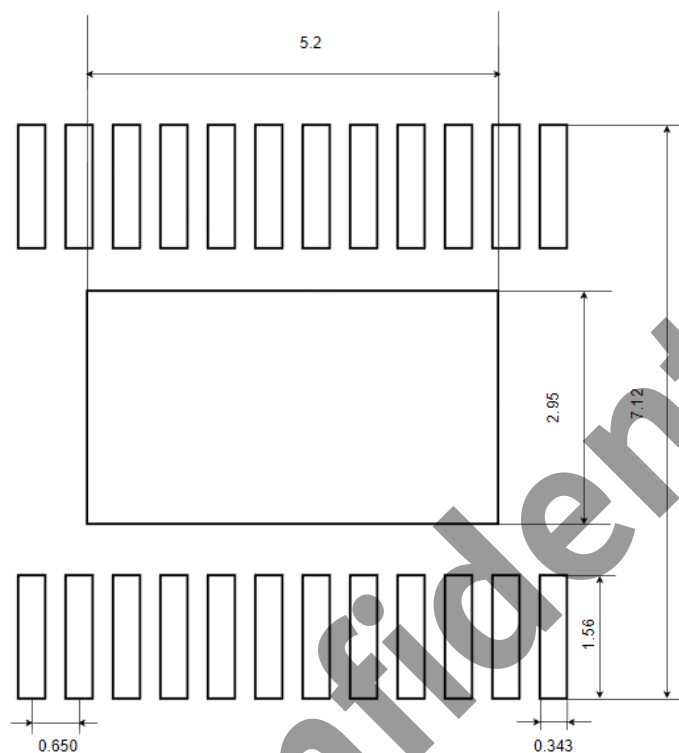


Figure 5-1 Mechanical data

6 Recommended pad size



Unit: mm

Figure 6-1 Recommended pad size

7 Customer Service

KKChips

9th Floor, Grandtek Building B2, No. 18 Kexue Avenue, Huangpu District, Guangzhou.

KKC Confidential

8 Revision history

Date	Revision	Changes
2025/04/23	0.0.1	1. Initial release
2025/12/15	0.0.2	1. Design target version
2026/05/12	0.0.3	1. Update the series of products

KKC Confidential